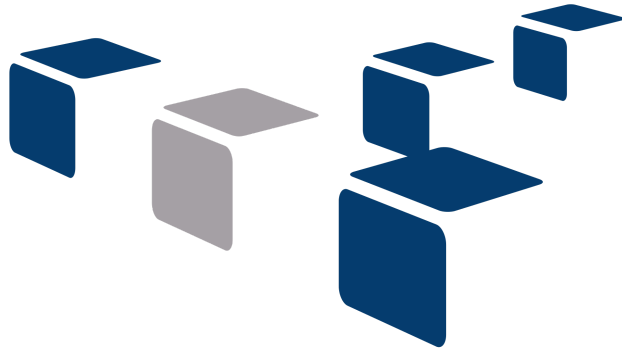


# InnoPick Manager Operations Manual

## Automated Order Preparation Solutions



# INNO PICK®

---

*Manual Revision 3.0: Published February 2026*

---

## Welcome

Welcome to the InnoPick Manager Technical Manual, designed to provide you with an in-depth understanding of the features, functionalities, and operational details of the InnoPick system.

This manual has been prepared to guide users of all levels through the setup, operation, and optimization of the InnoPick Manager software.

---

## Documentation Sections

### Getting Started

Learn how to access and navigate InnoPick Manager for the first time.

### Daily Operations

Standard procedures for starting up, monitoring, and shutting down the system.

### Main Screens

Detailed reference for all screens and features in InnoPick Manager.

### Troubleshooting

Alert resolution procedures and problem-solving guidance.

### Quick Reference

Quick lookup tables for alert codes and system status indicators.

---

## User Roles

Access to certain features depends on your user role:

- **Admin:** Full access to all controls and settings
- **Manager:** Access to operational settings, but not site configuration
- **Clerk:** Access to operate the system for production
- **Guest:** View-only access to browse screens

See [User Manager](#) for more information.

---

## System Overview

InnoPick Manager (IPM) is a web-based software interface for controlling and monitoring the InnoPick automated case picking system. The system manages:

- Automated case storage and retrieval across multiple levels
  - Replenishment scheduling and tracking
  - Case sequencing for order fulfillment
  - Real-time inventory management
  - Alert monitoring and resolution
- 

## Getting Help

- For operational questions, consult the [Troubleshooting](#) section
  - For alert resolution, see the [Alert Reference Page](#)
  - For system configuration, see [Administration Pages](#)
- 

**Navigation:** [Next: Getting Started](#) →

# Getting Started

[Home](#) > **Getting Started**

---

## Overview

This section will guide you through your first interactions with InnoPick Manager, from logging in to understanding the basic interface layout.

---

## What You'll Learn

- [Introduction](#) - Overview of InnoPick Manager and this manual
  - [Logging In](#) - How to access InnoPick Manager at your terminal
  - [Interface Overview](#) - Understanding the main navigation and screen layout
- 

## Prerequisites

Before you begin:

- You need a valid username and password for InnoPick Manager
  - The system must be accessible via your facility's network
- 

## Quick Start

1. **First time logging in?** Start with [Logging In](#)
  2. **Want to understand the interface?** See [Interface Overview](#)
  3. **Ready to operate the system?** Move to [Daily Operations](#)
- 

## Access Levels

Your user account will have one of the following roles, which determines what you can view and control:

- **Guest:** Browse screens but cannot make changes
- **Clerk:** Operate the system for production
- **Manager:** Access operational settings
- **Admin:** Full access to all controls and configuration

If you're unsure of your access level, ask your supervisor or system administrator.

---

**Navigation:** [← Home](#) | [Introduction →](#)

# Introduction

[Home](#) > [Getting Started](#) > **Introduction**

---

## About InnoPick Manager

InnoPick Manager is a web-based software interface designed to control and monitor the InnoPick automated case picking system. This powerful system manages the storage, retrieval, and sequencing of cases across multiple levels to efficiently fulfill orders.

---

## What InnoPick Manager Does

InnoPick Manager provides operators with tools to:

- Monitor real-time system status and inventory levels
  - Control operation modes and settings
  - Track and manage replenishments
  - View and resolve system alerts
  - Monitor case sequences, inventory, and production progress
  - Configure system parameters (for authorized users)
- 

## About This Manual

This manual has been prepared to guide users of all levels through the setup, operation, and optimization of the InnoPick Manager software.

## Manual Organization

The manual is organized into logical sections that match how you'll actually use the system:

- **Getting Started:** Initial login and interface orientation
- **Daily Operations:** Routine startup, monitoring, and shutdown procedures
- **Main Screens:** Detailed reference for each screen and feature
- **Troubleshooting:** Alert resolution and problem-solving
- **Reference:** Quick lookup information for codes and indicators

Within each section, content is arranged by the menu structure in InnoPick Manager, making it easy to follow along as you learn the system.

---

## Who Should Use This Manual

This manual is designed for:

- **Operators:** Daily startup, monitoring, and basic troubleshooting
- **Supervisors:** Advanced troubleshooting and system configuration
- **Maintenance Personnel:** Understanding system behavior during repairs
- **Administrators:** Complete system configuration and user management

Different sections will be relevant depending on your role and access level.

---

## System Architecture Overview

InnoPick consists of:

- **Multiple Levels:** Independent operating levels, each with storage lanes and sequence conveyors
- **Storage Lanes:** Buffer lanes that hold cases until needed

- **Sequence Conveyors:** Conveyors that move cases from infeed to storage and from storage to outfeed
  - **Infeed System:** Receives replenishment cases from upstream conveyor system and depalletizers
  - **Outfeed System:** Releases sequenced cases to downstream conveyor system and palletizers
  - **InnoPick Manager:** The software interface you'll use to control it all
- 

## Getting the Most from This Manual

### For New Users:

- Read through Getting Started and Daily Operations sections first
- Practice navigating the screens in Guest mode if available
- Ask questions and take notes as you learn

### For Experienced Users:

- Use this as a reference when encountering unfamiliar alerts or features
- Review the Troubleshooting section for efficient problem-solving
- Bookmark frequently-used pages in your browser

### For Administrators:

- Pay special attention to the Administration section
  - Review security and access control features
  - Understand the implications of configuration changes
- 

## Next Steps

Now that you understand what InnoPick Manager does, proceed to [Logging In](#) to access the system.

---

**Navigation:** [← Getting Started](#) | [Logging In](#) →

# Logging In to InnoPick Manager

[Home](#) > [Getting Started](#) > **Logging In**

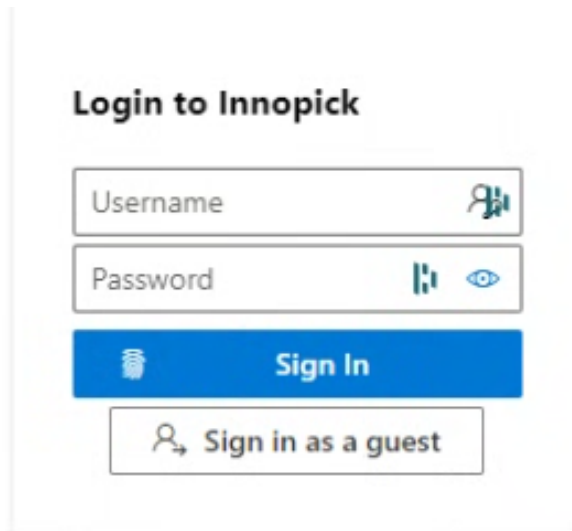
---

## Accessing InnoPick Manager

InnoPick Manager is accessed through a standard web browser at your operator terminal.

### Step-by-Step Login

1. **Open a web browser** (Chrome, Firefox, or Edge recommended)
2. **Navigate to the InnoPick Manager address**
  - Enter the web address provided by your system administrator
3. **The login page will appear**



**Login to Innopick**

Username 

Password  

 **Sign In**

 Sign in as a guest

#### 1. Enter your credentials

- Username: Your assigned username

- Password: Your assigned password

## 2. Click the Sign In **button**

---

## After Logging In

Once you successfully log in:

- You will remain logged in until the session expires or you manually log out
- The system will remember your session across page refreshes
- Your access level determines which screens and features you can use

## First Login

If this is your first time logging in, you'll see the [Home Page](#) with the main interface. Take a moment to familiarize yourself with the layout described in [Interface Overview](#).

---

## Access Levels

Depending on your assigned user role, some menu options may not be accessible:

- **Guest:** Can browse screens but cannot make changes to configuration or equipment status
- **Clerk:** Can operate the system for production but cannot change settings
- **Manager:** Can access operational settings but not site configuration
- **Admin:** Full access to all controls and configuration options

If you need different access permissions, contact your supervisor or system administrator.

---

## Session Management

## Session Timeout

For security, your session will automatically expire after a period of inactivity. If this happens:

1. You'll be redirected to the login page
2. Simply log in again to resume your session
3. Any unsaved changes may be lost

## Logging Out

To manually log out:

1. Click on your user avatar or the logout option (in the upper right corner of the page)
2. Confirm that you want to log out
3. You'll be returned to the login page

**Best Practice:** Always log out when leaving your workstation, especially if you have elevated access privileges.

---

## Troubleshooting Login Issues

### Cannot Access the Login Page

- Verify your network connection
- Confirm you're using the correct web address
- Contact IT support if the page won't load

### Invalid Credentials

- Double-check your username and password (case-sensitive)
- Ensure Caps Lock is not enabled
- Contact your supervisor if you've forgotten your password

## Account Locked

- After multiple failed login attempts, your account may be locked for security
  - Contact your supervisor or system administrator to unlock it
- 

## Next Steps

Now that you're logged in, proceed to [Interface Overview](#) to learn about the main navigation and screen layout.

---

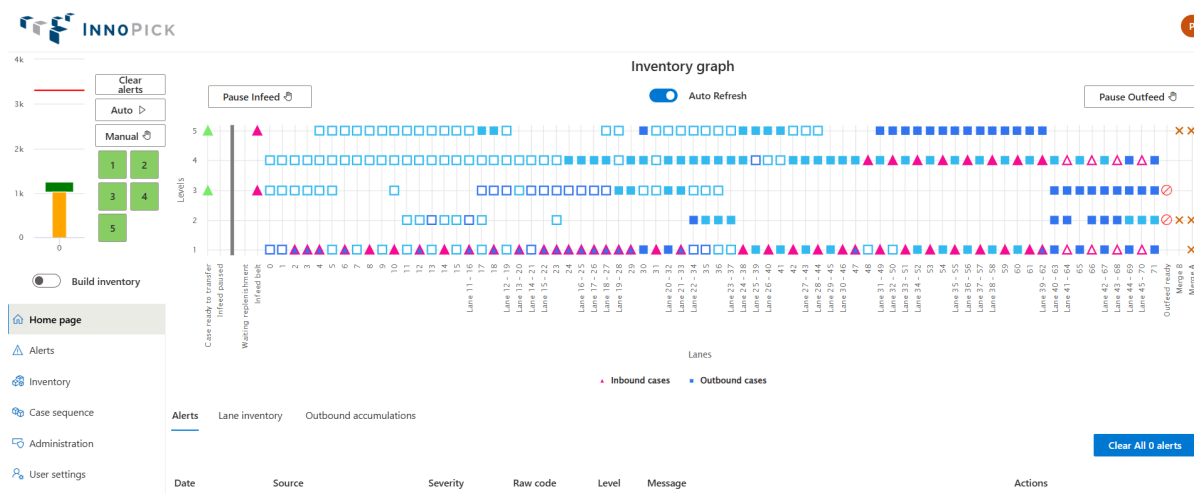
**Navigation:** [← Introduction](#) | [Interface Overview](#) [→](#)

# Interface Overview

[Home](#) > [Getting Started](#) > **Interface Overview**

## Main Navigation

After logging in, you'll see the InnoPick Manager interface with a navigation menu on the left side of the screen. This menu is available from any page within the application.



## Primary Navigation Buttons

Six main navigation buttons are always accessible on the left side:

### 1. [Home Page](#)

- System overview and real-time status
- Visual representation of cases and conveyors
- Current alerts and production progress

### 2. [Alerts](#)

- Current and historical alert listing

- Alert filtering and search functionality

### 3. **Inventory**

- Replenishments management
- Product configuration
- Lane inventory tracking
- Accumulation monitoring
- Product statistics

### 4. **Case Sequence**

- Order of cases exiting InnoPick
- Case tracking and filtering

### 5. **Administration**

- System configuration (authorized users only)
- Manual operations
- User management
- Recovery actions
- Operations Settings

### 6. **User Settings**

- Personal preferences
- Account settings

---

## Screen Layout

Each screen in InnoPick Manager follows a consistent layout:

### Top Section

- User information and logout option
- Current system date/time
- Navigation breadcrumbs (on some pages)

## Left Sidebar

- Main navigation buttons
- Quick access to all major sections

## Main Content Area

- Primary information display
- Interactive controls and buttons
- Data tables and visualizations

## Bottom Section

- Action buttons (Clear Alerts, Save, Cancel, etc.)
  - Status indicators
- 

# Common Interface Elements

## Status Indicators

### Level Status Colors:

- **Green:** Automatic mode - system is running or ready
- **Yellow:** Manual mode - system is paused
- **Red:** Fault mode - system has an active alert
- **Dark Red:** E-Stop active - safety system engaged
- **Blue:** Communication Loss

## Interactive Elements

### Clickable Icons and Buttons:

- Colored level numbers can be clicked to change mode

- Three-dot menus (⋮) reveal additional actions
- Blue hyperlinks navigate to related information
- Hover tooltips provide additional details

#### **Data Tables:**

- Column headers can often be clicked to sort
  - Search bars filter table contents
  - Action columns contain buttons for specific operations
- 

## Navigation Tips

### Getting Around Efficiently

1. **Use the Home button** to quickly return to the main overview
2. **Bookmark frequently-used pages** in your web browser
3. **Use browser Back/Forward** buttons to navigate your history
4. **Open links in new tabs** (right-click, "Open in new tab") to keep your place

### Finding Information Quickly

- Use the search features in data tables
  - Check the Alerts page for system status issues
  - Use the Home Page for at-a-glance system overview
  - Reference this manual's index for specific topics
- 

## Understanding Access Restrictions

Based on your user role, you may see some menu options grayed out or missing entirely:

- **Guest users:** Can view most screens but cannot modify settings or control equipment

- **Clerk users:** Can operate the system but cannot access configuration
- **Manager users:** Can access operational controls but not site setup
- **Admin users:** Have full access to all features

If you need access to a restricted feature, contact your supervisor or system administrator.

---

## Next Steps

Now that you understand the interface layout, you're ready to:

- Learn [Daily Operations](#) procedures
- Explore the [Main Screens](#) in detail
- Review [Troubleshooting](#) procedures

For routine operation, proceed to [Daily Operations: Startup](#).

---

**Navigation:** [← Logging In](#) | [Daily Operations](#) [→](#)

# Daily Operations

[Home](#) > Daily Operations

---

## Overview

This section covers the standard procedures for operating InnoPick Manager on a daily basis, including startup, monitoring, and shutdown procedures.

---

## Startup

Complete checklist for beginning operations each day, including visual inspection, login, and enabling automatic mode.

### Pre-Startup Visual Inspection:

Before powering up or enabling the system, conduct a thorough visual inspection of InnoPick:

**Check for:**

- Cases that are out of place or fallen
- Maintenance tools or equipment left on the machine
- Damaged cases or debris on conveyors
- Any unusual conditions or spills
- All safety guards and doors are properly closed
- E-Stop buttons are in the released (pulled out) position

**Action:** Remove any foreign objects and correct any unsafe conditions before proceeding.

---

## Startup Procedure

### Access InnoPick Manager

1. Open a web browser at your operator terminal
2. Navigate to the InnoPick Manager web address
3. Log in using your username and password

See [Logging In](#) for detailed instructions if needed.

---

### 2. Review System Status

Once logged in, you'll see the [Home Page](#):

#### 1. Check for existing alerts

- Review the Alerts List on the Home Page
- If alerts are present, see [Resolving Alerts](#)
- Clear any alerts from previous shift

#### 2. Check level statuses

- All levels should show their current mode (Manual/Auto/Fault)
- Verify the system is in the expected state
- Each level must be in Manual mode (Yellow) in order for it to be put into Auto mode.

#### 3. Review disabled lanes

- Check for disabled lanes on all levels
  - Check with maintenance staff if any disabled lanes can be put re-enabled following maintenance work.
- 

### 3. Put InnoPick into Automatic Mode

Once you've verified the system is ready:

1. Click the **Auto** button to put all levels into Auto mode



1. **Verify the level(s) turn green** indicating Automatic mode

**Important:** Only enable Automatic mode if:

- Visual inspection is complete
- No unresolved alerts exist
- All safety devices are functional
- Nobody is inside the InnoPick safety fence
- You're ready to begin production

---

#### 4. Monitor Initial Operation

After enabling Automatic mode:

1. **Watch the first few cycles** to ensure smooth operation
2. **Listen for unusual sounds** that might indicate problems
3. **Check the Inventory Graph** to verify cases are moving as expected on all levels
4. **Monitor for new alerts**

See [Monitoring Production](#) for ongoing monitoring procedures.

---

## Startup After Extended Downtime

If the system has been shut down for more than 24 hours:

1. Perform a more thorough visual inspection
  2. Verify all utilities are functioning (power, air pressure, network)
  3. Check with maintenance that all scheduled work is complete
  4. Manually jog any lanes or sequence conveyor chains that have had maintenance work done on them
  5. Allow a few moments for systems to initialize
  6. Test one level in Automatic mode before enabling all levels
- 

## Troubleshooting Startup Issues

### System Won't Enter Automatic Mode

#### Possible Causes:

- Active alerts must be cleared first
- If no alerts appear at first, go to Alerts Page and remove all filters to see *all* alerts.
- Because InnoPick automatically filters out alerts it thinks are not critical, sometimes an important alert is not immediately visible.
- E-Stop or door is open
- Upstream or downstream systems not ready

**Solution:** Check the [Alerts Page](#) for details

---

# Shutdown

Proper procedures for ending the shift and securing the system.

## 1. Verify Production is Complete

Check that there are no:

- **In Progress Cases:** Cases currently being processed on sequence conveyors
- **Incomplete Cases:** Cases scheduled but not yet completed
- **Pending Cases:** Cases in the case sequence waiting to be processed
- **Active Replenishments:** Replenishments currently inducting or in progress

**If production is not complete:**

- Coordinate with supervision about completing or postponing remaining orders
  - Allow system to finish current work before proceeding
  - Document any incomplete work for the next shift
- 

## 2. Put InnoPick into Manual Mode

- On the [Home Page](#), click the **Manual** button



- Verify all level indicators turn **Yellow** (Manual mode)

#### Why Manual Mode?

- Prevents the system from automatically starting new moves
  - Allows safe inspection and shutdown
  - Ensures system won't start unexpectedly
- 

### 3. Conduct Visual Inspection

Perform a walk-around inspection of InnoPick:

#### Check for:

- Cases out of place or fallen
- Cases left on conveyors
- Any damage to equipment or cases
- Spills or debris that need cleanup

#### Document any issues:

- Report safety concerns or mechanical issues to supervision
- 

## 4. Log Out

1. Click on your username or logout button
2. Confirm logout
3. Close the browser

**Security Note:** Always log out when leaving your workstation to prevent unauthorized access.

---

## 5. Suggested: Open an InnoPick Door

- It is recommended to request access to an InnoPick door, as this removes all latent electrical and pneumatic energy from the system
  - This puts the system into a state equivalent to E-Stop
  - Reduces wear on certain components
- 

# Essential Skills for Operators

To effectively operate InnoPick, you should be familiar with:

### 1. System Status Monitoring

- Understanding level status colors (Green/Yellow/Red)
- Reading the inventory graph
- Monitoring production progress

### 2. Alert Management

- Recognizing common alerts
- Basic alert resolution

- When to escalate to maintenance or support

### 3. Replenishments Awareness

- Understanding replenishment status
- 

## Safety Reminders

- Always perform visual inspections before starting equipment
  - Never bypass safety interlocks or E-stops
  - Ensure all personnel are clear before resuming automatic operation
  - Report any unsafe conditions to your supervisor
- 

**Navigation:** [← Interface Overview](#) | [Monitoring Production →](#)

# Monitoring Production

[Home](#) > [Daily Operations](#) > **Monitoring**

---

## Overview

During production, operators should actively monitor InnoPick Manager to ensure smooth operation, catch problems early, and maintain production efficiency.

---

## Primary Monitoring Dashboard

The [Home Page](#) provides all essential information for monitoring production at a glance.

### Key Elements to Monitor

#### 1. [Level Status Indicators](#)

- **Green:** Level is in Automatic mode and operating normally
- **Yellow:** Level is in Manual mode (paused)
- **Red:** Level has an active fault
- **Dark Red:** E-Stop condition is active
- **Blue:** Communication Loss

#### 2. [Inventory Graph](#)

- Visual representation of all cases on sequence conveyors
- Squares = Outbound cases (for customer pallets)
- Triangles = Inbound cases (replenishments being stored)
- Filled shapes = Physical cases present

- Hollow shapes = Scheduled case positions

### 3. Alerts List

- Shows current system alerts
- Alerts appear when issues require attention
- Go to the Alerts page for more details.

### 4. Production Graph

- Shows progress of current production run
- Green bar = In Progress cases
- Orange bar = Completed cases
- Red line = Total cases in current orders

### 5. Lane Inventory

- Select a level to view storage lane status
  - Hover over lanes for detailed inventory info
  - Monitor for abnormal low inventory levels
- 

## What to Watch During Production

### Continuous Monitoring Tasks

#### **Every 15-30 Minutes:**

- Glance at level status indicators (all should be green)
- Check for new alerts
- Verify production progress is advancing

#### **Every 1-2 Hours:**

- Check replenishment status
- Monitor case sequence progress

#### **Throughout the Shift:**

- Listen for unusual mechanical sounds

- Watch for stopped or hesitating conveyors
  - Be aware of downstream system status
- 

## Recognizing Normal Operation

When InnoPick is operating correctly:

- All levels show **green** status
  - Cases move smoothly through the system
  - **Inventory Graph** shows cases advancing position by position
  - **Production Graph** shows steady progress
  - **Alerts List** is empty or shows only minor warnings
  - Replenishments are being inducted as needed
  - Output cases are feeding downstream without backing up
- 

## Recognizing Problems Early

### Warning Signs to Watch For

#### **Immediate Attention Required:**

- Any level turns **Red** (fault condition)
- E-Stop condition (all levels **Dark Red**)
- Critical alerts appear
- Cases stopped on conveyors
- Unusual sounds or movements

#### **Monitor Closely:**

- Repeated alerts of the same type
- Accumulation building up at outfeed

- Replenishments not arriving as scheduled
- 

## Responding to Alerts

When an alert occurs:

### 1. Note the alert details

- Which level is affected
- What type of alert (code and message)
- Time of occurrence

### 2. Assess the situation

- Is it a single occurrence or repeating?
- Is production stopped or continuing?

### 3. Take appropriate action

- For common alerts: Follow [Alert Reference](#)
- For safety issues: Use E-Stop if necessary.
- For unknown issues: Contact supervisor, maintenance, or remote support.

See [Alert Guidelines](#) for detailed alert resolution procedures.

---

## Monitoring Replenishments

Replenishments ensure InnoPick has adequate inventory to fulfill orders.

### What to Monitor

Access the [Replenishments Page](#) to check:

- **Pending:** Replenishments requested but not yet started
- **In Progress:** Replenishments being prepared for induction

- **Inducting:** Replenishments currently entering InnoPick
- **Completed:** Replenishments fully inducted

## When to Take Action

**Normal:** Replenishments progress from Pending → In Progress → Inducting → Completed

### **Problem Indicators:**

- Inducted Quantity doesn't match expected quantity
- Replenishments backing up at infeed

**Action:** Coordinate with upstream depalletizing operations or see [Troubleshooting](#)

---

## Understanding System Settings That Affect Operations

Be aware of these settings and their effects:

### **Build Inventory Mode**

- When ON: Prioritizes storing replenishment cases over outputting cases
- When OFF: Balances input and output for maximum throughput

### **Pause Infeed / Pause Outfeed**

- Controls whether cases enter or exit InnoPick
  - Check: [Home Page](#)
- 

## Shift Handoff Checklist

When ending your shift, review with the next operator:

- Current system status and any active alerts
- Any recurring issues or unusual behavior observed
- Replenishments in progress or pending

- Any lanes disabled or products having problems
- Upcoming scheduled maintenance or system changes
- Any instructions from supervision

---

**Navigation:** [← Daily Operations](#) | [Main Screens](#) [→](#)

# Main Screens

[Home](#) > Main Screens

---

## Overview

This section provides detailed documentation for each screen and feature in InnoPick Manager. Use this as a reference guide to understand what information is available and how to use each feature.

---

## Screen Reference Guide

### Home Page

The main dashboard showing system status at a glance, including:

- Inventory graph with real-time case positions
- Level status indicators and mode controls
- Current alerts listing
- Lane inventory display
- Production progress graph

### Alerts Page

Comprehensive alert monitoring and history, including:

- Current and past alerts
- Alert filtering and search
- Alert details and severity levels

- Common alert descriptions

## Inventory Section

Complete inventory management tools:

- **Replenishments Page:** Track & manage incoming replenishments
- **Products Page:** Configure product settings and parameters
- **Lane Inventory Page:** View and edit storage lane contents
- **Accumulations Page:** Monitor outfeed accumulation space
- **Product Statistics Page:** Analyze product usage and performance

## Case Sequence Page

View the sequence of outgoing cases:

- Current case sequence
- Case status
- Case information (HUID, level, case #, etc)
- Filtering and search tools
- Case cancellation options

## Administration Pages

System configuration and advanced features (authorized users only):

- **Manual Motion Control:** Manual jog controls
  - **Setup InnoPick:** System configuration parameters
  - **Algorithm Parameters:** Algorithm tuning and optimization
  - **User Manager:** Create and manage user accounts
  - **Recovery Actions:** Advanced troubleshooting tools
  - **Operations:** Level and Merge statuses / controls
  - **User Settings:** Personal preferences
-

## How to Use This Section

### For Daily Operation

Focus on these screens:

- [Home Page](#) - Your primary monitoring dashboard
- [Alerts Page](#) - When issues occur
- [Replenishments Page](#) - Monitor replenishment statuses

### For Troubleshooting

Refer to these screens:

- [Alerts Page](#) - Identify problems
- [Lane Inventory](#) - Adjust lane counts
- [Case Sequence Page](#) - Track specific output cases
- [Manual Motion Control](#) - Jog lanes and conveyors
- Also, see the [Troubleshooting section of this manual](#)

### For Administration

Authorized users should understand:

- [Setup InnoPick](#) - System configuration
  - [Recovery Actions](#) - Advanced tools
- 

## Screen Navigation Tips

### Getting to Screens:

- Click the navigation buttons on the left sidebar
- Use browser bookmarks for frequently-accessed pages

- Some screens have sub-sections accessed via tabs

**Working with Data Tables:**

- Click column headers to sort
- Use search boxes to filter results
- Look for action menus (three dots :) for additional options

**Understanding Visual Elements:**

- Green/Yellow/Red color coding indicates status
- Hover tooltips provide additional information
- Blue hyperlinks navigate to related pages

---

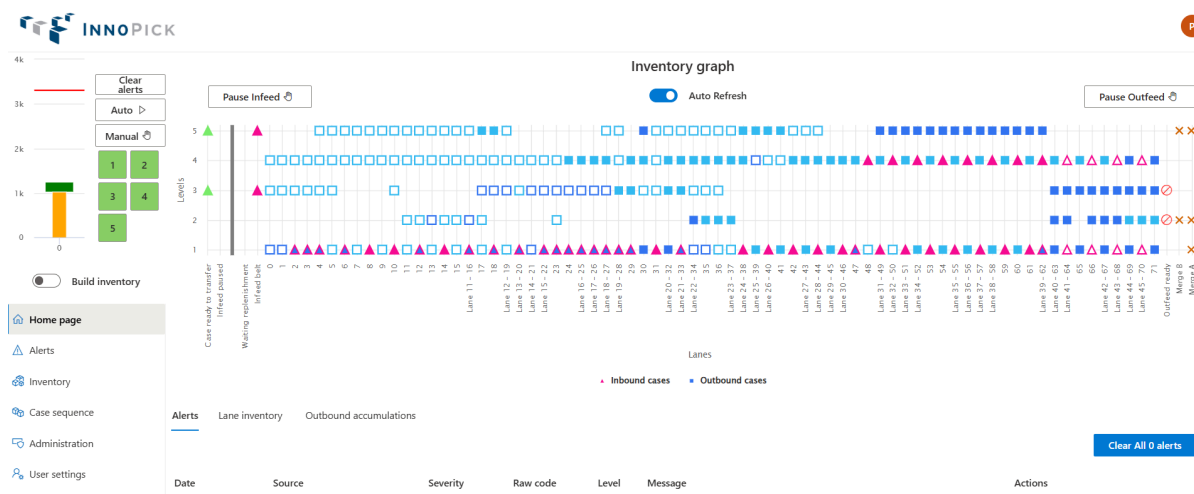
**Navigation:** [← Monitoring Production](#) | [Home Page →](#)

# Home Page

[Home](#) > [Main Screens](#) > [Home Page](#)

## Overview

The Home Page is your primary dashboard for monitoring InnoPick operations. It provides a real-time snapshot of system status, case locations, alerts, and production progress.



## Accessing the Home Page

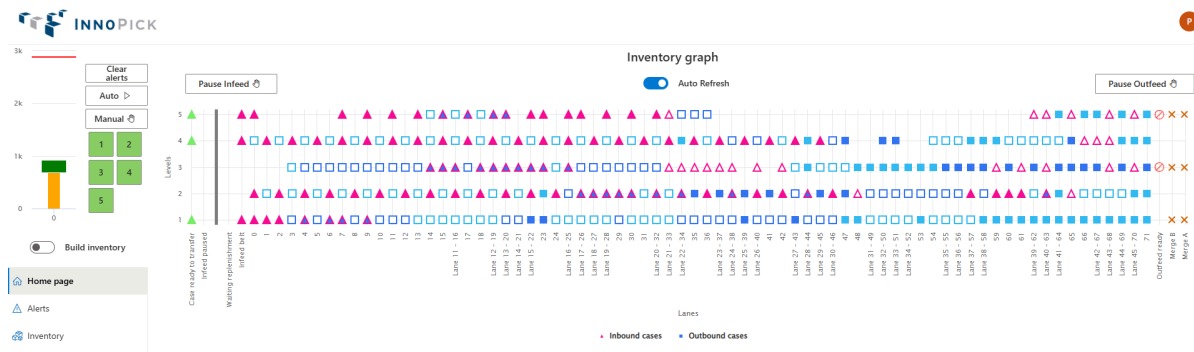
- Click the **Home Page** button in the left navigation menu
- The Home Page appears automatically after logging in
- Available from any screen in InnoPick Manager

# Home Page Components

The Home Page displays the following key information:

1. [Inventory Graph](#) - Visual representation of cases on sequence conveyors
2. [Level Statuses & Controls](#) - Mode indicators and control buttons
3. [Alerts List](#) - Current system alerts
4. [Lane Inventory](#) - Storage lane contents by level
5. [Production Graph](#) - Progress of current production run

## Inventory Graph



The main visual grid in the center-right of the Home Page provides a quick summary of case positions on InnoPick sequence conveyors.

## Understanding the Symbols

### Shapes:

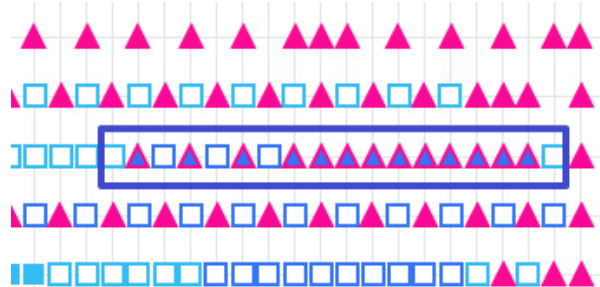
- **Squares** = Outbound cases (scheduled for customer pallets)
- **Triangles** = Inbound cases (replenishments being stored in lanes)

### Fill Status:

- **Filled shapes** = Physical case is present at that position
- **Hollow shapes** = Position is empty or case is expected

- If a triangle, it means the case has been stored
- If a square, it means the case is scheduled to be dispensed on this position

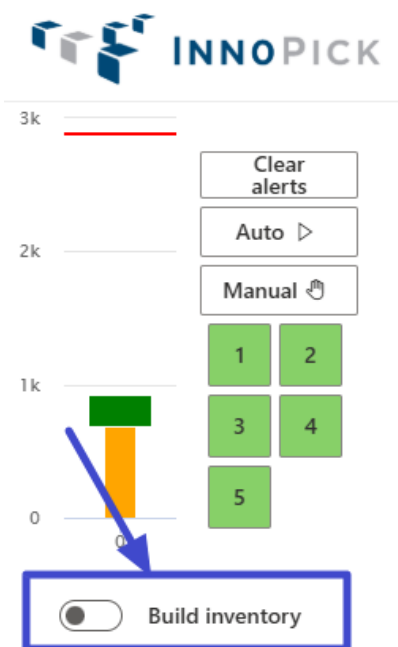
**"Gender Bender" cases:**



- Some Inbound case positions can be re-used as Outbound case positions when the system determines that the inbound case will be stored in a lane upstream of the outbound case's lane.
- This is shown via a special border around the inbound case on the Inventory graph (see screenshot above - cases highlighted in blue rectangle)
- When this case is stored, the triangle will be replaced with a square.

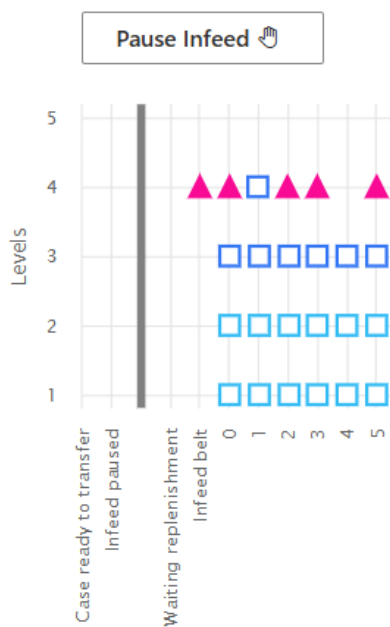
## Operational Controls on the Inventory Graph

### Build Inventory Mode

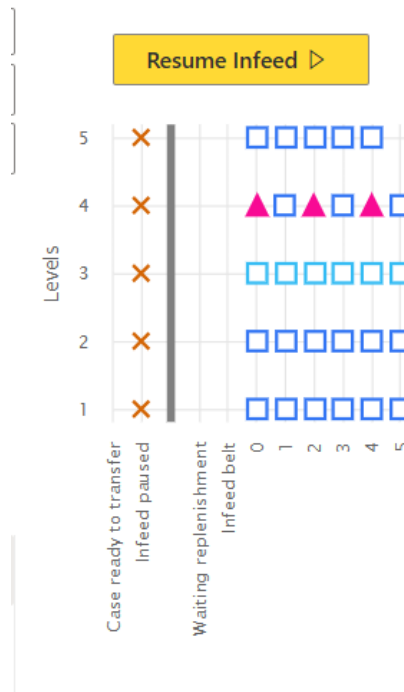


- Build Inventory mode prioritizes letting replenishment cases come into InnoPick over scheduling output cases.
- [More information on Build Inventory Mode](#)

### Pause / Resume Infeed

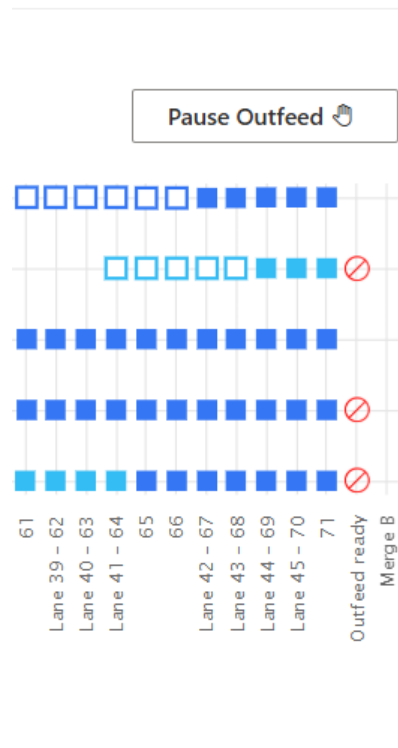


- Click the **Pause Infeed** button to pause incoming cases
- Cases will stop and wait on the first infeed position of InnoPick

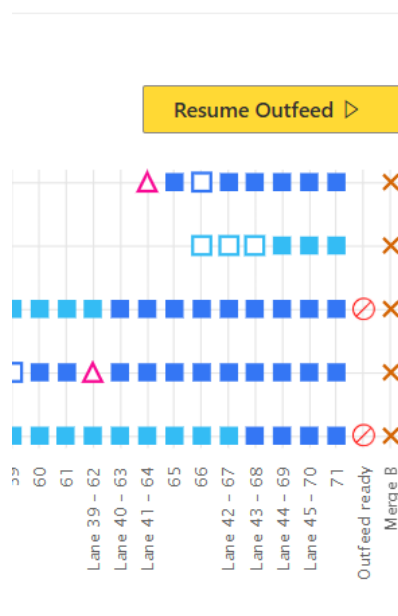


- Click the **Resume Infeed** button to resume incoming cases.

### Pause / Resume Outfeed



- Click the **Pause Outfeed** button to pause outgoing cases
- Cases will stop and wait on the last position of InnoPick



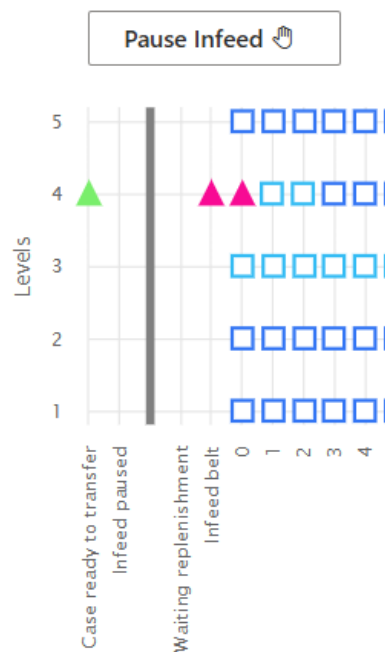
- Click the **Resume Outfeed** button to resume outgoing cases.

## Interpreting the Display

The inventory graph shows:

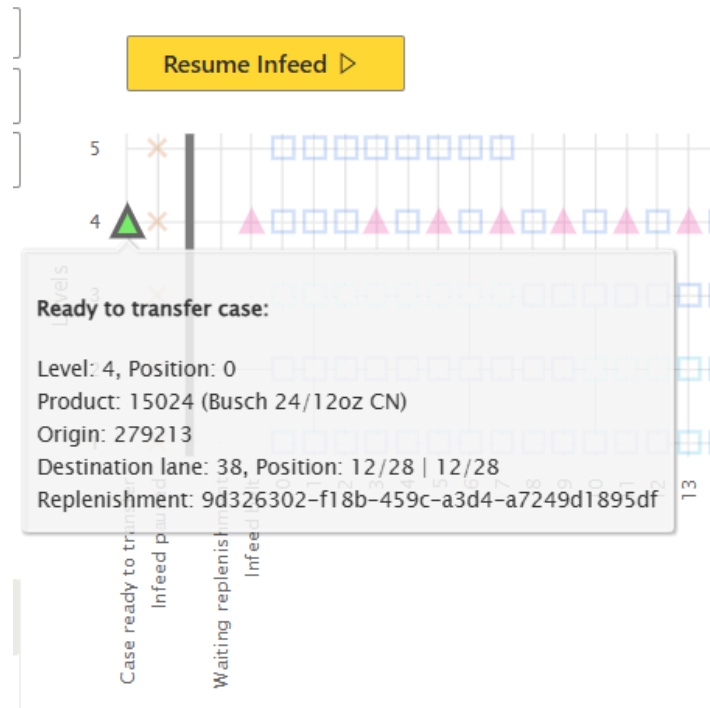
- Cases currently on all sequence conveyor positions, with the storage lanes displayed below the graph, indicating their position vertically.
- Various statuses of the transition points at both Infeed and Outfeed:

### Statuses at the Infeed

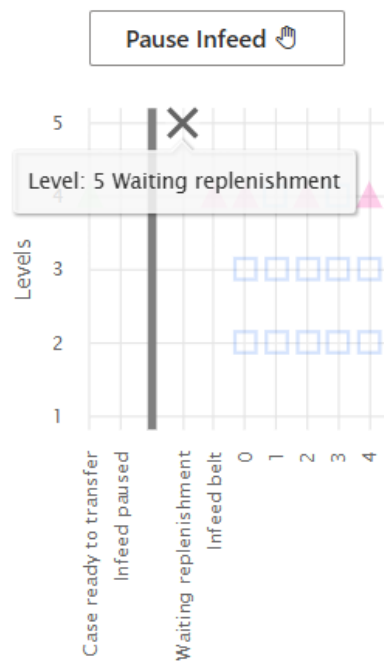


### Case ready to transfer

- When a case is to be transferred from the upstream conveyor system, it appears as a coloured triangle on that level.
- Hover over the triangle to get more information:

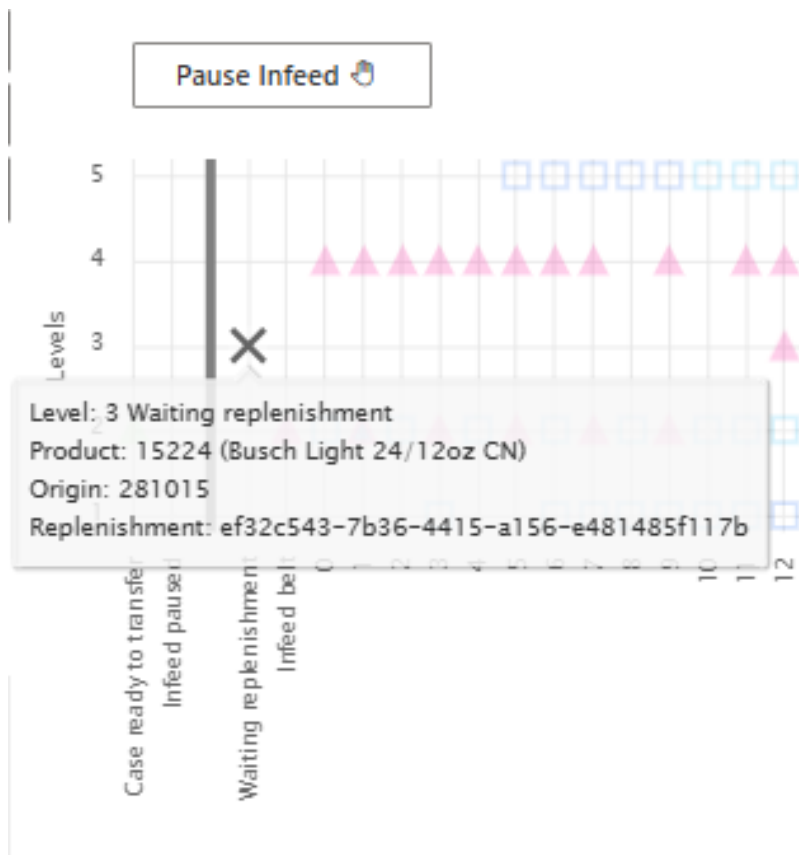


### Waiting replenishment



- When a level InnoPick is not able to schedule output cases because it is waiting for a replenishment, an x appears at that level.

- This allows an operator to know immediately why a given level is not scheduling output cases.
- Hover over the **x** to see which replenishment the level is waiting for.



### Infeed paused

- [See above](#)

### Statuses at the Outfeed

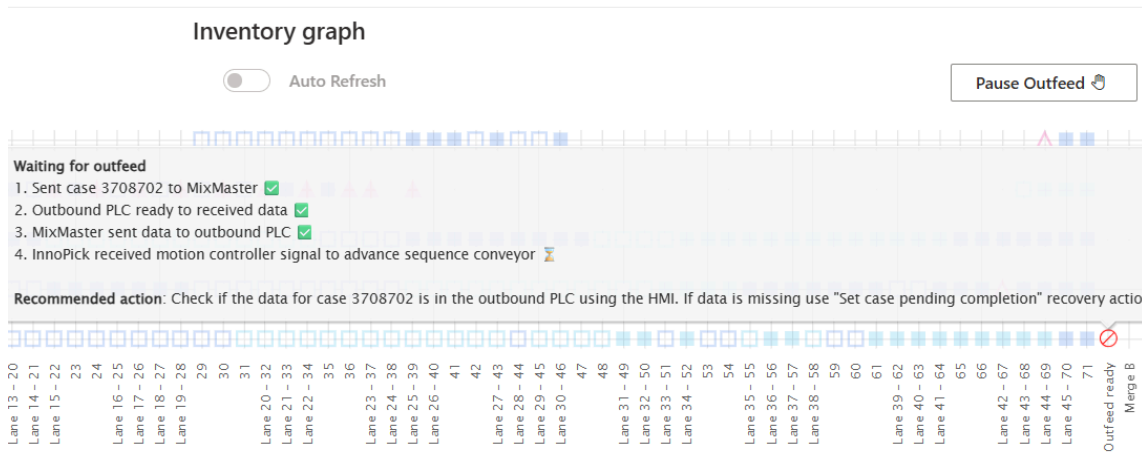


### Merge A / B

- If InnoPick calculates that a certain destination (Merge A or Merge B) cannot receive any more cases, it will stop scheduling output cases for that destination and indicate this with an **X** on the given level and over the given destination.

### Outfeed ready

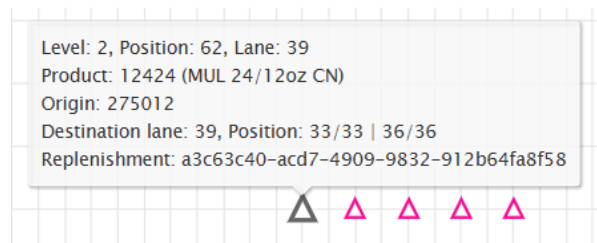
- If there is a case on the last position of InnoPick and the downstream conveyor system is not ready to receive it, a stop logo appears.
- Hover over the stop logo to get more information:



## Getting More Information

### Hover for Details:

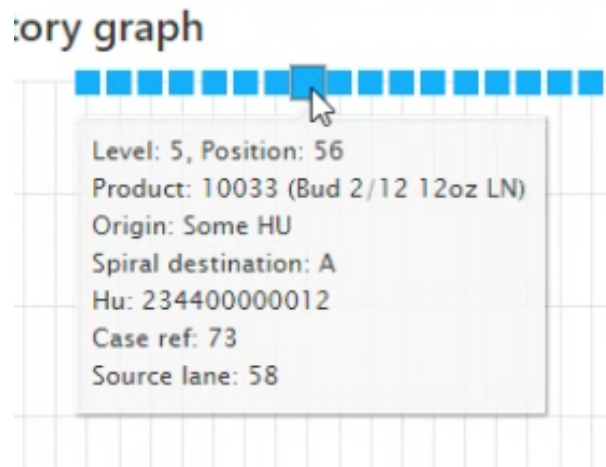
Hover your cursor over any case icon to see additional information:



Information includes:

- Level #, current position #, and lane # (if applicable).
- Product SKU and description
- Origin PalletID / HUID
- If Inbound case:
  - Destination lane
  - Position: the position of the case in the storage lane; followed by the position of the case in terms of the total replenishment quantity. In the example in the screenshot above, it is case 33 out of 33 in lane 39; and it was the last case of a 36-case replenishment.

- Replenishment ID (unique GUID); this can be used to identify the replen in InnoPick and MixMaster.
- If Outbound Case:



- Merge / Spiral destination
- Client Order PalletID
- Case reference number
- Source lane (where the case was dispensed from)

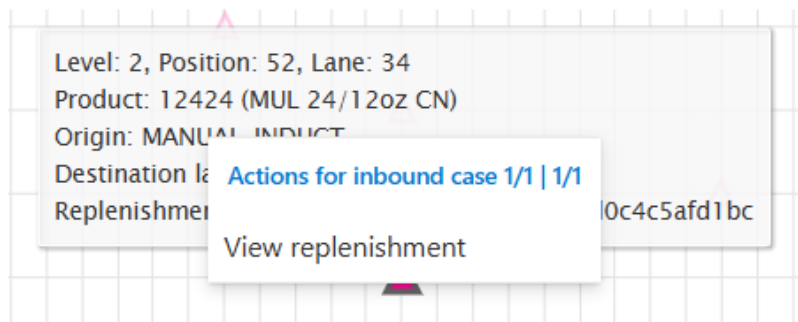
---

## Quick Links

Users can click on:

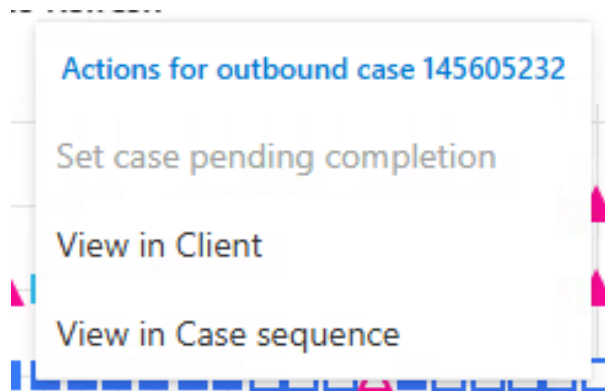
### 1. Replenishment Links:

For inbound cases (triangles), you can click on the case and see a clickable "View replenishment" link:



Clicking this link opens the [Replenishments Page](#) and displays details for that specific replenishment.

## 2. Outbound Case Links

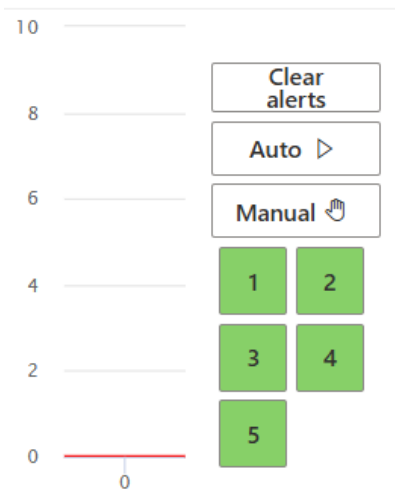


For outbound cases (squares), you can click on the case and see between 2 and 3 clickable links:

- **Set case pending completion:** This is only clickable if the case is on the last position of InnoPick. If clicked, a second confirmation will be required in order to execute the recovery action.
- For detailed information on what this recovery action does, see the [Commonly Used Recovery Actions](#).
- **View in Client:** This will redirect the user to the Case Sequence page in MixMaster where the case can be found.
- Tip: Right-click and "Open in New Tab" to keep this page open. You may need to login again in the new tab.
- **View in Case Sequence:** This will redirect the user to the Case Sequence page in InnoPick Manager, with a filter applied so only this case appears.
- Tip: Right-click and "Open in New Tab" to keep this page open. You may need to login again in the new tab.

---

## Level Statuses & Auto / Manual Controls



The numbered squares in the top left corner indicate the operational status of each InnoPick level.

---

## Status Color Meanings

- **Green:** Automatic mode - level is running or ready for production
  - **Yellow:** Manual mode - level is paused
  - **Red:** Fault mode - level has an active alert
  - **Dark Red:** E-Stop active - safety system engaged
  - **Blue:** Communication Loss
- 

## Controlling Level Modes

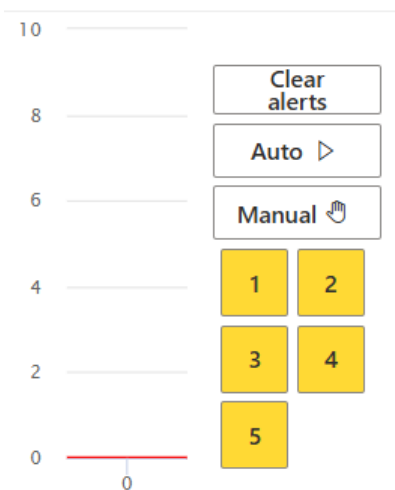
**To change a level's mode:**

1. Click on the colored level number
2. A pop-up may appear to confirm the mode change
3. The color will change to reflect the new mode

#### Mode Control Buttons:

- **Auto Button:** Puts InnoPick (all levels) into Automatic (production) mode
  - Use when ready to begin or resume production
  - System will autonomously manage case movements
  - **Manual Button:** Puts InnoPick (all levels) into Manual mode
  - System pauses automatic operation
  - Used for troubleshooting and maintenance
  - Required in order to jog lanes (manual control) or change certain settings, such as the inventory of a lane.
- 

#### When Levels Are in Manual Mode



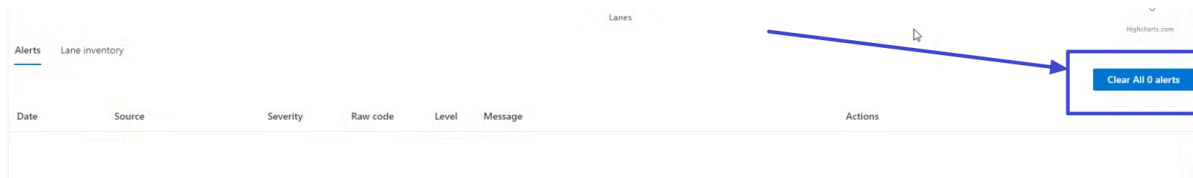
When InnoPick has an alarm, the affected level automatically enters Manual (Yellow) or Fault (Red) mode. This pauses production and allows the operator to:

- Assess the problem
- Resolve any issues

- Clear the alarm
- Resume production

---

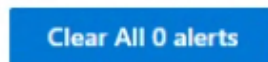
## Alerts List



The Alerts section shows current system alerts that require attention.

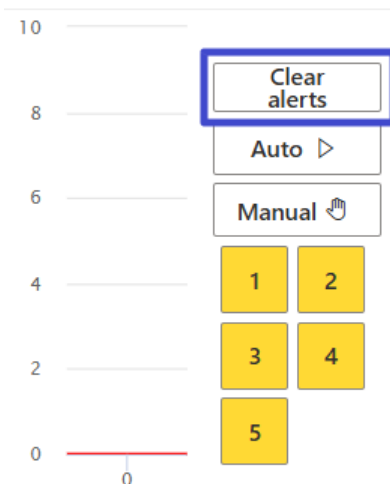
---

## Clear All Alerts Button



The **Clear All Alerts** button (bottom right of the inventory graph) attempts to clear all current alerts in the system.

There is also a **Clear Alerts** button always visible just above the Auto and Manual buttons:



### Important Notes:

- If the underlying issue has not been resolved, the alert **cannot** be cleared
- Common reasons alerts won't clear:
  - Photocell sensor still blocked
  - E-Stop button still pressed
  - Door still open
  - Physical case still in wrong position

### When Alerts Won't Clear:

1. Read the alert message carefully
  2. Resolve the underlying issue
  3. Try clearing alerts again
  4. See [Alert Guidelines](#) for specific resolution steps
  5. Try using the **Clear alarm on level** [recovery action](#).
  6. Open the [Alerts Page](#) and remove all filters to see if some other alert could be affecting the system.
    - Sometimes the system filters out an alert that is still active and cannot be cleared until the underlying issue has been fixed.
- 

## Viewing Complete Alert History

For a full list of all current and past alerts with search and filtering capabilities, see the [Alerts Page](#).

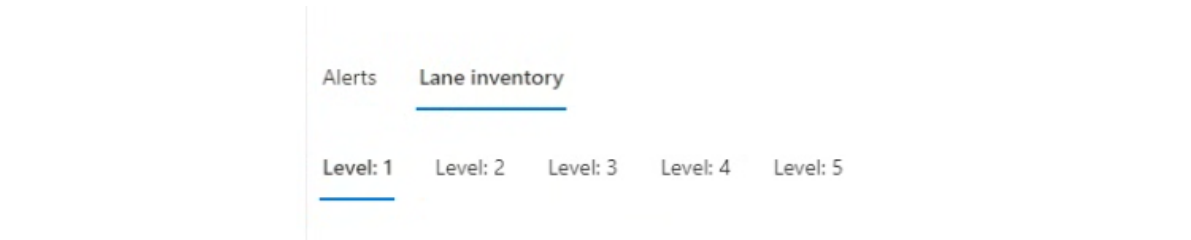
---

## Lane Inventory

The Lane Inventory display shows the contents of storage lanes for each level.

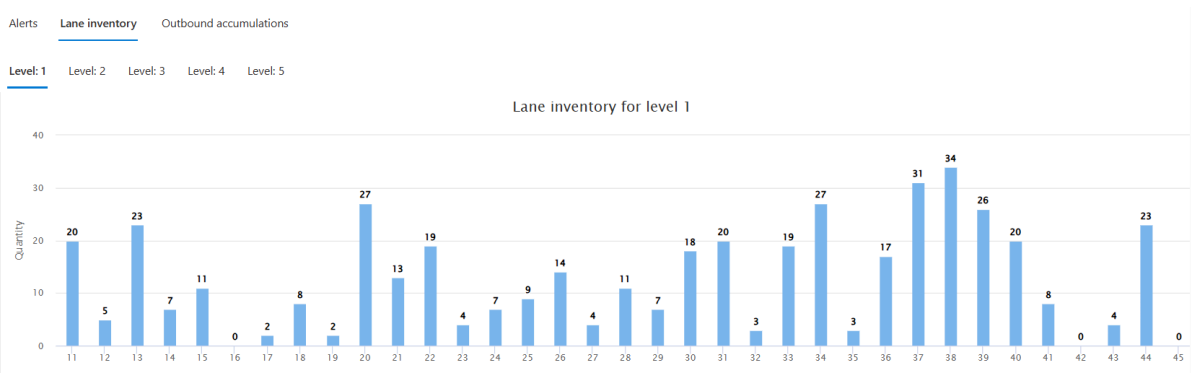
---

## Selecting a Level



Click on a level number to view its lane inventory.

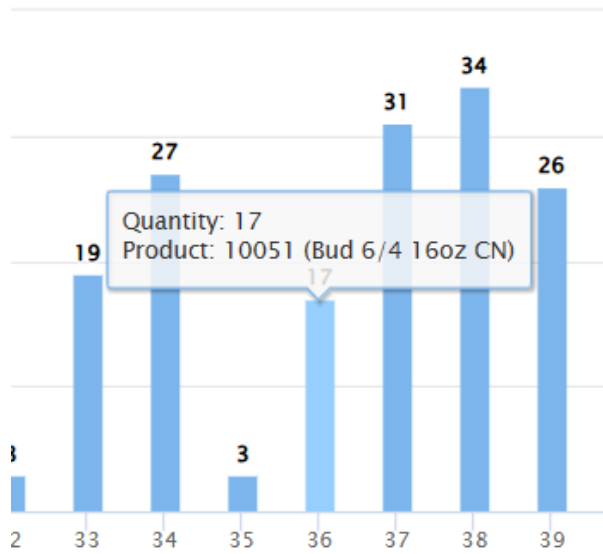
## Lane Display



Each storage lane is shown as a horizontal bar:

- Bar length represents relative fullness
- The display adjusts automatically to show all lanes for that level

## Getting Lane Details



Hover your cursor over any lane to see:

- Quantity of cases
- Product stored (SKU and description)

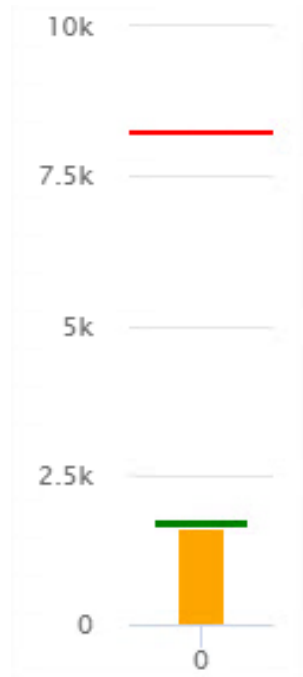
---

## Complete Lane Management

For more detailed lane inventory information and editing capabilities, see the [Lane Inventory Page](#).

---

## Production Graph



The Production Graph on the far left of the Home Page shows the current state of the production run.

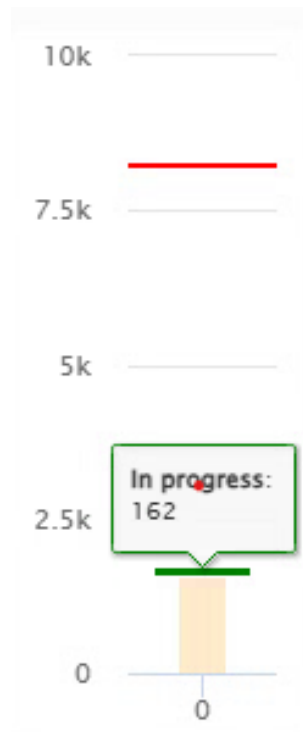
---

## Understanding the Graph

### Red Line at Top:

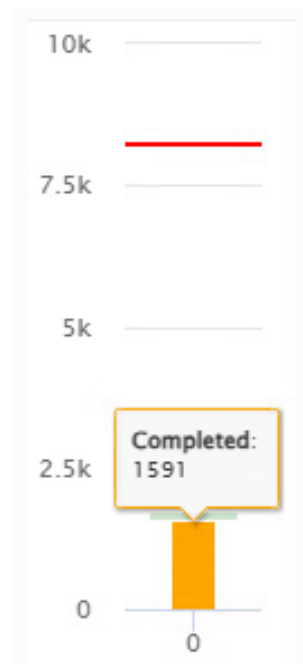
- Represents the total number of cases in currently launched orders
- This is the production target

### Green Bar:



- Represents **In Progress** cases
- These are cases currently on the sequence conveyors (outbound)
- Corresponds to the squares (filled and hollow) on the inventory graph
- Hover to see exact count

#### Orange Bar:



- Represents **Completed** cases
  - These cases have been output from InnoPick
  - This counter resets periodically as routes/truckloads are completed
  - Hover to see exact count
- 

## Using the Home Page Effectively

---

### For Daily Monitoring

The Home Page should be your primary screen during production:

1. **Glance regularly at level statuses** - All should be green during production
  2. **Watch for new alerts** - Address immediately when they appear
  3. **Monitor production progress** - Verify steady advancement
- 

### Quick Troubleshooting

When an alert occurs:

1. **Check which level** has the issue (red or yellow indicator)
  2. **Read the alert message** in the Alerts List
  3. **Look at the inventory graph** - does it match physical reality?
  4. **Address the issue** per the [Troubleshooting Guidelines](#)
  5. **Clear the alert** once resolved
-

## Navigation from Home Page

### To investigate issues further:

- Click level numbers → Changes mode or shows level details
- Click "See replenishment" links → View replenishment details
- Click alerts → Navigate to Alerts page for more information

### To access other features:

- Use left sidebar navigation buttons
  - See [Main Screens Index](#) for all available screens
- 

## Related Topics

- [Monitoring Production](#) - How to use the Home Page during operation
  - [Alerts Page](#) - Detailed alert information and history
  - [Alert Guidelines](#) - How to resolve alerts
  - [Lane Inventory Page](#) - Detailed lane management
- 

**Navigation:** [← Main Screens](#) | [Alerts Page](#) [→](#)

# Alerts Page

[Home](#) > [Main Screens](#) > Alerts Page

## Overview

The Alerts page provides a comprehensive view of current and historical alerts, with powerful search and filtering capabilities to help you identify and resolve issues quickly.

**Note:** If you're looking for detailed info on specific Alert Codes, see the [Alert Reference Page](#)

Clear alerts

Auto

Manual

Build inventory

Home page

**Alerts**

Inventory

Case sequence

Administration

User settings

Search

Filters

| Date                   | Source                              | Severity | Raw code | Level | Message                                                                                                                                                                                                                                              |
|------------------------|-------------------------------------|----------|----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1/23/2026, 12:23:02 PM | MotionController                    | Error    | 512      | 5     | Level 5 Infeed Middle Photocell was not Blocked during an Infeed Case Move on lane n0<br>Infeed Belt: MJL 24/12oz LN-PP;<br>Position 0: empty;<br>Position 1: empty;                                                                                 |
| 1/23/2026, 11:35:47 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 377mm. Measured 341.1393mm on lane n0<br>Infeed Belt: MJL 18/12oz LN;<br>Position 0: MJL 18/12oz LN;<br>Position 1: empty;                                              |
| 1/23/2026, 11:33:21 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 377mm. Measured 341.9036mm on lane n0<br>Infeed Belt: MJL 18/12oz LN;<br>Position 0: MJL 18/12oz LN;<br>Position 1: MJL 18/12oz LN;                                     |
| 1/23/2026, 11:32:12 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 377mm. Measured 340.8214mm on lane n0<br>Infeed Belt: MJL 18/12oz LN;<br>Position 0: MJL 18/12oz LN;<br>Position 1: empty;                                              |
| 1/23/2026, 11:29:14 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 377mm. Measured 341.3821mm on lane n0<br>Infeed Belt: MJL 18/12oz LN;<br>Position 0: MJL 18/12oz LN;<br>Position 1: empty;                                              |
| 1/23/2026, 11:20:42 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 342mm. Measured 229.7696mm on lane n0<br>Infeed Belt: Corona Extra 24/7oz LN-LSE;<br>Position 0: Corona Extra 24/7oz LN-LSE;<br>Position 1: Corona Extra 24/7oz LN-LSE; |
| 1/23/2026, 11:03:29 AM | InventoryManagementService.Inductio | Error    |          | 3     | Cannot accept induction of replenishment 'ddb776cb-73b7-4f69-aa44-b4121e66c104' because the currently inducing replenishment '45ebb790-e6bd-4129-b113-310b02af888f' has too many cases left (10) to induct on level 3                                |
| 1/23/2026, 10:56:21 AM | MotionController                    | Error    | 517      | 3     | Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 377mm. Measured 340.8786mm on lane n0<br>Infeed Belt: MJL 18/12oz LN;                                                                                                   |

1

100/page

(82 results)

## Accessing the Alerts Page

- Click the **Alerts** button in the left navigation menu
- Available from any screen in InnoPick Manager

## Default Display

When the Alerts page first loads:

- **Only uncleared alerts** are displayed by default
  - Alerts are shown in chronological order (most recent first)
- 

## Alerts Page Columns

The Alerts page displays detailed information about each alert:

---

### Date

- The date and time when the alert first occurred
  - Useful for tracking when issues began
- 

### Source

- Indicates which portion of the software generated the alert
  - Examples: Motion Controller, PLC, Core Logic, Buffer Management
  - Helps identify which system component detected the issue and what aspect of InnoPick is the problem
- 

### Severity

Alerts are categorized by severity level:

| Severity |
|----------|
| Critical |
| Error    |
| Warning  |
| Warning  |

**Warning:** Minor alert that may not require immediate action

- System can often continue operating
- Should be monitored for patterns
- No audible sound

**Error:** A standard Alert which stops the affected level and puts it into Manual (Faulted/Red) mode.

- Other levels continue operating
- The panel Buzzer will sound to alert the operator

**Critical:** Major system alert requiring attention

- Production on all levels may be stopped
- Requires operator intervention
- The panel Buzzer will sound to alert the operator

---

## Raw Code

- The numeric code associated with the alert
- Used to identify specific alert types
- Reference these codes in the [Alert Reference Page](#)
- Examples: 119, 121, 515, 555, 709

---

## Level

- The InnoPick level where the alert occurred
  - May be blank for system-wide alerts
- 

## Message

- Descriptive text explaining the nature of the alert
- Written to help operators understand what happened
- May include specific details like lane numbers or photocell locations, as well as contextual information (nearby cases, replenishment details, etc).
- Examples:

Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 342mm. Measured 229.7696mm on lane n0  
Infeed Belt: Corona Extra 24/7oz LN-LSE;  
Position 0: Corona Extra 24/7oz LN-LSE;  
Position 1: Corona Extra 24/7oz LN-LSE;

Cannot accept induction of replenishment 'ddb776cb-73b7-4f69-aa44-b4121e66c104' because the currently inducing replenishment '45ebb7f0-e6bd-4129-b113-310b02afd88f' has too many cases left (10) to induct on level 3

Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move on lane n27 (Quantity = 24 ClientProductReference = 10224 Description = BDL 24/12oz CN) (ClientProductReference = 10224 Description = BDL 24/12oz CN in front of lane)

Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move on lane n38 (Quantity = 9 ClientProductReference = 60424 Description = Modelo Especial 24/12oz CN) (ClientProductReference = 47646 Description = Stella Artois 24/11.2oz LN US in front of lane)

---

## Using Search and Filters

The screenshot shows a 'Filters' dialog box with a dropdown arrow and a small '3' in a circle. It is divided into two columns: 'Alert types' and 'Levels'. Under 'Alert types', there are four checkboxes: 'E-stop' (checked), 'Not Initial' (unchecked), 'Warning' (checked), and 'Error' (checked). Under 'Levels', there are five checkboxes numbered 1 to 5, all of which are unchecked. At the bottom, there are four buttons: 'Apply', 'Cancel', 'Clear all', and 'Select all'.

| Alert types                                 | Levels                     |
|---------------------------------------------|----------------------------|
| <input checked="" type="checkbox"/> E-stop  | <input type="checkbox"/> 1 |
| <input type="checkbox"/> Not Initial        | <input type="checkbox"/> 2 |
| <input checked="" type="checkbox"/> Warning | <input type="checkbox"/> 3 |
| <input checked="" type="checkbox"/> Error   | <input type="checkbox"/> 4 |
|                                             | <input type="checkbox"/> 5 |

Buttons: Apply, Cancel, Clear all, Select all

## Searching Alerts

Use the search bar to find specific alerts by:

- Alert code (e.g., "555")
  - Keywords in message text (e.g., "photocell", "clutch")
  - Level number
  - Date/time information
- 

## Common Filtering Strategies

### To find all alerts for a specific level:

- Sort by the Level column
- All alerts for each level will be grouped together

### To find recent critical alerts:

- Filter by Severity = Critical
- Sort by Date (most recent first)

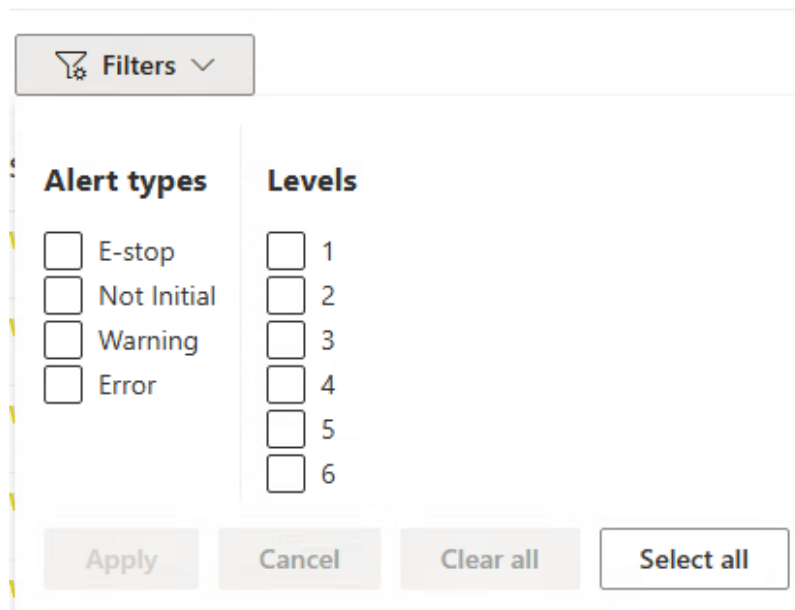
### To track a specific type of fault:

- Search for the alert code or key message terms
  - Review the pattern and frequency
- 

## Viewing All Alerts

To see all alerts:

- Remove all filters:



The screenshot shows a 'Filters' dialog box with a title bar containing a funnel icon and the text 'Filters' with a dropdown arrow. Below the title bar, there are two columns of checkboxes. The first column is titled 'Alert types' and contains four items: 'E-stop', 'Not Initial', 'Warning', and 'Error'. The second column is titled 'Levels' and contains six items: '1', '2', '3', '4', '5', and '6'. At the bottom of the dialog box, there are four buttons: 'Apply', 'Cancel', 'Clear all', and 'Select all'.

| Alert types                          | Levels                     |
|--------------------------------------|----------------------------|
| <input type="checkbox"/> E-stop      | <input type="checkbox"/> 1 |
| <input type="checkbox"/> Not Initial | <input type="checkbox"/> 2 |
| <input type="checkbox"/> Warning     | <input type="checkbox"/> 3 |
| <input type="checkbox"/> Error       | <input type="checkbox"/> 4 |
|                                      | <input type="checkbox"/> 5 |
|                                      | <input type="checkbox"/> 6 |

Buttons: Apply, Cancel, Clear all, Select all

- This shows all alerts, including some Warnings and "Not Initial" alerts that are filtered out by default.
  - This can be useful when a level seems to be faulted but there is no obvious reason why.
- 

## Why View Historical Alerts?

- Identify patterns of recurring issues
- Verify that alerts were properly resolved
- Troubleshoot intermittent problems
- Document issues for maintenance review

---

## Common Alert Codes

Here are some frequently encountered alerts (see [Alert Reference](#) for complete list):

### 119 - Photocell still blocked after dispensing

- Photocell for a lane remained blocked after dispense move completed
- Usually indicates a case issue (broken, wrong size, poorly positioned)

### 121 - Photocell above lane never detected case during storing move

- InnoPick tried to store a case but photocell was never blocked
- Usually means no case was present to store

### 123 - Photocell never detected case during dispense move

- Case expected but not detected during dispense
- Possible causes: missing case, inventory count error

### 310 - Clutch Fault

- Clutch assembly failed to engage properly after multiple attempts
- Usually requires mechanical inspection

### 515 - Unexpected case entering infeed

- Infeed photocell blocked when no case was expected
- Check upstream conveyor system

### 555 - E-Stop condition active

- Safety system engaged (E-Stop pressed or door open)
- Clear E-Stop condition, then clear alert to resume

## 709 - Photocell blocked (PLC I/O Check)

- Photocell unexpectedly blocked during routine monitoring
  - Check for debris, damaged cases, or sensor malfunction
- 

## Working with Alerts

### Understanding Alert Progression

#### **Typical Alert Lifecycle:**

1. Issue occurs (sensor detects problem)
  2. Alert generated and logged
  3. System puts affected level in Manual or Fault mode
  4. Alert appears on Home Page and Alerts Page
  5. Operator resolves underlying issue
  6. Operator clears alert
  7. System resumes operation
- 

### When to Clear Alerts

#### **You should clear alerts only when:**

- The underlying physical issue has been resolved
- Cases are in correct positions
- Sensors are unblocked
- Mechanical issues are fixed
- Safety devices are reset

#### **Do not clear alerts if:**

- The problem still exists
  - You're unsure what caused the alert
  - Physical conditions don't match expectations
  - Multiple related alerts are occurring
- 

## If Alerts Won't Clear

Some alerts cannot be cleared until specific conditions are met:

1. **Read the alert message carefully**
2. **Check physical conditions**
  - Is a photocell still blocked?
  - Is an E-Stop still pressed?
  - Is a case in the wrong position?
3. **Resolve the root cause**
4. **Try clearing again**
  - Use the [Recovery Action: Send Clear Alarm on Level](#)
5. The Alert in question is being filtered (not visible).
  - [See how to View All Alerts](#)
6. **Consult troubleshooting guide** if issue persists

See [Alert Guidelines](#) for best Alert resolution practices.

---

## Alert Troubleshooting Strategy

### For Single Alerts

If an alert occurs once:

1. Note the alert type and details

2. Resolve the issue
  3. Clear the alert
  4. Monitor to ensure it doesn't recur
- 

## For Repeated Alerts

If the same alert occurs multiple times:

1. **There's likely an underlying problem** that wasn't fully resolved
  2. Check for:
    - Damaged cases causing repeated issues
    - Incorrect inventory quantities
    - Mechanical problems
    - Configuration issues
  3. Resolve the root cause, not just the symptom
  4. Document the issue for maintenance review
- 

## For Multiple Different Alerts

If multiple alert types occur in sequence:

1. **They may be related** - one issue causing cascading effects
  2. Review the [Alert Guidelines](#)
  3. Focus on resolving the earliest alert first
  4. Verify physical conditions match the inventory graph
  5. May indicate a larger system issue requiring maintenance
- 

## Best Practices

## During Production

- Keep the Alerts page open in a second browser tab
- Check regularly for new alerts
- Address warnings before they become critical
- Document recurring issues

## For Troubleshooting

- Use search to find similar past alerts
- Note the frequency and pattern
- Check if alerts correlate with specific products or lanes
- Share alert history with maintenance personnel

---

## Related Topics

- [Alert Reference Page](#) - Complete list of all alert codes
- [Alert Guidelines](#) - Step-by-step resolution procedures
- [Home Page Alerts](#) - Quick alert overview
- [Troubleshooting](#) - General problem-solving guidance

---

**Navigation:** [← Home Page](#) | [Inventory Section →](#)

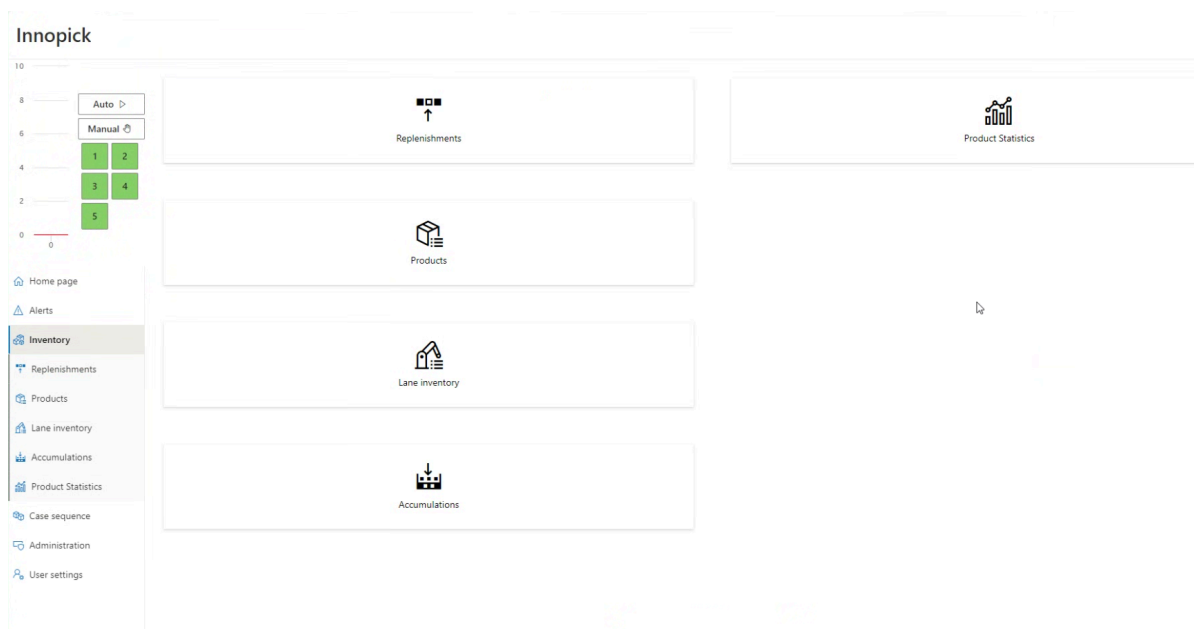
# Inventory Section

[Home](#) > [Main Screens](#) > **Inventory Section**

---

## Overview

The Inventory section provides comprehensive tools for managing product replenishments, configuring product parameters, monitoring storage lane contents, tracking accumulation, and analyzing product statistics.



## Inventory Section Pages

The Inventory section contains five pages:

1. [Replenishments](#) - Track incoming product replenishments

2. [Products](#) - Configure product settings and parameters
3. [Lane Inventory](#) - View and edit storage lane contents
4. [Outbound Accumulations](#) - Monitor outfeed accumulation space
5. [Product Statistics](#) - Analyze product usage

---

## Replenishments Page

[Back to Top - Inventory Pages Overview](#)

| <div><input type="text" value="Search"/></div> <div> Filters</div> |          |                   |                    |           |       |        |         |                                |                  |                    |                   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------|--------------------|-----------|-------|--------|---------|--------------------------------|------------------|--------------------|-------------------|---|
| Id                                                                                                                                                  | Sequence | Initiator Case... | Initiator Palle... | State     | Level | Lanes  | Product | Product description            | Desired quantity | Quantity to induct | Inducted quantity |   |
| e491782a-2986-4bf1-9d7c-45096800b72                                                                                                                 | ...70020 | 1                 | 1825342            | Completed | 5     | 60, 59 | 10033   | Bud 2/12 12oz LN               | 56               | 56                 | 56                | ⌵ |
| acbd6444-85ce-4b9b-8932-23a09309aac5                                                                                                                | ...70021 | 2                 | 0010829000018251   | Completed | 4     | 60     | 61021   | Modelo Chel Limon 2/12 12oz CN | 81               | 31                 | 31                | ⌵ |
| 521c8693-c34c-47d1-62ce-a7febf0c0de                                                                                                                 | ...70022 |                   |                    | Completed | 2     |        | 13127   | Busch Light 15/25oz CN         | 1                | 1                  | 1                 | ⌵ |
| 1c1ddda-9d47-48ac-99ba-8da109595bcb                                                                                                                 | ...70023 |                   |                    | Completed | 2     |        | 13127   | Busch Light 15/25oz CN         | 1                | 1                  | 1                 | ⌵ |
| 69769199-3d81-4ca1-b003-897153dfaca7                                                                                                                | ...70024 |                   |                    | Completed | 1     |        | 97026   | Franziskaner Hefe 4/6 12oz LN  | 1                | 1                  | 1                 | ⌵ |
| 28276d5f-3aa3-49ab-bd57-d14befb0e8bb                                                                                                                | ...70025 |                   |                    | Completed | 1     |        | 97026   | Franziskaner Hefe 4/6 12oz LN  | 1                | 1                  | 1                 | ⌵ |
| 900f13a0-75c1-48a6-ad66-e43426039092                                                                                                                | ...70026 |                   |                    | Completed | 1     |        | 97026   | Franziskaner Hefe 4/6 12oz LN  | 1                | 1                  | 1                 | ⌵ |

## What Are Replenishments?

Replenishments occur when InnoPick doesn't have sufficient inventory for a particular order. When this happens:

1. InnoPick generates a replenishment request to MixMaster
2. A pallet of the needed product is sent to the replenishment section
3. The pallet is depalletized
4. Cases are inducted into InnoPick

---

## Replenishment Quantities

A replenishment can range from:

- **Minimum:** A single layer
- **Maximum:** An entire pallet

Replenishment quantities are determined automatically by InnoPick's algorithms unless manually overridden. See [Products Page](#) for manual configuration options.

---

## Replenishments Page Columns

### Id

- The GUID (unique identifier) for the replenishment
- Useful for tracking in MixMaster or other systems

### Sequence

- Number that increases incrementally based on when the need was identified
- Indicates desired order of replenishments for a given level
- Used to prioritize replenishment requests for depalletizing

### Initiator Case / Pallet

- Details about what triggered the replenishment need
- Shows which customer order created the demand

### State

Replenishments progress through these states:

- **Pending:** Replenishment is planned but not yet called for depalletizing
- **In Progress:** Anywhere in process from being requested to just before induction
- **Inducting:** Currently being inducted into InnoPick
- **Completed:** Fully inducted into storage lanes

Replenishments can also be **Canceled**

### Level

- Which InnoPick level the replenishment is assigned to

### Lanes

- Which storage lanes will receive the replenishment cases

**Product**

- Product SKU code

**Product Description**

- Configured name of the product

**Desired Quantity**

- How many cases were requested in the replenishment

**Quantity to Induct**

- Actual number of cases being sent from depalletizer to InnoPick
- May be less than Desired Quantity if:
  - Source pallet has fewer layers than requested
  - Replenishment was shorted by 1 or more layers during depalletization
  - Cases were rejected at inbound QC station
- If quantity is insufficient to fulfill the client orders, another replenishment is automatically created

**Inducted Quantity**

- Number of cases successfully inducted to date
  - Updates in real-time during induction
- 

**Replenishment Actions**

| Desired quantity | Quantity to induct | Inducted quantity |   |
|------------------|--------------------|-------------------|---|
| 56               | 56                 | 56                | ⋮ |
| 81               | 31                 | 31                | ⋮ |
| 1                | 1                  | 1                 | ⋮ |
| 1                | 1                  | 1                 | ⋮ |
| 1                | 1                  | 1                 | ⋮ |
| 1                | 1                  | 1                 | ⋮ |
| 1                | 1                  | 1                 | ⋮ |
| 1                | 1                  | 1                 | ⋮ |

In Progress  
Induct  
Alter Quantity

Click the three vertical dots (⋮) on the right side of any replenishment to access these options:

### In Progress

- Manually changes replenishment status from **Pending** to **In Progress**
- Only works if the replenishment is in status **Pending**
- Normally set automatically by MixMaster (WCS)
- Use only in exception scenarios when automatic update failed

### Induct

13127 Busch Light 15/25oz CN

**Induct Replenishment**

Quantity to induct

56

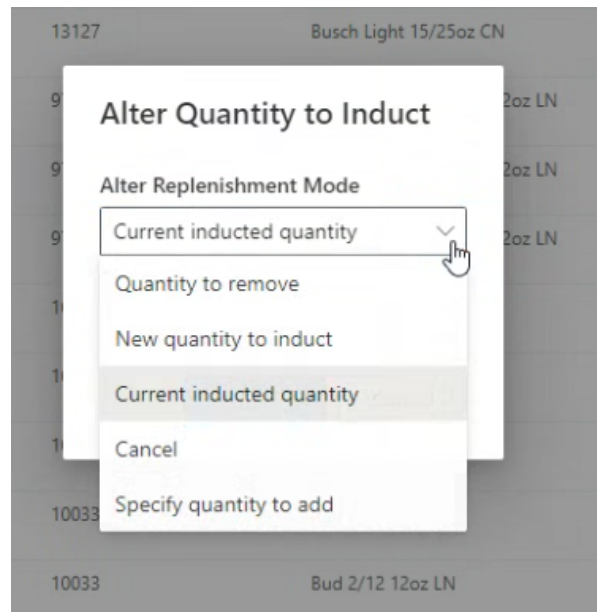
Case Origin \*

Confirm Cancel

- Changes status from **In Progress** to **Inducting**

- Only works if:
- Replenishment is already in "In Progress" state
- No other replenishment is currently "Inducting" on that level
- Use if there was a communication problem with MixMaster.

### Alter Quantity



- Allows manual adjustment of replenishment quantity
- Use when cases need to be manually removed before reaching InnoPick
- **Note:** Cases rejected at Inbound QC are automatically deducted

---

## Products Page

[Back to Top - Inventory Pages Overview](#)

Search

Filters

| Reference | Description                    | Dynamic R... | Level Num...  | Auto assig...                       | Repl Qty | Repl Qty ... | Repl Qty ... | Repl Qty set | Max Qty/L... | Max Qty/L... | Acceleration | Length (m... | Width (mm) | Height (m... | Last seen date  |
|-----------|--------------------------------|--------------|---------------|-------------------------------------|----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|------------|--------------|-----------------|
| 70012     | SW Daytrip 2/12 12oz CN        | 20           | 1             | <input checked="" type="checkbox"/> | 20       | 20           |              |              | 29           |              | 0.1          | 411          | 277        | 127          | Fri Sep 19 2025 |
| 70002     | SW 420 Extra 2/12 12oz CN      | 20           | 1             | <input checked="" type="checkbox"/> | 20       | 20           |              |              | 29           |              | 0.1          | 411          | 277        | 127          | Fri Sep 19 2025 |
| 61054     | Modelo Chel Mango 12/24oz CN   | 28           | 3             | <input checked="" type="checkbox"/> | 28       | 28           |              |              | 33           |              | 0.1          | 308          | 228        | 194          | Tue Jan 27 2026 |
| 61024     | Modelo Chel Limon 12/24oz CN   | 28           | 2             | <input checked="" type="checkbox"/> | 28       | 28           |              |              | 33           |              | 0.1          | 308          | 228        | 194          | Tue Jan 27 2026 |
| 61022     | Modelo Chel VP 2/12 12oz CN    | 27           | 2             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 411          | 277        | 127          | Tue Jan 27 2026 |
| 61021     | Modelo Chel Limon 2/12 12oz CN | 27           | 5             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 411          | 277        | 127          | Tue Jan 27 2026 |
| 61005     | Modelo Chel Pina 12/24oz CN    | 28           | 2             | <input checked="" type="checkbox"/> | 28       | 28           |              |              | 33           |              | 0.1          | 308          | 228        | 194          | Tue Jan 27 2026 |
| 61004     | Modelo Chellada 12/24oz CN     | 28           | 5             | <input checked="" type="checkbox"/> | 28       | 28           |              |              | 33           |              | 0.1          | 308          | 228        | 194          | Tue Jan 27 2026 |
| 60833     | Pacifico 2/12 12oz LN          | 27           | 5             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 30           |              | 0.1          | 401          | 265        | 247          | Tue Jan 27 2026 |
| 60821     | Pacifico 2/12 12oz CN          | 27           | 4             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 401          | 277        | 127          | Tue Jan 27 2026 |
| 60632     | Modelo Negra 2/12 12oz BTL     | 21           | 3             | <input checked="" type="checkbox"/> | 14       | 14           |              | 28           | 27           |              | 0.1          | 460          | 299        | 201          | Tue Jan 27 2026 |
| 60480     | Modelo Especial 12/32oz BTL    | 40           | 5             | <input checked="" type="checkbox"/> | 16       | 16           |              | 24           | 27           |              | 0.1          | 393          | 298        | 266          | Tue Jan 27 2026 |
| 60475     | Modelo Especial 12/24oz CN     | 28           | 5             | <input checked="" type="checkbox"/> | 28       | 28           |              |              | 33           |              | 0.1          | 308          | 228        | 194          | Tue Jan 27 2026 |
| 60451     | Modelo Especial 6/4 16oz CN    | 28           | 4             | <input checked="" type="checkbox"/> | 14       | 14           |              | 28           | 29           |              | 0.1          | 414          | 274        | 161          | Tue Jan 27 2026 |
| 60437     | Modelo Especial 24/13oz LN     | 21           | 5             | <input checked="" type="checkbox"/> | 14       | 14           |              | 28           | 27           |              | 0.1          | 448          | 299        | 201          | Tue Jan 27 2026 |
| 60436     | Modelo Especial 18/12oz LN     | 40           | 2             | <input checked="" type="checkbox"/> | 20       | 20           |              | 30           | 34           |              | 0.1          | 448          | 225        | 201          | Tue Jan 27 2026 |
| 60432     | Modelo Especial 2/12 12oz BTL  | 56           | 1             | <input checked="" type="checkbox"/> | 14       | 14           |              | 21           | 27           |              | 0.1          | 456          | 305        | 199          | Tue Jan 27 2026 |
| 60430     | Modelo Especial 4/6 12oz BTL   | 28           | 5             | <input checked="" type="checkbox"/> | 14       | 14           |              | 28           | 27           |              | 0.1          | 447          | 299        | 201          | Tue Jan 27 2026 |
| 60424     | Modelo Especial 24/12oz CN     | 45           | 5             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 400          | 268        | 124          | Tue Jan 27 2026 |
| 60422     | Modelo Especial 18/12oz CN     | 90           | 3, 5          | <input checked="" type="checkbox"/> | 20       | 20           |              | 30           | 37           |              | 0.1          | 398          | 201        | 124          | Tue Jan 27 2026 |
| 60421     | Modelo Especial 2/12 12oz CN   | 45           | 1, 3, 2, 5, 4 | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 401          | 277        | 127          | Tue Jan 27 2026 |
| 60420     | Modelo Especial 4/6 12oz CN    | 27           | 1             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 29           |              | 0.1          | 401          | 277        | 127          | Wed Nov 12 2025 |
| 60400     | Modelo Especial 24/7oz LN-LSE  | 27           | 1             | <input checked="" type="checkbox"/> | 18       | 18           |              | 27           | 30           |              | 0.1          | 390          | 256        | 176          | Tue Jan 27 2026 |

2

100/page

(144 results)

The Products page displays all products in the InnoPick Manager database with their configuration settings.

## Main Products Page Columns

### Reference

- The product's SKU code
- Primary identifier for the product

### Description

- Configured product name
- Human-readable description

### Dynamic Replen Quantity

- Number of cases in a standard replenishment as calculated by IPM
- Dynamically adjusted based on demand patterns

### Level Number(s)

- Which levels InnoPick is allowed to use for this product

- Products can be restricted to specific levels

### **Auto Assign Levels**

- True/False setting
- When True: InnoPick automatically determines which levels to use
- **Note:** It is possible to change the level assignment(s) when a Auto Assign is still enabled. This temporarily overrides the algorithm
- When False: Levels must be manually assigned

### **Replenishment Quantity**

- Default replenishment size calculated by InnoPick's algorithm
- Based on historical usage patterns

### **Replenishment Quantity Set**

- User-defined override for default replenishment quantity
- Use to manually control replenishment size

### **Replenishment Quantity Minimum**

- Minimum cases allowed in a replenishment (calculated by InnoPick)
- Must be an even multiple of cases per layer

### **Replenishment Quantity Minimum Set**

- User-defined minimum replenishment quantity
- Override automatic calculation if needed

### **Max Quantity / Lane**

- Maximum cases that can fit in a storage lane (calculated by InnoPick)
- Based on: lane length, configured gaps, case width

### **Max Quantity / Lane Set**

- User-defined maximum quantity per lane
- **Caution:** Do not exceed automatic calculation or cases may go past lane end

### **Acceleration**

- Controls how storage lanes handle the product during store/dispense operations

- Measured in g's (gravitational force)
- **Typical range:** 0.07g to 0.15g
- **Default:** 0.1g
- **Setting guideline:** Choose highest value that doesn't cause storage faults
- Higher acceleration = faster operations but may disturb unstable cases or cause cases to slide on chains and lose the gaps between cases.

#### **Length (mm)**

- Case length in millimeters
- Always the longer horizontal dimension

#### **Width (mm)**

- Case width in millimeters
- Always shorter than length

#### **Height (mm)**

- Case height in millimeters

#### **Last Seen Date**

- Last time this product was present in InnoPick
- Useful for identifying inactive products

---

## Editing Products

## Corona Light 2/12 12oz LN



### General

Acceleration \*

Max Quantity per lane set (cases)

Replenishment Quantity Set (cases)

Replenishment Quantity Min Set (cases)

Assigned levels

☐

Level: 1

☐

Level: 2

☐

Level: 3

☒

Level: 4

☐

Level: 5



Auto assign sequence conveyors

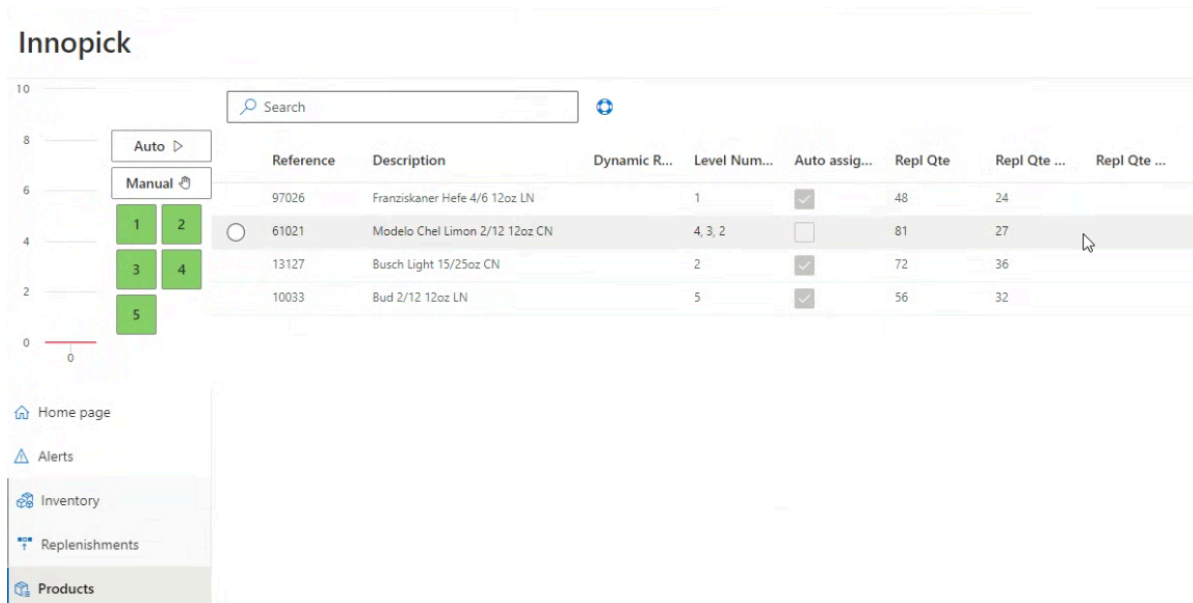
### Actions

Delete 

To edit a product:

1. Click on the product row in the table
2. Edit panel appears on the right side of the screen
3. Modify desired parameters
4. Save changes (button at the bottom of the sidebar)

### Example: Manual Level Assignment



When **Auto Assign** is disabled, you can manually assign the product to specific levels:

1. Uncheck "Auto Assign Levels"
2. Select which levels can store this product
3. Save the configuration

**Note:** It is possible to change the level assignment(s) when a Auto Assign is still enabled. This temporarily overrides the algorithm

---

## Lane Inventory Page

[Back to Top - Inventory Pages Overview](#)

| Level: 1      | Level: 2 | Level: 3          | Level: 4                      | Level: 5        |                      |                           |         |                  |                |            |
|---------------|----------|-------------------|-------------------------------|-----------------|----------------------|---------------------------|---------|------------------|----------------|------------|
| Lane number ↑ | Quantity | Product reference | Product description           | Origins         | Sources              | Entered inventory dates   | Actions | Disable/Enabl... | Ignore sensors | Lane Types |
| 11            | 0        |                   |                               | --              | --                   | --                        |         |                  |                |            |
| 12            | 17       | 47622             | Stella Artois 2/12 12oz CN    | 276441          | Replenishment        | 1/27/2026, 10:11 PM +16   |         |                  |                |            |
| 13            | 9        | 47637             | Stella Artois 18/11.2oz LN US | 275767          | Replenishment        | 1/26/2026, 11:19:58 AM +8 |         |                  |                |            |
| 14            | 21       | 26824             | Select 55 24/12oz Ch          | 276782          | Replenishment        | 1/27/2026, 2:29:13 PM +20 |         |                  |                |            |
| 15            | 8        | 15236             | Busch Light 18/12oz LN        | 276409          | Replenishment        | 1/26/2026, 10:04:10 PM +7 |         |                  |                |            |
| 16            | 0        |                   |                               | --              | --                   | --                        |         |                  |                |            |
| 17            | 26       | 47646             | Stella Artois 24/11.2oz LN US | MANUAL_INDUCT + | laneCaseSources.1 +1 | 1/27/2026, 8:13:06 AM +25 |         |                  |                |            |
| 18            | 2        | 60421             | Modelo Especial 2/12 12oz Ch  | 276809          | Replenishment        | 1/27/2026, 2:05:46 PM +1  |         |                  |                |            |
| 19            | 0        |                   |                               | --              | --                   | --                        |         |                  |                |            |
| 20            | 26       | 12431             | MUL 4/6 12oz LN               | 276824          | Replenishment        | 1/27/2026, 2:56:36 PM +25 |         |                  |                |            |
| 21            | 5        | 10251             | BOL 6/4 16oz CN               | 276833          | Replenishment        | 1/27/2026, 3:02:55 PM +4  |         |                  |                |            |
| 22            | 23       | 12433             | MUL 2/12 12oz LN              | 276815          | Replenishment        | 1/27/2026, 2:48:16 PM +22 |         |                  |                |            |
| 23            | 3        | 48833             | Becks Pilsner 2/12 12oz LN    | 276798          | Replenishment        | 1/27/2026, 2:35:55 PM +2  |         |                  |                |            |
| 24            | 13       | 10051             | Bud 6/4 16oz CN               | 276837          | Replenishment        | 1/27/2026, 3:02:25 PM +12 |         |                  |                |            |
| 25            | 10       | 60432             | Modelo Especial 2/12 12oz BT  | 276823          | Replenishment        | 1/27/2026, 2:59:32 PM +9  |         |                  |                |            |

The Lane Inventory page is an expanded version of the lane display available on the [Home Page](#).

## Main Lane Inventory Page Columns

### Lane Number

- Identifies the specific storage lane

### Quantity

- Current number of cases in the lane

### Product Reference

- SKU code of product stored in lane

### Product Description

- Configured name of the product

### Origins

- Pallet ID of the source replenishment
- Tracks where cases came from

### Sources

- Indicates how cases entered the lane:
- **Replenishment:** Normal replenishment process
- **Manual Replenishment:** Manually created replenishment
- **Change Quantity Operation:** Manually added by operator

**Entered Inventory Dates**

- When the product was inducted into the lane
- May show multiple dates if lane has cases from different replenishments

**Actions**

- Click the icon to edit lane contents
- Only available in Manual mode

---

**Enable/Disable Controls**

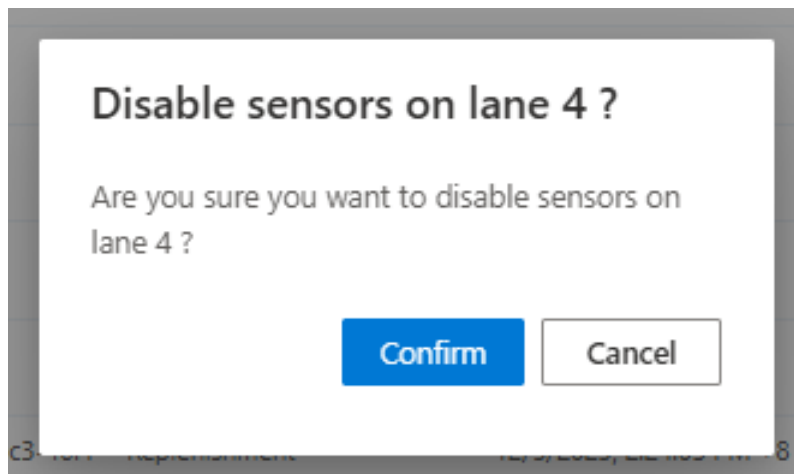
| Actions                                                                             | Disable/Enabl...                    | Ignore sensors                      |
|-------------------------------------------------------------------------------------|-------------------------------------|-------------------------------------|
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |
|  | <input checked="" type="checkbox"/> | <input type="checkbox"/>            |

### Disable / Enable Lane Toggle

- Controls whether a specific lane can be used in production
- **Warning:** Disabling a lane during production may require manual intervention if:
  - Products were already scheduled to store in that lane
  - Products need to be dispensed from that lane
- See the [Disabled Lane Troubleshooting Guide](#) for information on how to deal with with disabling a lane that has scheduled inbound or outbound cases.

### Ignore Sensors

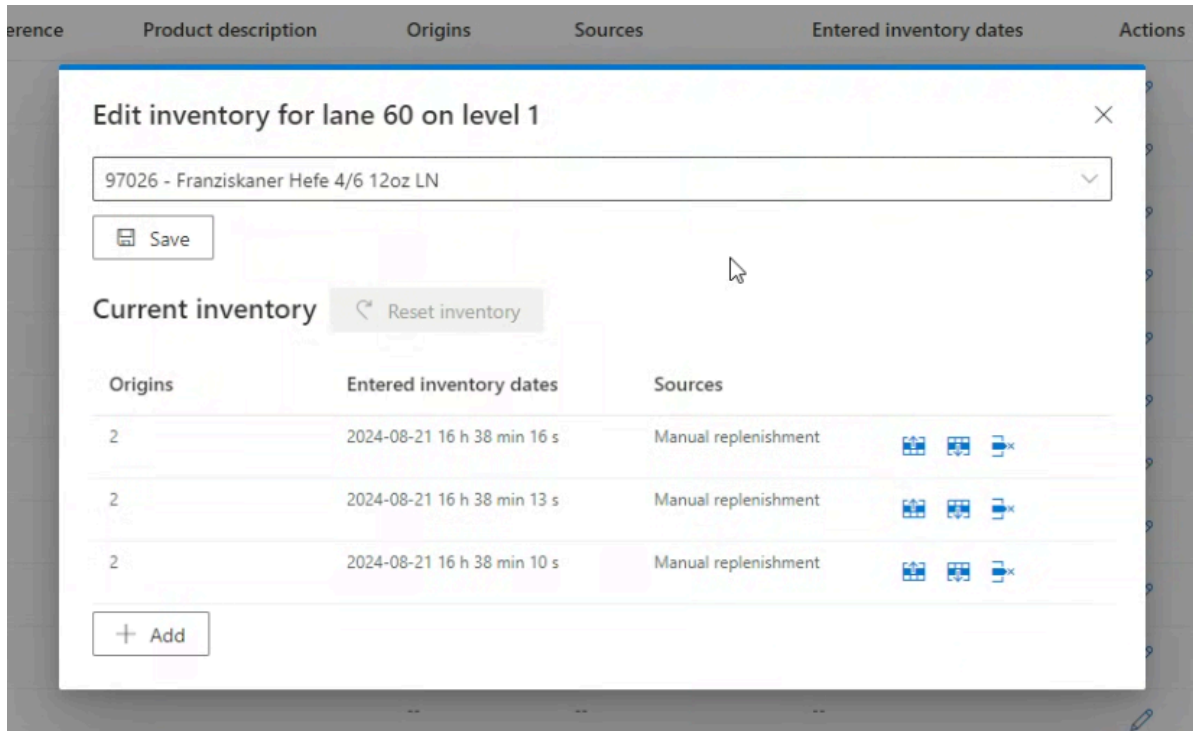
- This option can be enabled only on a disabled lane.
- Normally, a sensor in an unexpected state causes an alert continually on that level of InnoPick.
- When this option is used, the system ignores the unexpected state of the sensors associated with that lane.



- This option should be used in coordination with maintenance staff responsible for mechanical interventions.

---

### Editing Lane Inventory



### Prerequisites for Editing

You can only edit lane contents when the system is in **Manual mode** because:

- Lane contents may be changing in Automatic mode
- Editing during Automatic could cause synchronization issues

### If you try to edit in Automatic mode:

- Error message appears: "No modifications are allowed during automatic mode. Please switch to manual mode to update lane inventories."

### Edit Lane Dialog

When editing a lane, you can:

#### Change Product Type:

- Use the drop-down menu at top to select different product

#### Manage Individual Cases:

- Each case shown as a horizontal entry
- Icons on right allow you to:

- Move case up in sequence
- Move case down in sequence
- Remove case from lane

#### Add Cases:

**Edit inventory for lane 60 on level 1**

97026 - Franziskaner Hefe 4/6 12oz LN

Save

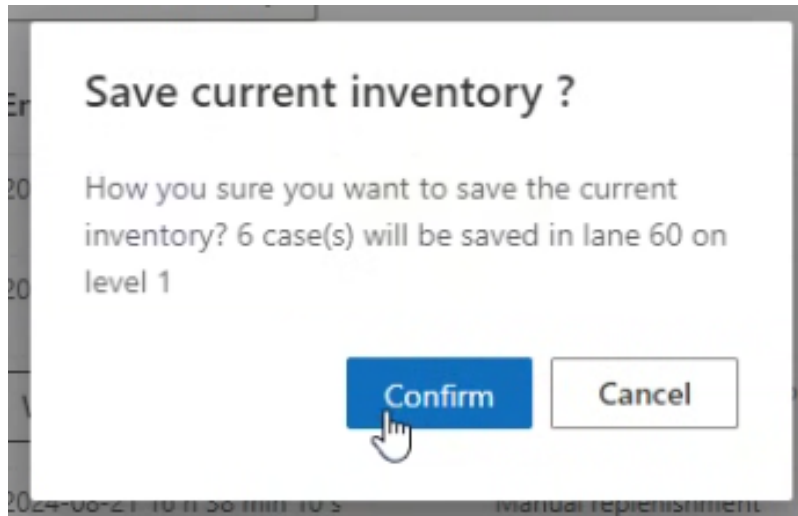
**Current inventory** Reset inventory

| Origins | Entered inventory dates     | Sources                   |
|---------|-----------------------------|---------------------------|
| 2       | 2024-08-21 16 h 38 min 16 s | Manual replenishment      |
| 2       | 2024-08-21 16 h 38 min 13 s | Manual replenishment      |
| 2       | 2024-08-21 16 h 38 min 10 s | Manual replenishment      |
|         | Wed Aug 21 2024             | Change quantity operation |
|         | Wed Aug 21 2024             | Change quantity operation |

+ Add

- Click **+ Add** button on bottom left
- Creates new cases with "Change quantity operation" as source
- Useful for manual inventory corrections

#### Save Changes:



- Click **Save** button when edits are complete
- Confirm the changes in the pop-up dialog

#### Important Considerations

##### Use caution when changing lane contents during production:

- Inventory may already be committed to the case sequence
- InnoPick may not always accommodate sudden changes
- Inventory changes can trigger a [Deadlock](#) -- see [here for a troubleshooting guide](#).

---

## Outbound Accumulations Page

[Back to Top - Inventory Pages Overview](#)



The Accumulations page displays values related to InnoPick's management of the space between InnoPick's exit and the downstream merge point (Merges or equivalent).

## How Accumulation Works

InnoPick tracks:

- How many millimeters of cases it has released (if using low-pressure accumulation) or the number of cases (if using zone accumulation)
- Cases exiting the downstream merge point (Merges)
- Available accumulation space remaining

Based on these factors, InnoPick decides whether to schedule more output cases.

## Accumulations Page Columns

## Level

- Which InnoPick level and outbound conveyor (A vs B)
- Shared: Accumulation space that is shared across all levels

## In Progress mm A / B

- Millimeters of cases currently scheduled to be released
- Separate tracking for each destination (A and B)

## Completed mm A / B

- Millimeters of cases that have been released
- Represents space currently occupied in accumulation zone

## Accum % A / B

- What the Completed mm represents as percentage of total configured accumulation space
- Example: 75% means accumulation zone is three-quarters full

## Zone A & Zone B

- How many cases are currently scheduled to be released compared to the number of accumulation zones.

## Clear

- Button to clear the accumulation tracking
- Tells InnoPick the accumulation space is now empty
- Use when:
  - Downstream conveyors have been cleared
  - Cases have been manually removed
  - Accumulation tracking is out of sync with reality

**Note: The degree to which InnoPick schedules output cases is affected by, among other things, whether the *Build Inventory Mode* is active or not.**

For more information on the Build Inventory Mode, see [Setup InnoPick > Buffer](#) section

---

# Product Statistics Page

[Back to Top - Inventory Pages Overview](#)

Search

| Reference | Description                   | Completed Case Cou... | CCC Today | CCC Yesterday | CCC Week | CCC Month | Completed Case Perc... | Lane Count | Lanes Cases | Youngest Case Date     |
|-----------|-------------------------------|-----------------------|-----------|---------------|----------|-----------|------------------------|------------|-------------|------------------------|
| 10031     | Bud 4/6 12oz LN               | 0                     | 0         | 0             | 0        | 0         | 0 %                    | 0          | 0           |                        |
| 203205    | NUTRL Fruit VP 3/6 12oz CN    | 4414                  | 54        | 89            | 460      | 3053      | 0.5120091312548284 %   | 1          | 22          | 1/27/2026, 12:23:16 PM |
| 10050     | Bud 15/16oz CN                | 4759                  | 55        | 155           | 622      | 3198      | 0.5520279691077771 %   | 1          | 3           | 1/27/2026, 11:07:49 AM |
| 47637     | Stella Artios 16/11.2oz LN US | 1619                  | 5         | 14            | 172      | 1340      | 0.1677965463302146 %   | 1          | 9           | 1/26/2026, 11:19:56 AM |
| 26824     | Select 55 24/12oz Cn          | 2102                  | 58        | 82            | 363      | 1905      | 0.2438249183243428 %   | 1          | 21          | 1/27/2026, 2:29:13 PM  |
| 60233     | Corona Light 2/12 12oz LN     | 4712                  | 53        | 118           | 600      | 3355      | 0.5465761274262594 %   | 1          | 16          | 1/27/2026, 1:00:32 PM  |
| 60133     | Corona Premier 2/12 12oz LN   | 3599                  | 37        | 68            | 377      | 2181      | 0.3548336956294789 %   | 1          | 24          | 1/27/2026, 12:45:1 PM  |
| 10071     | Bud 15/25oz CN                | 1813                  | 24        | 53            | 266      | 1228      | 0.2103018928333213 %   | 1          | 3           | 1/27/2026, 11:23:07 AM |
| 16251     | Natural Ice 6/4 16oz CN       | 4602                  | 75        | 170           | 728      | 3190      | 0.533816497968899 %    | 1          | 6           | 1/27/2026, 2:57:51 PM  |
| 10233     | BDL 2/12 12oz LN              | 5095                  | 70        | 116           | 622      | 3439      | 0.5910028372776054 %   | 1          | 11          | 1/27/2026, 1:02:42 PM  |
| 15621     | BudCh NA 2/12 12oz CN         | 2230                  | 41        | 39            | 277      | 1549      | 0.2575125218363659 %   | 1          | 22          | 1/27/2026, 12:52:40 PM |
| 60080     | Corona Familiar 12/22oz BTL   | 6717                  | 74        | 303           | 1024     | 4247      | 0.7791493735021935 %   | 1          | 10          | 1/27/2026, 2:06:59 PM  |
| 60621     | Pacifico 2/12 12oz CN         | 1836                  | 22        | 40            | 216      | 1145      | 0.21320180861947798 %  | 1          | 21          | 1/27/2026, 12:12:38 PM |

The Product Statistics page provides detailed information about how InnoPick uses various products.

## Product Statistics Columns

Innopick

Search

Auto

Manual

1

2

3

4

5

| CCC Month | Completed Case Perc... | Lane Count | Lanes Cases | Youngest Case Date          | Oldest Case Date            | SC In | SC Out | Replen. Cases | Case Seq. Total |
|-----------|------------------------|------------|-------------|-----------------------------|-----------------------------|-------|--------|---------------|-----------------|
| 0         | 0 %                    | 2          | 8           | 2024-08-21 16 h 57 min 50 s | 2024-08-21 16 h 38 min 10 s | 0     | 0      | 0             | 0               |
| 23        | 88.46153846153845 %    | 1          | 33          | 2024-08-21 16 h 23 min 59 s | 2024-08-21 16 h 23 min 50 s | 0     | 0      | 0             | 0               |
| 0         | 0 %                    | 1          | 2           | 2024-08-21 16 h 38 min 07 s | 2024-08-21 16 h 38 min 03 s | 0     | 0      | 0             | 0               |
| 3         | 11.538461538461538 %   | 1          | 28          | 2024-08-21 16 h 33 min 38 s | 2024-08-21 16 h 33 min 29 s | 0     | 0      | 0             | 0               |

Home page

Alerts

Inventory

Replenishments

Products

Lane inventory

Accumulations

Product Statistics

Case sequence

Administration

User settings

1

100/page

(4 results)

### Reference

- Product SKU code

### Description

- Configured product name

#### **Completed Case Count**

- Total number of completed cases for this product (all time)

#### **CCC Today / Yesterday / Week / Month**

- Completed case count within the specified time period
- Useful for tracking demand patterns

#### **Completed Case Percentage**

- What percentage of total production this product represents
- Helps identify high-volume products

#### **Lane Count**

- How many storage lanes currently contain this product

#### **Lanes Cases**

- Total number of cases of this product currently in InnoPick storage

#### **Youngest Case Date**

- Date of the most recently inducted case
- Indicates how fresh the inventory is

#### **Oldest Case Date**

- Date of the oldest case in inventory
- Helps identify slow-moving or stale inventory

#### **SC In**

- Cases on sequence conveyor going to storage (inbound)
- Currently being inducted from replenishment

#### **SC Out**

- Cases on sequence conveyor going to outfeed (outbound)
- Currently being output to palletizers

### Replen. Cases

- Cases scheduled to be inducted as replenishments
- Pending or in-progress replenishments

### Case Seq Total

- How many cases of this product remain in the current case sequence
  - Still needed to complete active orders
- 

## Using Product Statistics

### For Inventory Management:

- Find slow-moving products (old Oldest Case Date)
- Track high-volume products (high Completed Case Percentage)

### For Optimization:

- Products with high CCC should have adequate lane assignments
  - Products with frequent replenishments may need quantity adjustments
- 

## Related Topics

- [Home Page Lane Inventory](#) - Quick lane overview
  - [Monitoring Production](#) - Using inventory data during operations
- 

**Navigation:** [← Alerts Page](#) | [Case Sequence](#) [→](#)

# Case Sequence Page

[Home](#) > [Main Screens](#) > [Case Sequence](#)

## Overview

The Case Sequence page displays the order in which cases will exit InnoPick and arrive at the palletizers. This powerful tool allows operators to track individual cases, verify sequencing, and troubleshoot order fulfillment issues.

| CaseRef | PalletRef    | Product | Product description | State   | Origin | Exit level | Enter inventory | Exit inventory | Destination | Actions |
|---------|--------------|---------|---------------------|---------|--------|------------|-----------------|----------------|-------------|---------|
| 27      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 28      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 29      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 30      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 31      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 32      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 33      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 34      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 35      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 36      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 37      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 38      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 39      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 40      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 41      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |
| 42      | 234400000012 | 10033   | Bud 2/12 12oz UN    | Pending |        | 5          | --              | --             | A           |         |

## Understanding the Case Sequence

The case sequence represents:

- The exact order cases will exit InnoPick (per level)
- Which cases belong to which customer pallets

- Current status of each case (pending, in progress, completed, cancelled)
- Which downstream destination each case is destined to

This information is critical for:

- Verifying correct order fulfillment
  - Tracking specific cases or pallets
  - Troubleshooting sequencing issues
  - Monitoring production progress
- 

## Case Sequence Columns

Each case in the sequence has the following information:

### Case Ref

- Internal case tracking number
- Unique identifier for each individual case
- Used for detailed case tracking and troubleshooting

### PalletRef

- Pallet ID for the customer pallet this case belongs to
- Multiple cases with the same PalletRef are part of the same order
- Useful for tracking pallet completion

### Product

- SKU code of the product

### Product Description

- Configured product name

## State

Cases progress through these states:

- **Pending:** Case is scheduled but not yet scheduled
- **In Progress:** Case is scheduled, which means currently visible on the sequence conveyor as an empty or full square (output case).
- **Completed:** Case has been successfully output from InnoPick
- **Cancelled:** Case has been removed from the sequence (see [Actions](#) below)

## Origin

- Pallet ID of the replenishment that brought this case into InnoPick
- Tracks the source of the case
- Useful for tracing product batches or investigating quality issues

## Exit Level

- The InnoPick level the case is assigned to
- Indicates which physical level will output this case

## Enter Inventory

- Date and time the case entered InnoPick
- Useful for tracking case age and identifying stale inventory

## Destination

- Which downstream equipment will receive the case
- Typical values: Merge A, Merge B, or other merge point designation
- Used by InnoPick to route cases correctly

## Actions

The only action available on this page is:

Cancel Case

| Exit le... | Enter inventory      | Exit inventory       | Destination | Actions                                      |
|------------|----------------------|----------------------|-------------|----------------------------------------------|
| 2          | 1/27/2026, 3:02:34 P | 1/27/2026, 3:04:45 P | B           | <div><div></div><div>Cancel case</div></div> |
| 2          | 1/27/2026, 3:02:40 P | 1/27/2026, 3:04:53 P | B           |                                              |
| 2          | 1/27/2026, 3:02:46 P | 1/27/2026, 3:05:02 P | B           |                                              |

- Removes the case from the active sequence
- Changes state to "Cancelled"
- **Warning:** During normal operations, it should not be necessary to cancel cases from the case sequence. This should only be done as part of a larger recovery effort in coordination with system experts or remote support.

Using Search and Filters

Auto ▾

Manual ▾

1

2

3

4

5

41

Filters ▾

Actions ▾

| CaseRef | PalletRef    | Product | Product description | State   | Origin | Exit level | Enter inventory | Exit inventory | Destination | Actions     |
|---------|--------------|---------|---------------------|---------|--------|------------|-----------------|----------------|-------------|-------------|
| 41      | 234400000012 | 10033   | Bud 2/12 12oz LN    | Pending |        | 5          | ---             | ---            | A           | <div></div> |
| 141     | 234400000036 | 10033   | Bud 2/12 12oz LN    | Pending |        | 5          | ---             | ---            | A           | <div></div> |
| 241     | 234400000043 | 10033   | Bud 2/12 12oz LN    | Pending |        | 5          | ---             | ---            | B           | <div></div> |

Home page

Alerts

Inventory

Case sequence

Administration

User settings

The Case Sequence page becomes particularly powerful when filters are applied.

## Search Bar

Use the search bar to quickly find cases by:

- Case reference number
  - Pallet ID
  - Product SKU
  - Product description
  - Any keyword in the displayed fields
- 

## Common Filtering Scenarios

### **Find All Cases for a Specific Pallet**

1. Enter the Pallet ID in the search bar
2. All cases for that pallet will be displayed
3. Verify the sequence is correct
4. Check that all expected cases are present

### **Find Cases by Level**

1. Filter by Exit Level
2. See all cases scheduled for a specific level
3. Useful for level-specific troubleshooting

### **Find Cancelled Cases**

1. Filter by State = Cancelled
2. Review which cases have been removed from sequence

### **Find Cases from a Specific Replenishment**

1. Search or filter by Origin (Pallet ID of replenishment)
2. See all cases from that replenishment
3. Useful for tracking product batches or investigating quality issues

## Find Cases by Destination

1. Filter by Destination (Merge A, Merge B, etc.)
  2. Verify routing is correct
- 

## Using Case Sequence for Troubleshooting

### Tracking a Missing Case

If a case doesn't arrive at the palletizer when expected:

1. **Search for the case** by Pallet ID or Case Ref
  2. **Check the State:**
    - Pending: Still waiting to be processed (or was somehow missed because of a system glitch)
    - Often, a pending case indicates the system is still waiting for a replenishment to provide that case
    - In Progress: On conveyor or in lane - check [Home Page Inventory Graph](#)
    - Completed: InnoPick released it - check downstream tracking
    - Cancelled: Was removed - determine why
  3. **Verify the case details:**
    - Correct product?
    - Correct destination?
    - Correct level?
  4. **Take appropriate action** based on findings.
    - If a case is stuck in pending mode and no replenishment for that product is on its way, contact support.
- 

### Verifying Pallet Completion

To confirm all cases for a pallet have been output:

1. **Search for the Pallet ID**

2. **Check the State of all cases:**

- All should be "Completed" for a finished pallet
  - If any are Pending or In Progress, pallet is not complete
  - A pending case usually indicates the system is still waiting for a replenishment to provide that case
  - Note any Cancelled cases
- 

## Investigating Sequencing Problems

If cases are arriving in wrong order:

1. **Compare actual sequence** to expected order

2. **Filter by Exit Level and Destination**

3. **Check for:**

- Cases from wrong level
- Cases routed to wrong destination
- Cancelled cases affecting sequence

4. **Correct the sequence** if sequence is incorrect.

5. *If InnoPick appears to be building an incorrect sequence, or the sequence in InnoPick does not match that of MixMaster, contact support.*

---

## Related Topics

- [Home Page Inventory Graph](#) - Visual representation of In Progress cases
  - [Replenishments](#) - Source of cases entering sequence
-

**Navigation:** [← Inventory Section](#) | [Administration](#) [→](#)

# Administration

[Home](#) > [Main Screens](#) > **Administration**

## Overview

The Administration section contains system configuration, advanced troubleshooting tools, and user management features. Access to these pages is typically restricted to Admin and Manager roles.



10

8

6

4

2

0

0

Clear alerts

Auto ▶

Manual ◀

1

2

3

4

5

☐ Build inventory

Home page

Alerts

Inventory

Case sequence

Administration

Manual motion control

Setup Innopick

Algo Parameters

User Manager

Recovery actions

Operations

User settings

  
Manual motion control

  
Setup Innopick

  
Algo Parameters

  
User Manager

  
Recovery actions

  
Operations

## Administration Pages

1. [Manual Motion Control](#) - Manual jog controls for troubleshooting
  2. [Setup InnoPick](#) - System configuration parameters
  3. [Algorithm Parameters](#) - Algorithm tuning
  4. [User Manager](#) - Create and manage user accounts
  5. [Recovery Actions](#) - Advanced troubleshooting tools
  6. [Operations](#) - Level and Merge statuses / controls
  7. [User Settings](#) - Personal preferences
- 

**Navigation:** [← Case Sequence](#) | [Manual Motion Control](#) [→](#)

# Manual Motion Control

[Home](#) > [Main Screens](#) > [Administration](#) > **Manual Motion Control**

Level operations

Level

1

Jog sequence conveyor

558.8 mm

At acceleration

0.1 g

◀ Towards lane 11

Towards lane 45 ▶

Lane operations

Level

1

Lane

11

Pop Up

Pop Down

Fix distance

☒

Jog lane

274 mm

At acceleration

0.1 g

▲ Storage direction

▼ Dispense direction

Level cycle operations

Jog sequence conveyor

558.8 mm

Iterations

1

At acceleration

0.1 g

Level

1

to

1

☒ Use range

Level

1

+ Add

☐ Use custom selection

Delay (seconds)

2

Jog levels

Lane cycle operations

Level

1

to

1

☒ Use range

Level

1

+ Add

☐ Use custom selection

Iterations

1

Jog lane

274 mm

At acceleration

0.1 g

Lane

11

to

11

☒ Use range

Lane

11

+ Add

☐ Use custom selection

Delay (seconds)

2

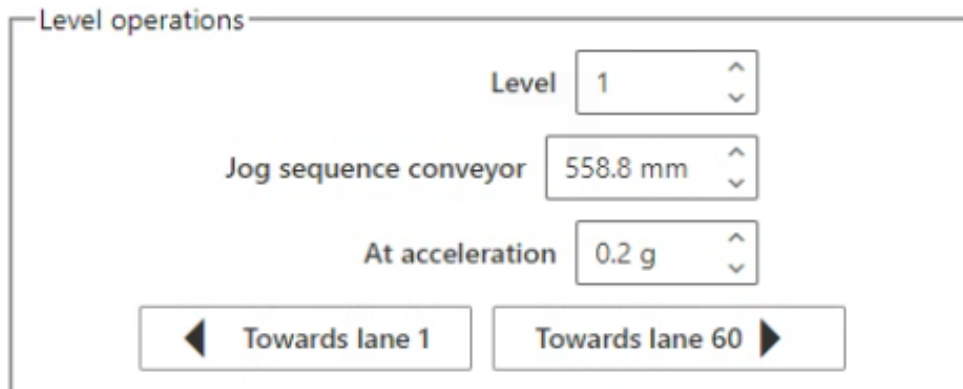
Jog lanes

The Manual motion control page provides manual control of InnoPick conveyors for troubleshooting and maintenance purposes.

## Four Main Sections

1. Level Operations - Control sequence conveyors
  2. Lane Operations - Control individual buffer lanes
  3. Level Saga Operations - Pre-program multiple sequence conveyor moves
  4. Lane Saga Operations - Pre-program multiple buffer lane moves
- 

## Level Operations



The screenshot shows a control interface titled "Level operations". It contains three input fields with up/down arrows: "Level" set to 1, "Jog sequence conveyor" set to 558.8 mm, and "At acceleration" set to 0.2 g. At the bottom, there are two buttons: "Towards lane 1" with a left-pointing arrow and "Towards lane 60" with a right-pointing arrow.

Manually jog the sequence conveyor of a selected level.

### Controls

#### Select Level:

- Choose which level to control

#### Jog Distance:

- Default: 558mm (22 inches) - standard move
- Can be customized as needed
- Measured in millimeters

#### Acceleration:

- Set in g's (gravitational force)

- Controls how quickly the conveyor accelerates and decelerates

#### Direction:

- Forward or reverse
- Typically moves toward outfeed (forward)

#### When to Use

- Troubleshooting positioning issues
- Manual case positioning after alerts
- Testing conveyor operation
- Maintenance and setup

**Important:** Only use in Manual mode when production is stopped.

## Lane Operations

Lane operations

Level

1

^

v

Lane

1

^

v

Pop Up

Pop Down

Fix distance

☒

Jog lane

0 mm

^

v

At acceleration

0.2 g

^

v

Storage direction

▲

Dispense direction

▼

Manually control individual storage lanes and clutch assemblies.

## Controls

### Level and Lane Selection:

- Drop-down menus to select specific lane

### Pop Up / Down:

- Forces the lane's pop-up assembly to move up or down
- Used for manual case handling

### Distance:

- Default: Width of product in lane (if present), otherwise 0mm
- If "Fix Distance" option is unchecked, the system will automatically set the Jog lane distance to according to the product in the lane (if applicable)

### Directions:

- **Storage:** Move into the lane (backward)
- **Dispense:** Move toward sequence conveyor (forward)

### When to Use

- Clearing jammed cases
- Manual case positioning
- Testing lane operation
- Maintenance procedures

**Caution:** Be aware of case positions before jogging lanes.

---

## Level & Lane Saga Operations

#### Level cycle operations

Jog sequence conveyor 558.8 mm

Iterations 1

At acceleration 0.1 g

Level 1 to 1 ☒ Use range

Level 1 ☐ + Add ☐ Use custom selection

Delay (seconds) 2

Jog levels

#### Lane cycle operations

Level 1 to 1 ☒ Use range

Level 1 ☐ + Add ☐ Use custom selection

Iterations 1

Jog lane 274 mm

At acceleration 0.1 g

Lane 11 to 11 ☒ Use range

Lane 11 ☐ + Add ☐ Use custom selection

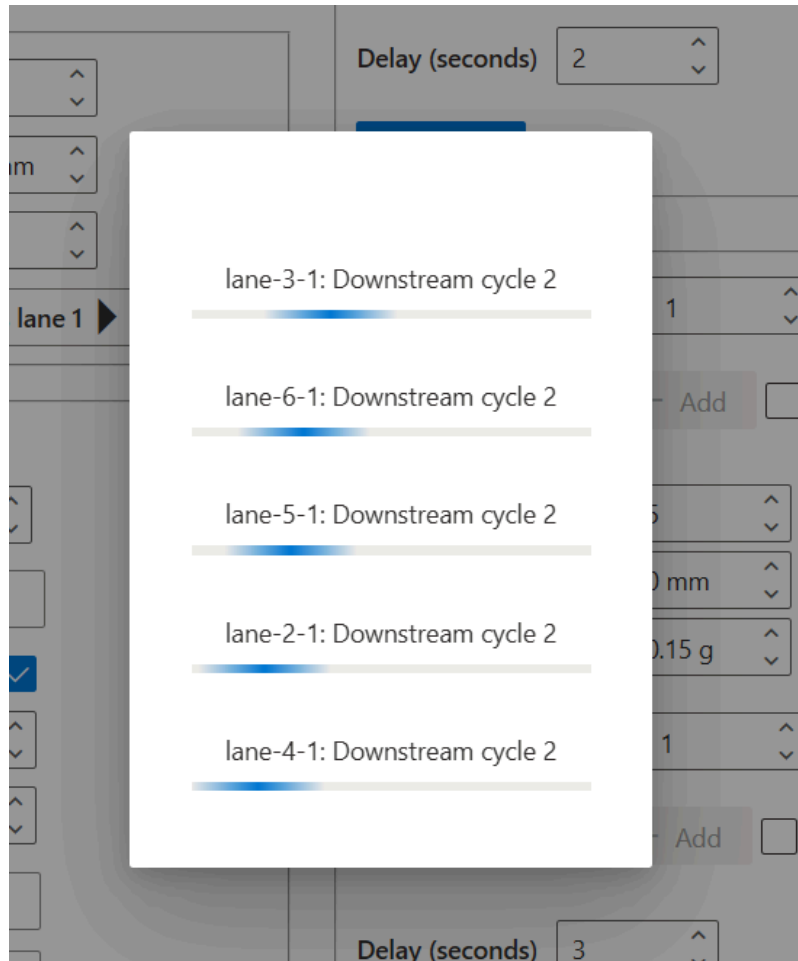
Delay (seconds) 2

Jog lanes

These interfaces allow pre-programming multiple moves in sequence.

To use these controls:

- Enter in the # of iterations and specify the level(s) & lane(s), as well as a delay.
- The delay is necessary to allow the system to complete the manual move before the next move in the sequence is commanded.
- While the manual moves are being executed, the page will display something like this:



### Purpose

- Execute a series of moves
- Testing and validation after mechanical work
- Break-in of equipment at installation
- Advanced troubleshooting scenarios

### Usage

Primarily used by:

- System integrators
- Advanced maintenance personnel
- During commissioning and testing

---

**Navigation:** [← Administration](#) | [Setup InnoPick](#) [→](#)

# Setup InnoPick

[Home](#) > [Main Screens](#) > [Administration](#) > **Setup InnoPick**

Cancel

Submit

Site Settings

Site Levels

Packages

Masking

Motion Controller

Buffer

Simulator Setup

Core setup

Length of individual lane in Millimeters

10260

Lane Pitch In Millimeters

558.8

Gap Between Cases In Millimeters

75

Manual Replenishment Empty Lanes Threshold

4

Maximum Acceptable Products Per Level

30

☐ Use Dynamic Replenishment Quantities

Max Consecutive Output Cases Per Destination

50

Sequence Conveyor Acceleration In Gs

## Setup InnoPick Sub-Sections:

- [Site Settings](#)
- [Site Levels](#)
- [Packages](#)

- [Masking](#)
- [Motion Controller](#)
- [Buffer](#)
- [Simulator Setup](#)
- [Core Setup](#)

The Setup InnoPick page contains configuration settings for InnoPick's internal operation. Access is typically restricted to Admin users and system integrators.

---

## Site Settings

[Back to Setup InnoPick Sub-Sections](#)

Cancel

Submit

Site Settings

Site Levels

Packages

Masking

Motion Controller

Buffer

Simulator Setup

Core setup

Length of individual lane in Millimeters

10260

Lane Pitch In Millimeters

558.8

Gap Between Cases In Millimeters

75

Manual Replenishment Empty Lanes Threshold

4

Maximum Acceptable Products Per Level

30

☐ Use Dynamic Replenishment Quantities

Max Consecutive Output Cases Per Destination

50

Sequence Conveyor Acceleration In Gs

Critical system-wide configuration parameters.

---

## Gap Between Cases in Millimeters

The gap InnoPick maintains between cases stored in buffer lanes.

Purpose:

- Prevents cases from pressing together
  - Allows smooth dispensing
  - Accommodates case size variations
  - **Note:** Do not change this parameter unless directed to system expert / remote support.
- 

## Manual Replenishment Empty Lanes Threshold

Minimum number of empty lanes required before IPM allows manual replenishment creation.

Purpose:

- Ensures sufficient space for manual replenishments
  - Prevents overfilling storage capacity
- 

## Use Dynamic Replenishment Quantities

Enable/disable IPM's ability to dynamically adjust replenishment quantities based on demand patterns.

When enabled:

- IPM analyzes usage patterns
- Adjusts quantities automatically

- Optimizes inventory levels

When disabled:

- Uses static configured quantities
- More predictable but less adaptive

**Note:** Should be managed by system integrators.

---

## Max Consecutive Output Cases Per Destination

Limits consecutive cases sent to a single destination (Merge A, B, etc.).

Purpose:

- Balances output across destinations
- Prevents overwhelming single destination
- Contingent on other function being enabled

**Note:** Should be managed by system integrators.

---

## Sequence Conveyor Acceleration / Deceleration in Gs

Acceleration rate for sequence conveyor indexing moves.

**Measured in:** g's (gravitational force)

Considerations:

- Higher = faster moves but more case disturbance
  - Lower = gentler but slower throughput
  - Balance between speed and stability
-

## Additional Configuration Options

Other settings visible on this page are for:

- Initial system setup
- System integration
- Advanced configuration

**Warning:** Do not change these without consulting system integrators.

---

## Site Levels

[Back to Setup InnoPick Sub-Sections](#)

## Levels Configuration

Level Number: 1

**Initialization String :** 1:(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11 0 0 12 13 14 15 0 0 16 17 18 19 0 0 0 20 21 22 0 0 23 24 25 26 0 0 27 28 29 30 0 0 31 32 33 34 0 0 35 36 37 38 0 0 0 39 40 41 0 0 42 43 44 45 0)

**Motion Controller Address :** 10.7.5.141

**Zone Number :** 1

Lane Number:  
0000000000000000001100121314150016171819000202122002324252600272829300031323334003536373800039404100424344450

Merge A

Merge B

Lane Pitch In Millimeters

558.87

Center Placement in Millimeters

70

Center offset In Millimeters

-5

Length offset In Millimeters

0

Target offset In Millimeters

0

Send variables to Galil

This page contains level-specific configuration settings:

---

### Lane Pitch in Millimeters

- Value (in mm) representing the standard distance between storage lanes

- **Note:** Values may differ slightly per level due to:

- Chain stretch over time
- Tensioning adjustments
- Mechanical variations

- See the [Lane Pitch Adjustment Guide](#) for guidance on how to adjust these parameters to ensure good centering on all lanes

---

#### **Center Placement in millimeters**

- Controls the initial positioning of a case as it first enters InnoPick infeed.
  - See the [Lane Pitch Adjustment Guide](#) for guidance on how to adjust these parameters to ensure good centering on all lanes
- 

#### **Center offset in millimeters**

- Adjusts the expected centerpoint of infeed cases.
  - If there are midpoint alarms being generated despite the cases looking nicely centered, then this parameter needs to be adjusted.
  - Contact system integrators if adjustment is needed.
- 

#### **Length offset in millimeters**

- Adjusts the expected length by this amount
- 

#### **Target offset in millimeters**

- Adjusts the final case placement when dispensing a case
- 

#### **Send variables to Galil**

- Press this after making a change to one of the variables in this section, followed by the 'Submit' button at the top of the page.
-

## Packages

[Back to Setup InnoPick Sub-Sections](#)

Product configuration page for system setup.

**Access:** System integrators during initial setup

---

## Masking

[Back to Setup InnoPick Sub-Sections](#)

Configuration for masking certain system moves.

**Access:** System integrators during initial setup

---

## Motion Controller

[Back to Setup InnoPick Sub-Sections](#)

Connection and configuration for the motion controller.

**Access:** System integrators during initial setup

---

## Buffer

[Back to Setup InnoPick Sub-Sections](#)

#### Number Of Cases To Exit Merge In The Accumulation LookAhead

- ☒ Buffer Enabled
- ☐ Build Inventory
- ☒ Destination Decoupling Mode Enabled
- ☐ Destination interleaving mode enabled

#### Destination decoupling mode max pallet offset

Define how many pallets can be skipped by the destination decoupling mode when the destination of those pallets is paused until it reaches the limit or finds a pallet with unpaused destination

#### Destination decoupling mode max case offset

Define how many cases can be skipped by the destination decoupling mode when the destination of those cases is paused until it reaches the limit or finds a pallet with unpaused destination

Configuration for accumulation management and output buffering.

## Number Of Cases To Exit Merge In The Accumulation LookAhead

How many cases IPM considers when forecasting and scheduling output. This is only applicable if Destination Decoupling is Disabled. This # is meant to represent how many cases can exit the Merge during the time it takes a case to travel the entire length of InnoPick.

Impact:

- **Higher number:** More aggressive scheduling, uses more accumulation
- **Lower number:** More conservative, leaves accumulation space unused
- **Too high:** Risk of misjudging available space, may block output more often
- **Too low:** Underutilizes accumulation, reduces throughput

*Recommendation:* Determine empirically based on system performance

## Buffer Enabled

(Default = On)

When enabled, InnoPick keeps track of the cases in the outbound accumulation to optimize outbound scheduling when space is available and pause outbound scheduling to allow inbound to keep entering when the outbound is full.

**When enabled:**

- IPM actively manages accumulation
- Schedules output based on available space
- Prevents overloading downstream equipment

**When disabled:**

- IPM schedules output independently of accumulation status
  - Simpler but may cause downstream congestion issues
- 

## Build Inventory

InnoPick can operate in two modes regarding accumulation:

**Normal Operation Mode**

- IPM schedules as many output cases as possible, using Buffer logic if that option (see above) is enabled
- Assumes downstream conveyors advance at nominal rate
- Maximizes throughput
- May occasionally have to stop if outbound flow stops unexpectedly

**Build Inventory Mode**

- IPM prioritizes building lane inventory over outputting cases
- Only schedules output cases that will definitely fit in current accumulation space
- More conservative approach
- Level should not have to stop due to full accumulation
- Allows continuous replenishment induction

**When to enable:**

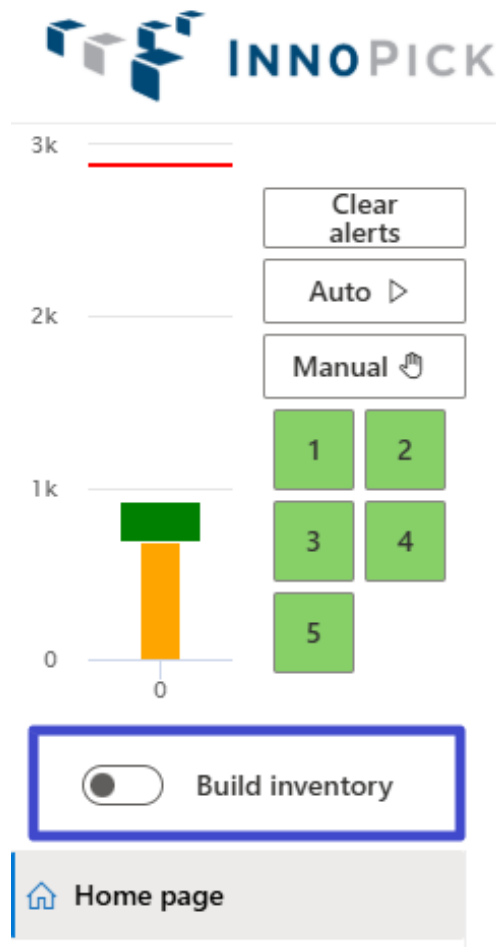
- Starting production with low inventory
- Recovering from downstream stoppage

- Building to nominal inventory levels

**When to disable:**

- Normal production operations
- When inventory is at nominal or above
- To maximize throughput

**Note:** Build Inventory Mode can be turned on and off directly from the set of controls that are always visible in the top left corner of the InnoPick Manager screen:



---

## Destination Decoupling Modes

**Enabled**

- Allows InnoPick to schedule cases for multiple destinations in parallel based on downstream accumulation availability and inventory.
- If InnoPick cannot schedule the next outbound case for the next priority pallet, it is allowed to look for pallets further down the sequence for other destinations and attempt scheduling them if accumulation and inventory allows it.
- This mode is affected by the two parameters below:

#### **Destination decoupling mode max pallet offset**

- This is configuration parameter for the "Destination Decoupling Mode Enabled" above.
- See the description beneath the setting.
- Note: The decoupling level is limited by whichever limit is reached first: # of pallets or # of cases.

#### **Destination decoupling mode max case offset**

- This is configuration parameter for the "Destination Decoupling Mode Enabled" above.
- See the description beneath the setting.
- Note: The decoupling level is limited by whichever limit is reached first: # of pallets or # of cases.

---

## **Destination A / B Accumulation**

Destination A      Destination B

Shared zone accumulation (# of zones)

100

Level 1

Accumulation Length In Mm

0

Zone Accumulation (# of zones)

14

Level 2

Accumulation Length In Mm

0

Zone Accumulation (# of zones)

14

Level 3

Accumulation Length In Mm

0

Zone Accumulation (# of zones)

14

Level 4

Accumulation Length In Mm

0

Zone Accumulation (# of zones)

14

Level 5

Accumulation Length In Mm

0

Zone Accumulation (# of zones)

14

These settings teach InnoPick about various accumulation zones for its algorithmic calculations.

**Shared zone accumulation (# of zones)**

- This setting teaches InnoPick how many cases can fit between the exit of the merge and the palletizer.

**Level 1 to 5: Accumulation Length in Mm**

- This setting teaches InnoPick the length of low pressure accumulation where cases are accumulated in a way that they bunch up together with no gaps.

#### **Level 1 to 5: Zone Accumulation (# of zones)**

- This setting teaches InnoPick how many individual cases can be accumulated between the exit of InnoPick and the merge.
- 

## Simulator Setup

[Back to Setup InnoPick Sub-Sections](#)

Configuration page for developers and system testing.

**Access:** System integrators and developers

---

## Core Setup

[Back to Setup InnoPick Sub-Sections](#)

Core system parameters for developers.

**Access:** System integrators and developers

---

**Navigation:** [← Manual Motion Control](#) | [Algorithm Parameters →](#)

# Algorithm Parameters

[Home](#) > [Main Screens](#) > [Administration](#) > [Algorithm Parameters](#)

Innopick

Auto

Manual

1

2

3

4

5

0

Save

| Name                                                 | Description                                                                         | Default Value                        | Value                                                             |
|------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------|
| InnoPick Settings (20)                               |                                                                                     |                                      |                                                                   |
| CopiedSku_IsActive                                   | Activate the Copied sku algorithm                                                   | <input checked="" type="checkbox"/>  | <input checked="" type="checkbox"/>                               |
| CopiedSku_CaseSequenceSampleSize                     | Set the number of cases the algorithm will run with                                 | 10000                                | <input type="text" value="10000"/>                                |
| CopiedSku_CaseSequenceMinSampleSize                  | Set the minimum case sequence size to run the algorithm                             | 2000                                 | <input type="text" value="2000"/>                                 |
| CopiedSku_SlotQuantity                               | Set the maximum number of product copied on levels                                  | 20                                   | <input type="text" value="20"/>                                   |
| CopiedSku_SlotSizeInPercentage                       | The required percentage of the product in the volume to add one slot to the product | 2.5                                  | <input type="text" value="2.5"/>                                  |
| DynamicSlottingByHeight_IsActive                     | Activate the DynamicSlottingByHeight algorithm                                      | <input checked="" type="checkbox"/>  | <input checked="" type="checkbox"/>                               |
| DynamicSlottingByHeight_CaseSequenceMinSampleSize    | Set the minimum case sequence size to run the algorithm                             | 2000                                 | <input type="text" value="2000"/>                                 |
| DynamicSlottingByHeight_CaseSequenceSampleSize       | Set the number of cases the algorithm will run with.                                | 10000                                | <input type="text" value="10000"/>                                |
| DynamicSlottingByHeight_ProductHeightForGroups       | Set the product group height separate by comma                                      | 123,140,160,171,194,228,239,247,2... | <input type="text" value="123,140,160,171,194,228,239,247,2..."/> |
| DynamicSlottingByHeight_MaximumProductsAssignedToS   | Set the maximum number of products assigned to a level                              | 40                                   | <input type="text" value="40"/>                                   |
| DynamicSlottingByHeight_MaximumSequenceConveyorLa    | Set the maximum number of lanes used on a level                                     | 55                                   | <input type="text" value="55"/>                                   |
| DynamicSlottingByHeight_MaxToleranceForConsideringCa | To be determined                                                                    | 0.05                                 | <input type="text" value="0.05"/>                                 |
| DynamicSlottingByHeight_MinimumDeltaBetweenProduct   | Set the minimum products delta between two levels to allow product permutation      | 3                                    | <input type="text" value="3"/>                                    |
| DynamicReplenishmentQuantity_IsActive                | Activate the DynamicReplenishmentQuantity algorithm                                 | <input checked="" type="checkbox"/>  | <input checked="" type="checkbox"/>                               |
| DynamicReplenishmentQuantity_CaseSequenceMinSample   | Set the minimum case sequence size to run the algorithm                             | 2000                                 | <input type="text" value="2000"/>                                 |
| DynamicReplenishmentQuantity_CaseSequenceSampleSiz   | Set the number of cases the algorithm will run with                                 | 10000                                | <input type="text" value="10000"/>                                |

Home page

Alerts

Inventory

Case sequence

Administration

Manage site

Setup Innopick

Algo Parameters

User Manager

Recovery actions

User settings

The Algo Parameters page lists parameters that control various elements of InnoPick's embedded algorithms.

## Purpose

- Fine-tune system performance
- Adjust algorithm behavior
- Optimize for specific products or operations

## Access Level

Parameters are meant to be altered by:

- System architects
  - System administrators
  - Personnel with deep understanding of InnoPick operation
- 

## Documentation

Each parameter includes a description on the page explaining its purpose and effect.

**Note:** Changes to algorithm parameters can significantly affect system behavior.

---

**Navigation:** [← Setup InnoPick](#) | [User Manager](#) [→](#)

# User Manager

[Home](#) > [Main Screens](#) > [Administration](#) > **User Manager**

|  Create new user |                     |  Search |         | <input type="button" value="Show inactive users"/>                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                   | Name                | Access name                                                                              | Role    | Actions                                                                                                                                                                                                                                                                                                                                            |
| PC                                                                                                | Phil Collins        | 4654564                                                                                  | Admin   |             |
| TC                                                                                                | Tim Cross           | tim                                                                                      | Manager |             |
| BD                                                                                                | Bernad Desnoy       | 428962                                                                                   | Admin   |             |
| SP                                                                                                | Samuel Porter       | 0949227                                                                                  | Admin   |         |
| FM                                                                                                | Figen Mouna         | a428067                                                                                  | Admin   |     |
| EA                                                                                                | Eric Asenault       | 157229                                                                                   | Admin   |     |
| FL                                                                                                | Felix Oliver Landry | 8008                                                                                     | Admin   |     |
| MP                                                                                                | Michael Peladeau    | 7784616                                                                                  | Admin   |     |
| DT                                                                                                | David Thout         | 200635                                                                                   | Clerk   |     |
| EJ                                                                                                | Emmanuel Jean-Pier  | 2746266                                                                                  | Admin   |     |

The User Manager page allows Admin-level users to create and manage user accounts for InnoPick Manager.

## User Roles

Each user account is assigned one of these roles:

## Admin

- Full access to all controls and options
- Can create and manage other user accounts
- Can modify system configuration
- Can access all pages and features

## Manager

- Access to all operational settings
- Cannot modify site settings or core configuration
- Can manage production operations
- Can resolve alerts and manage inventory

## Clerk

- Access necessary to operate system for production
- Cannot change configuration settings
- Can monitor status and clear alerts
- Limited administrative access

## Terminal User

- Access only to MixMaster Terminals
- There are no Terminals within InnoPick Manager, but since User Accounts are shared between the systems, the option appears here.

---

## Creating a New User

---

 Create new user

First name \*

John

Last name \*

Doe

Phone number

( ) - -

Second phone number

( ) - -

Access name \*

1234567

Access card number \*

1234567

Employee id \*

1234567

Role

Admin

Password \*

\*\*\*\*\*

Description

Add some extra description

Go back

Save

To create a new user account:

1. Click "Create New User" or similar button
2. Enter required information:
  - **Username:** Login identifier
  - **Password:** Initial password
  - **Full Name:** User's actual name
  - **Email:** Contact email address
  - **Role:** Select from Admin, Manager, Clerk, Terminal User
3. Save the new user

---

## Managing Existing Users

From the User Manager page, you can:

- View all user accounts
- Edit user information

- Change user roles
  - Disable or delete accounts
  - Reset passwords
- 

## Security Best Practices

- Assign the minimum necessary role for each user
  - Review user accounts periodically
  - Disable accounts for personnel no longer needing access
  - Use strong passwords
  - Change default passwords immediately
- 

**Navigation:** [← Algorithm Parameters](#) | [Recovery Actions →](#)

# Recovery Actions

[Home](#) > [Main Screens](#) > [Administration](#) > [Recovery Actions](#)

Innopick

Auto

Manual

1

2

3

4

5

Home page

Alerts

Inventory

Case sequence

Administration

Manage site

Setup Innopick

Algo Parameters

User Manager

Recovery actions

User settings

| Name                                   | Description                                                                                                                                                                                      | Action                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| General (8)                            |                                                                                                                                                                                                  |                                                                             |
| Reload Cache                           | Force Innopick to reload data from the database to rebuild its cache if ever the data was modified within the database while the system was live                                                 | <div>Execute</div>                                                          |
| Cancel in progress case                | Cancel a case that is currently 'In Progress'                                                                                                                                                    | <div>Case reference</div> <div>Execute</div>                                |
| Force in progress case to be completed | Force a case that is currently 'In Progress' to be completed                                                                                                                                     | <div>Case reference</div> <div>Execute</div>                                |
| Set inducing replenishment to in prog  | Set a replenishment that is currently 'Inducting' to 'In Progress'                                                                                                                               | <div>Replenishment id</div> <div>Execute</div>                              |
| Cancel in progress case on level       | Cancel a case that is currently 'In Progress' on specified level                                                                                                                                 | <div>Level number</div> <div>Execute</div>                                  |
| Trim database aggressively             | Trim anything operational in the database (alerts, orders and replenishments) that is completed and more than X hours old                                                                        | <div>Hours</div> <div>Execute</div>                                         |
| Migrate data from legacy server        | Migrate data from innopick legacy database to the new database                                                                                                                                   | <div>Server url</div> <div>Database name</div> <div>Execute</div>           |
| Migrate data to legacy server          | Migrate data innopick 2.0 to legacy database                                                                                                                                                     | <div>Server url</div> <div>Database name</div> <div>Execute</div>           |
| Buffer (2)                             |                                                                                                                                                                                                  |                                                                             |
| Get buffer accumulation                | Fetch the buffer spiral accumulation from the Database                                                                                                                                           | <div>Execute</div>                                                          |
| Clear buffer accumulation              | Force clear the buffer spiral accumulation                                                                                                                                                       | <div>Level number</div> <div>Clear completed cases</div> <div>Execute</div> |
| Motion Controller (4)                  |                                                                                                                                                                                                  |                                                                             |
| Set InCastLen                          | Send the currently inducting product package length to the given motion controllers separated by ";" e.g. "2;4;5"                                                                                | <div>E.g. "2;4;5"</div> <div>Execute</div>                                  |
| Set OutCEna=1 for level                | Simulate the 'Ok to transfer' signal from Bastian to Innopick at the outfeed to compensate for a loss of handshake that sometimes happens after an Innopick restart or due to Bastian-side fault | <div>Level number</div> <div>Execute</div>                                  |
| Send raw command to level              | Send a command as raw text to a level's motion controller                                                                                                                                        | <div>Level number</div> <div>Command</div> <div>Execute</div>               |
| Clear alarm on level                   | Force Send CxAlarm=1 on specified levels separated by ";" e.g. "2;4;5"                                                                                                                           | <div>E.g. "2;4;5"</div> <div>Execute</div>                                  |
| Core (14)                              |                                                                                                                                                                                                  |                                                                             |

Recovery Actions are built-in features for dealing with specific issues that arise during operation.

## Purpose

Provide operators and support personnel with:

- Tools for recovering from unusual situations
- Options for resolving issues that can't be handled through normal procedures
- Advanced troubleshooting capabilities

# Recovery Action Groups

## General

- System-wide recovery actions
- General troubleshooting tools

## Buffer

- Actions related to accumulation management
- Output buffering issues

## Motion Controller

- Actions related to conveyor motion
- Drive and motor issues

## Core

- Core system recovery actions
  - Advanced system state management
- 

# Using Recovery Actions

Recovery actions include a description of what it does and some of them require the operator to put in some required information (in the Action column).

- If a recovery action has an input field for the operator to put in information, it typically will not work unless it has that information.

### **Important:**

- Understand the action before executing
- Document which recovery actions were used

- Consult with support if unsure

**Note:** The list of available Recovery Actions may change depending on:

- System version
  - Integrator customizations
  - Specific installation needs
- 

## Most Commonly Used Recovery Actions

### Set case pending completion

This recovery action resends the outfeed case data to the downstream conveyor system.

- This is used when recovering from issues at the outbound transition (handshake), when the downstream conveyor system does not have the tracking data for the case currently on the last InnoPick position.

### Set InCasLen

This recovery action re-sends the case length to the motion controller for the currently inducting product on a given level.

- This is used when after a system restart or some other abnormal stoppage, cases are not entering InnoPick infeed even though everything else is good, and the replenishment is in the *Inducting* state.

### Set OuXfEna=1 for level

This recovery action simulates the "OK to Receive" signal from the downstream conveyor system to InnoPick, which is the last step in the outfeed handshake process.

- This can be used when the downstream conveyor system has tracking data for the case, but InnoPick is not advancing the physical case from the last position.

### Clear alarm on level

This recovery action sends a 'clear alarm' signal to the motion controller for a given level.

- This is used when a fault or alert on the InnoPick system refuses to clear normally, but the underlying conditions that caused the alarm have been resolved (meaning it should be able to clear).

## Clear level processor error

This recovery action clears internal processor errors and restarts the level processor for a given level.

- This is used after certain critical faults whose descriptions indicate to the operator that the level processor has been stopped.

---

**Navigation:** [← User Manager](#) | [Operations →](#)

# Operations

Home > Main Screens > Administration > Operations

## Level status

| Level status                   |                          |                          |                          |                          |                     |                         |
|--------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|---------------------|-------------------------|
| Merges state                   |                          |                          |                          |                          |                     |                         |
| <div><div></div> Refresh</div> |                          |                          |                          |                          |                     |                         |
| Levels                         | PreInfeed ready          | Infeed ready             | Outfeed ready            | Waiting for outfeed      | Last case data sent | Last pending completion |
| 1                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 3453488             | 3453488                 |
| 2                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 3453342             | 3453342                 |
| 3                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 3453064             | 3453064                 |
| 4                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 3453433             | 3453433                 |
| 5                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | 3453489             | 3453489                 |

- This page provides an overview of each level's various statuses as well as:
- Last case data sent (CaseRef #)
- Last pending completion (CaseRef #)

## Merges state

Level status

Merges state

☒ Merge A Enabled

☒ Merge B Enabled

Destination assignment mode

Just in time

✕ Cancel

 Save

Configuration for downstream merge points (where the different levels combine to a single conveyor that feeds one of the palletizers).

---

## Merge A / B Enabled

Enable or disable individual downstream destinations.

### Purpose:

- Route all cases to operational Merge if one fails
- Handle downstream equipment failures
- Temporary operational adjustments

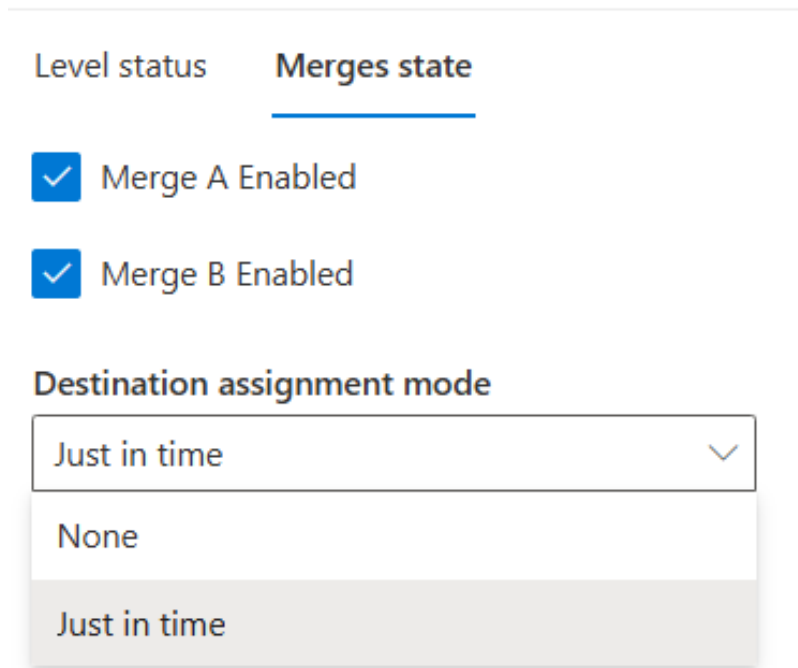
### Usage:

- Last resort when downstream fault cannot be resolved
- Will not affect cases already on sequence conveyors -- those cases that are already scheduled for a certain destination will retain their destination.
- High likelihood that remote support interventions will be required if this is changed during production.

**Caution:** Only use when absolutely necessary and coordinate with support. Can have unintended effects and complications.

---

## Destination assignment mode



Level status      **Merges state**

☒ Merge A Enabled

☒ Merge B Enabled

**Destination assignment mode**

Just in time

None

Just in time

There are two possible settings for the destination assignment mode:

1. None
2. Just-in-time

**None:** This is the regular destination mode:

- InnoPick will initially load balance pallet assignment when receiving a new case sequence
- Will never change a destination unless a destination is disabled

**Just in time:**

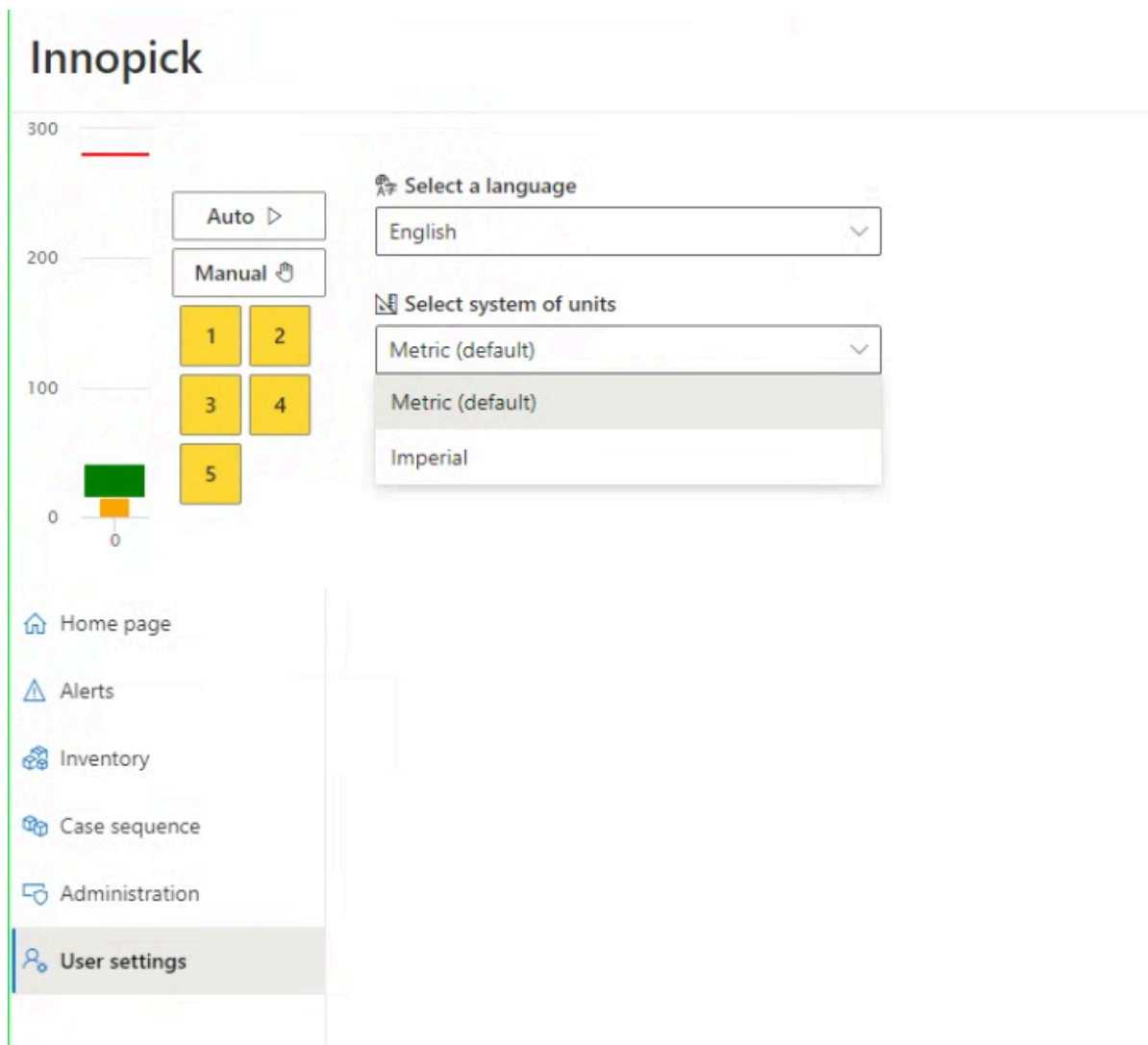
- InnoPick will initially load balance pallet assignment when receiving a new case sequence (like normal)

- InnoPick may change the destination when the first case of a pallet gets scheduled, based on the remaining accumulation available to load balance assignments "Just-in-Time"
  - In this mode, InnoPick responds dynamically to current accumulation levels and assigns pallets to the destination with the most capacity.
- 

**Navigation:** [← Recovery Actions](#) | [User Settings](#) [→](#)

# User Settings

[Home](#) > [Main Screens](#) > [Administration](#) > **User Settings**



The User Settings page contains personal preferences that individual users can configure.

## Available Settings

Settings include:

- Language preferences (if applicable)
  - System of units (metric vs imperial)
- 

## Accessing User Settings

- Click **User Settings** in the left navigation menu
  - Settings are specific to your user account
- 

## Related Topics

- [Monitoring Production](#) - Using admin tools during operation
  - [Troubleshooting](#) - When to use recovery actions
  - [Products Page](#) - Product configuration
- 

**Navigation:** [← Operations](#) | [Troubleshooting →](#)

# Troubleshooting

[Home](#) > Troubleshooting

---

## Overview

This section provides guidance for resolving alerts, troubleshooting common issues, and recovering from problems during InnoPick operation.

---

## Troubleshooting Resources

### [Guidelines for Resolving Alerts](#)

General principles and best practices for resolving alerts successfully, including:

- Understanding level status colors
  - General guidelines for operating InnoPick
  - When alerts indicate bigger issues
  - Important system settings that affect behavior
- 

### [Alerts Reference](#)

Complete listing of all InnoPick alerts with:

- Alert code and description organized by ranges
- Explanation and common causes

- Step-by-step resolution procedures
  - Quick navigation by alert code ranges
- 

## Common Issues

Frequently encountered problems and their solutions:

- Outfeed faults
  - Infeed issues
  - Sequencing problems
  - Performance issues
- 

## Case Replacement

Procedures for replacing broken, missing, or incorrect cases during operation.

---

## Case Centering Adjustments

Procedures for adjusting the centering of cases on a given level.

---

## Deadlocks

Explanations of and guidance for resolving deadlock situations.

---

## Disabling an Active Lane

Guidance on how to handle disabling a lane that has incoming replenishment cases or already-scheduled output cases.

---

## Quick Reference by Symptom

### **System won't start:**

- Check for [E-Stop condition](#)
- Verify no [unresolved or hidden alerts](#)

### **Alert won't clear:**

- See [Alert Guidelines](#)
- Verify physical issue is resolved
- Use [Recovery Action](#) to send a clear alarm to the motion controller.

### **Case missing from sequence:**

- See [Case Replacement](#)
- Check [Case Sequence Page](#)

### **Repeated same alert:**

- See [Alert Guidelines](#)
- Check for underlying issues

### **Photocell fault:**

- See [Alert Reference](#) codes 109, 119, 121, 123, 709

### **Clutch fault:**

- See [Alert Reference](#) codes 310, 333, 334, 340

### **Infeed problem:**

- See [Alert Reference](#) codes 512, 515, 517
-

# Troubleshooting Approach

## Step 1: Identify the Problem

1. **Check the [Alerts Page](#)** for current alerts
2. **Note the alert code and message**
3. **Identify which level** is affected
4. **Determine if this is** a new issue or recurring problem

## Step 2: Understand the Situation

1. **Read the alert description** in [Alert Reference](#)
2. **Check the [Home Page Inventory Graph](#)** - does it match physical reality?
3. **Consider recent events** - what was happening when the alert occurred?
4. **Look for patterns** - have similar alerts occurred recently?

## Step 3: Resolve the Issue

1. **Follow the resolution steps** in the Alert Reference
2. **Verify physical conditions** match expected state
3. **Make corrections** as needed
4. **Clear the alert** only when underlying issue is resolved

## Step 4: Verify Resolution

1. **Resume operation** (put level back in Auto)
2. **Monitor closely** for recurrence
3. **Report persistent problems** to maintenance

---

## Safety Reminders

- **Always use E-Stop** if there's immediate danger to personnel or equipment
  - **Never bypass safety interlocks** or disable safety devices
  - **Lock out equipment** before entering guarded areas
  - **Follow lockout/tagout procedures** for maintenance work
  - **Report safety concerns** immediately to supervision
- 

## Prevention Tips

### Reduce Alerts Through Good Practices

- Perform visual inspections during [startup](#)
- Monitor production actively as described in [Monitoring](#)
- Address minor warnings before they become critical
- Keep inventory graphs matched to physical reality
- Maintain clear communication during shifts
- Follow proper [shutdown procedures](#)

### Maintain System Health

- Report unusual sounds or movements immediately
  - Note patterns in alerts and report to maintenance
  - Keep documentation updated
  - Follow scheduled maintenance procedures
  - Address root causes, not just symptoms
- 

## Related Topics

- [Daily Operations](#) - Preventing issues through proper procedures

- [Alerts Page](#) - Viewing and filtering alerts
- [Home Page](#) - Monitoring system status
- [Administration](#) - Recovery actions for advanced issues

---

**Navigation:** [← User Settings](#) | [Alert Guidelines →](#)

# Common Issues

[Home](#) > [Troubleshooting](#) > **Common Issues**

---

## Overview

This page addresses frequently encountered issues that may not have specific alert codes but require operator attention and intervention.

---

## Outfeed Issues

---

### Cases Not Exiting InnoPick

#### **Symptoms:**

- Cases stuck at outfeed position
- Output has stopped
- Accumulation showing as full
- Levels remain in Auto but no output

#### **Common Causes:**

- Downstream conveyor system stopped or faulted
- Accumulation space completely full
- HOLD active on outbound conveyor system
- Communication error with downstream equipment

- Photocell at exit blocked

**Solution:**

**1. Check downstream equipment status:**

- Check for faults on outbound conveyor HMI
- Look for HOLDS or stopped conditions

**2. Check Inventory Graph Outfeed statuses:**

- On the [Home Page](#), look at the output statuses

**3. Coordinate with downstream operators:**

- Communicate the issue
- Verify when downstream will be ready
- May need to wait for accumulation to clear

**4. Check InnoPick settings:**

- Check if [Pause Outfeed](#) is active
- Verify [Buffer settings](#)

**5. If issue persists:**

- If no explanation can be found for the stoppage, contact support
- 

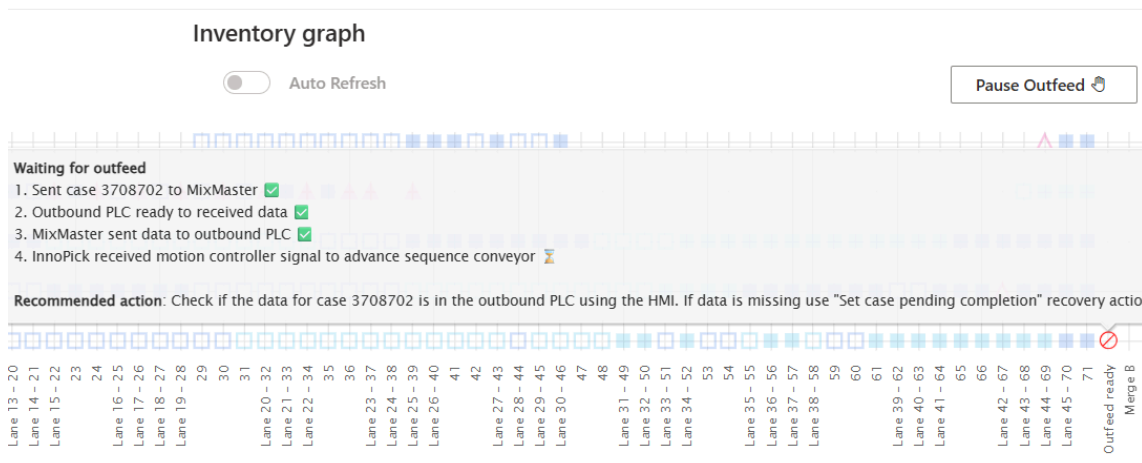
## Next Outfeed Case Data is not in Outbound Conveyor System

**Symptoms:**

- The case at the outfeed position of InnoPick is waiting for the "Ok to Receive" signal from the downstream conveyor system
- It is determined that the case's data has not been received by that system

**Solution:**

- Check the details of the transfer sequence by hovering over the case on the Inventory Graph:



The steps for each case transfer are as follows:

1. Send case to MixMaster
  - This is done automatically when the case arrives at the last InnoPick position
2. Outbound PLC ready to receive data
  - InnoPick checks to see if the outbound PLC is ready for new case data
3. MixMaster sends data to outbound PLC
  - MixMaster sends case data to the outbound PLC
4. Outbound PLC sends the "Ok to Receive" signal to InnoPick
5. InnoPick advances the case and transfers it to the outbound conveyor system.

#### Additional Info:

- The pop-up which appears when the cursor hovers over the case on the outfeed position shows what steps have been completed to date for this case.
- If step 3 has a checkmark next to it (as in the screenshot above), then MixMaster has already sent the case data to the outbound PLC, and it will not re-send it unless a recovery action is used.
- Therefore, if the outbound PLC does not have the case data, the operator can use the recovery action **"Set case pending completion"**

**Note:** This recovery action re-sends the case data to the outbound PLC. If the data was already there, the downstream system will have a duplicate data which can cause serious tracking issues. Use with caution and monitor downstream system for any issues after using.

## Next Outfeed Case not advancing despite all conditions met

### Symptoms:

- If the outbound PLC does have case data for the next outfeed case
- And the Inventory Graph shows that all steps have been completed for the case transfer
- And all other likely causes have been ruled out (system is not in manual, not faulted, downstream conveyor is not full, faulted, or held, etc)
- But the case still does not advance

### Solution:

- It may be necessary to force the case forward using the recovery action called "[Set OuXfEna=1 for level](#)"

**Note: This recovery action will force the sequence conveyor to move forward. If it is used incorrectly or in the wrong scenario, it will make the situation worse. Use with caution.**

---

## Infeed Issues

### Cases Not Being Inducted

#### Symptoms:

- Replenishments show "Inducting" but cases not entering InnoPick infeed
- Upstream conveyor has cases but won't release them

#### Common Causes:

- HOLD active on inbound conveyor system
- No Replenishment in the **Inducting** status
- InnoPick at capacity (all lanes full) on that level
- Communication / handshake error between systems
- **Pause Infeed** enabled in InnoPick
- If following a system reboot, a recovery action is needed

## **Solution:**

### **1. Check upstream conveyor system:**

- Verify no HOLD is active
- Check HMI for faults or stopped conditions
- Verify that cases are not physically jammed
- Confirm cases have correct tracking data and are ready to release

### **2. Check InnoPick capacity:**

- Review [Lane Inventory](#)
- Verify lanes available for induction

### **3. Check InnoPick settings:**

- Verify Pause Infeed is not active on the Home page

### **4. Check Replenishments Status on that level**

- Go to the Replenishments Page
- Filter for the level in question
- Make sure the replenishment associated with the cases at the infeed is in the state **Inducting**
- If not, investigate why, and then manually set the replenishment to **Inducting**
- [See here for more info on the Replenishments Page](#)

### **5. If a system or motion controller reboot occurred recently:**

- Use the recovery action [Set InCasLen](#) for the level in question.
- 

## **Wrong Product Being Inducted**

### **Symptoms:**

- [Alert 517](#) (incorrect case length)
- Cases don't match expected replenishment
- Lane contains mixed products

### **Common Causes:**

- Operator error in selecting pallet
- Tracking data mismatch in inbound conveyor system
- Upstream case routing mixup
- Cases were removed without updating upstream conveyor system or the InnoPick replenishment

**Solution:**

**1. Stop induction immediately:**

- Put level in Manual mode
- Prevent more wrong cases from entering

**2. Verify what was expected:**

- Check [Replenishments Page](#)
- Note expected product SKU
- Check inducted quantity
- Check recently filled lanes for incorrect inventory

**3. Coordinate with upstream operators:**

- Inform depalletizer operator or appropriate personnel

**4. Resolve the issue:**

- Remove wrong cases from InnoPick
- Adjust replenishment quantities as needed
- Adjust buffer lane inventory as needed
- Correct upstream tracking if needed

**5. Resume InnoPick production**

- Pay close attention to the first transitions between replenishments

---

## Inventory Tracking Not Matching Physical Reality

**Symptoms:**

- [Alert 121](#) or [Alert 123](#) (cases not found where expected)

- Outfeed alerts (expecting case but none present, or vice versa)
- Lane inventory counts don't match physical count

**Common Causes:**

- Improper alert recovery
- Skipped verification after clearing alerts
- Cases manually moved without updating system
- Motion controller thread crash ([Alert 791](#))

**Solution:**

**1. Put level in Manual mode**

**1. Conduct physical inventory check:**

- Compare each affected lane to [Lane Inventory Page](#)
- Compare sequence conveyor to [Inventory Graph](#)
- Fix all discrepancies

**2. Correct lane inventory:**

- Use [Edit Lane Inventory](#) feature
- Adjust quantities to match physical reality
- Update product types if wrong

**3. Correct sequence conveyor tracking:**

- May need to jog conveyor to align cases
- Remove or add cases as needed to match Inventory Graph
- Follow [Case Replacement](#) procedures

**4. Clear all alerts** after corrections

**5. Resume in Auto and monitor closely:**

- Watch first several cycles carefully
- Verify tracking stays accurate

**6. Prevent recurrence:**

- Always verify Inventory Graph after clearing alerts
- Never assume - always check
- Train operators on proper procedures

---

## Frequent Alerts on Specific Products

### Symptoms:

- Certain SKUs causing repeated alerts
- Same product repeatedly has storage/dispense issues
- Product-specific fault patterns

### Common Causes:

- Case dimensions incorrectly configured
- Case quality issues (weak, damaged, irregular)
- Acceleration settings too aggressive for product

### Solution:

#### 1. Review product configuration:

- Check acceleration settings in the [Products Page](#)
- Verify dimensions are correct in MixMaster product configurations (length, width, height)

#### 2. Inspect physical cases:

- Look for damage, weak boxes, irregular shapes

#### 3. Adjust settings:

- Lower acceleration if cases are unstable
- This is modified in the [Products Page](#)

#### 4. Test and monitor:

- Monitor results over several cycles

#### 5. Consider removing product from InnoPick

- If the product is very problematic such that it is slowing down the entire production and the problems cannot be resolved by parameter changes, it may be better to remove the product from automation.
-

## Related Topics

- [Alert Reference](#) - Specific alert code resolutions
  - [Alert Guidelines](#) - General troubleshooting principles
  - [Monitoring Production](#) - Catching issues early
  - [Case Replacement](#) - Replacing problem cases
- 

**Navigation:** [← Alert Reference](#) | [Deadlocks](#) [→](#)

# Alert Guidelines

[Home](#) > [Troubleshooting](#) > **Alert Guidelines**

---

## Overview

This page provides general principles and best practices for successfully resolving InnoPick alerts. Following these guidelines will help you avoid cascading faults and minimize downtime.

---

## Understanding Level Status Colors

When alerts occur, InnoPick changes the level status indicators to show the severity and type of issue.

---

## Level Color Meanings



- **Green:** Automatic mode - system is operating normally
  - **Yellow:** Manual mode - system is paused
  - **Red:** Fault mode - system has an active alert that stopped operation
  - **Dark Red:** E-Stop active - safety system engaged (door open or E-Stop pressed)
  - **Blue:** Communication Loss
- 

## Alert Progression

### When an alarm occurs:

1. InnoPick stops the affected level
2. Level enters Fault mode (Red) or E-Stop status (Dark Red)
3. Alert appears on [Home Page](#) and [Alerts Page](#)

### After clearing the alarm:

1. Level goes to Manual mode (Yellow)
2. Operator can verify conditions are correct
3. Level can be put back into Automatic mode (Green)

---

## Six General Guidelines for Operating InnoPick Successfully

Following these guidelines will prevent most cascading faults and reduce troubleshooting time.

---

### Guideline 1: Always Compare Physical Reality to the Inventory Graph

**When responding to an Alert, always verify physical cases match their position on the Inventory Graph.**

The [Inventory Graph](#) is the gold standard for what should be where. Physical cases should ALWAYS match the graph.

#### **Why This Matters**

- A good operator always double-checks visuals after clearing faults
  - This is the absolute best way to avoid knock-on faults from incorrect recoveries
  - Most cascading faults occur because this step was skipped
- 

### Guideline 2: After Store/Dispense Alerts, Check Physical Case Position

**Always verify where the case actually ended up.**

#### **After a Store Fault:**

- Case should be in the storage lane
- Verify it's not blocking the photocell
- Confirm it's properly positioned

#### **After a Dispense Fault:**

- Case should be on the sequence conveyor

- Verify it cleared the photocell
- Confirm proper spacing to adjacent cases

**If in doubt, check the lane case count (inventory).**

This is a general guideline, but remember Guideline #1 - the Inventory Graph is your ultimate reference.

---

### Guideline 3: Check Replenishment Status After Any Infeed Alarm

**Infeed alarms require extra vigilance about several details.**

Infeed alarms are confusing because they occur where two different systems connect. Problems can originate on either side, and if not properly identified and corrected, more issues will follow.

**After Every Infeed Alarm, Check:**

**1. Product (SKU) is correct**

- Case matches what was expected
- Replenishment is for the right product

**2. Number of cases inducted (or yet-to-be inducted) is correct**

- Go to [Replenishments Page](#)
- Verify inducted quantity
- Check that 'quantity to induct' is accurate

**3. Upstream conveyor system has correct tracking data**

- Cases are properly tracked upstream
- No missing or extra cases in upstream system

**4. If cases had to be moved, follow Guideline 1**

- Resume in Automatic mode for a few moves
  - Verify case placement on sequence conveyor
  - Ensure inventory graph matches physical reality
-

## Guideline 4: Determine if Alert is Stand-Alone or a Symptom

**Consider whether the alert is an isolated incident or indicates a bigger issue.**

### Symptom Alerts

Certain alerts are **rarely stand-alone problems** and usually indicate previously mishandled faults or bigger issues. Correctly identifying this makes **the difference between:**

- One 4-minute stoppage (properly handled)
- Five stoppages costing 25 minutes (symptom not recognized)

### Common Symptom Alerts:

These typically indicate inventory graph doesn't match physical reality:

- **Fault Code 121:** *Photocell above lane never detected case during storing move*
- **Fault Code 123:** *Photocell never detected case during dispense move*
- ***Expecting a case but there is no case present at the outfeed***
- ***Not expecting a case but there is a case at the outfeed***

These faults typically show up when someone has not followed Guideline #1.

### When You See Symptom Alerts:

1. Stop and assess the situation
  2. Review recent alert history
  3. Check if earlier fault was improperly resolved
  4. Verify that the inventory graph matches physical reality
  5. Correct any discrepancies before clearing alert and resuming production
- 

## Guideline 5: Be Aware of InnoPick Settings

**Always understand current system settings and their implications.**

A good operator is aware of settings that affect InnoPick's behavior. Failing to understand current settings leads to serious operator mistakes.

## **Key Settings to Monitor:**

### **Pause Infeed / Pause Outfeed**

- Controls whether cases enter or exit InnoPick
- Check: [Home Page](#)

### **Build Inventory Mode**

- When ON: Prioritizes storing replenishments over outputting cases
- When OFF: Balances input and output for maximum throughput
- Check: [Build Inventory](#)

### **Destination (Merge) A or B Enabled / Disabled**

- Controls which downstream destinations are active
  - If a Merge is disabled, all cases route to other destination(s)
  - Check: [Administration > Setup InnoPick > Merges](#)
- 

## **Guideline 6: Be Aware of Other System Settings**

**InnoPick is connected to other systems that directly affect its behavior.**

### **Systems That Affect InnoPick:**

#### **MixMaster (or equivalent WMS/WCS)**

##### **Impact Examples:**

- **HOLD on a pallet:** InnoPick won't schedule output cases for it or subsequent pallets
- **Order changes:** May affect case sequence and replenishment needs

#### **Inbound Conveyor System (Pre-InnoPick)**

##### **Impact Examples:**

- **HOLD enabled:** Cases won't be released to InnoPick
- **Tracking errors:** Wrong cases or quantities arrive

- **Photocell issues:** Cases arrive unexpectedly

## Outbound Conveyor System (Post-InnoPick)

### Impact Examples:

- **HOLD or fault active:** InnoPick cannot release cases at outfeed
- **Accumulation full:** Output stops even if InnoPick is ready
- **Downstream equipment down:** Cases back up into InnoPick

### When Troubleshooting:

1. **Check downstream first** if output issues occur
  2. **Check upstream first** if infeed issues occur
  3. **Coordinate with other system operators** to understand full picture
  4. **Verify communication** between systems is working
- 

## Clearing Alerts

### When to Clear Alerts

Clear alerts only when:

- ✓ Underlying physical issue is resolved
  - ✓ Cases are in correct positions
  - ✓ Inventory graph matches reality
  - ✓ Sensors are unblocked
  - ✓ Mechanical issues are fixed
  - ✓ Safety devices are reset
- 

### When Alerts Won't Clear

If alerts cannot be cleared, it usually means the underlying issue has not been resolved.

**Common reasons alerts won't clear:**

- Photocell sensor still blocked
- E-Stop button still pressed
- Door still open
- Case still in wrong position
- Mechanical issue not resolved
- The Alert in question is being filtered (not visible).
- [See how to View All Alerts](#)

**What to do:**

1. Read the alert message carefully
  2. Inspect physical conditions
  3. Resolve the actual problem
  4. Try clearing again
    - Use the [Recovery Action: Send Clear Alarm on Level](#)
  5. See [Alert Reference](#) for specific alert resolution steps
- 

## Recognizing Symptoms of Bigger Issues

### Single Occurrence vs. Pattern

**If an alert occurs once:**

- Likely an isolated incident
- Resolve and monitor
- Document but may not indicate systemic issue

**If same alert repeats:**

- Indicates underlying problem not fully resolved
- Stop and reassess

- Don't just keep clearing and restarting
  - Look for root cause
- 

## Multiple Related Alerts

**If different but related alerts occur in sequence:**

- May be cascading effects from one root cause
  - Example: Infeed issue → Storage fault → Dispense fault
  - Address the earliest/root issue first
  - Verify inventory graph accuracy
  - May need to stop and reset entire level
- 

## Best Practices Summary

1. **Inventory Graph is Truth** - Always verify physical matches visual
  2. **Verify Case Position** - After store/dispense, check where case ended up
  3. **Check Replenishments** - After any infeed alarm, verify quantities and tracking
  4. **Look for Root Causes** - Some alerts are symptoms, not standalone issues
  5. **Know Your Settings** - Understand how system is configured
  6. **Check Connected Systems** - Upstream and downstream affect InnoPick
  7. **When in Doubt, Ask** - Better to escalate than cause more problems
- 

## Related Topics

- [Alert Reference](#) - Specific resolution procedures for each alert code
- [Home Page Inventory Graph](#) - Understanding the visual display

- [Monitoring Production](#) - Catching issues early

---

**Navigation:** [← Troubleshooting](#) | [Alert Reference](#) [→](#)

# Alert Reference Page

[Home](#) > [Troubleshooting](#) > **Alert Reference**

---

## Overview

This page provides detailed information about every InnoPick alert code, including explanations, common causes, and step-by-step resolution procedures.

---

## How to Use This Reference

1. **Find your alert code** in the list below
  2. **Read the explanation** to understand what happened
  3. **Review the possible causes**
  4. **Follow the solution steps** to resolve the issue
  5. **Verify resolution** before clearing the alert
- 

## Alert Code Format

Each alert entry follows this format:

- **Alert Code - Description**
  - **Explanation & Causes**
  - **Solution**
-

## 98 - ServoDrive Faulted while Moving (Sys\_Flt)

### **Explanation:**

The servo drive faulted during a motion operation.

### **Common Causes:**

- Drive overload or overheating
- Electrical fault in drive
- Motor connection issue
- Drive parameter error
- Excessive load on axis

### **Solution:**

1. Check drive status indicators/LEDs
  2. Verify motor connections
  3. Check for mechanical binding or excessive load
  4. Review drive fault code in servo drive diagnostics
  5. Contact maintenance if drive hardware fault suspected
  6. Clear alert after resolving issue
- 

## 99 - ServoDrive Time-out Fault occurred during Start-up (Sys\_Flt)

### **Explanation:**

The servo drive did not respond in time during system start-up.

### **Common Causes:**

- Connection issue in plug or cabling
- Communication timeout
- Rarely occurs when clearing and resuming from faults

### **Solution:**

1. Attempt to clear the fault
  2. If it clears successfully, safe to put level back in automatic
  3. If fault persists or repeats, contact maintenance
  4. Notify supervisors for maintenance follow-up even if it clears
- 

## 109 - Photocell above lane was still blocked after the end of the storing move

### **Explanation:**

After storing a case into a buffer lane, the photocell remained blocked when it should be clear.

### **Common Causes:**

- Turned or skewed case
- Damaged case (torn cardboard, collapsed)
- Wrong product (different case dimensions)
- Case dimensions incorrectly referenced in system
- Case too far forward in lane

### **Solution:**

1. Visually inspect the lane and case
2. Identify why photocell is still blocked
3. Correct the issue:
  - Adjust case position if needed
  - Replace damaged case
  - Retrieve correct product if wrong case
  - Remove case entirely if necessary
4. Ensure photocell is clear
5. Verify other cases in lane are properly positioned
6. Clear the alert
7. Resume operation

---

## 113 / 117 - Buffer lane tried to dispense a case but the case was too far back

### **Explanation:**

InnoPick tried to dispense a case from buffer lane, but no case blocked the photocell during the dispense move, or case was detected later than expected.

### **Common Causes:**

- No case present in lane (inventory count wrong)
- Case too far back from front of lane
- Case fell or was removed
- Gaps between cases too large
- Lane position tracking error

### **Solution:**

#### **If no case is present:**

1. Find replacement case with correct product
2. Place it on the **sequence conveyor** in front of the lane (NOT in the buffer lane)
3. Verify position matches Inventory Graph
4. Clear alert

#### **If case is too far back:**

1. InnoPick reached maximum programmed advance distance
  2. Either:
    - Manually adjust cases in lane, OR
    - Jog the lane forward to fully dispense the case
  3. Ensure case is now on sequence conveyor
  4. Verify position on Inventory Graph
  5. Clear alert
-

115 - Buffer lane tried to dispense a case but the case was too close to the front

**Explanation:**

Dispense case was detected earlier than expected (case seen too early).

**Common Causes:**

- Case positioned too far forward in lane
- Case moved forward unexpectedly
- Lane position tracking error

**Solution:**

1. Verify actual case position in lane
  2. Check other cases in lane for proper spacing
  3. Adjust case position if needed
  4. Verify case dimensions match product configuration
  5. Clear alert once position is correct
- 

119 - Photocell still blocked after dispensing

**Explanation:**

The photocell for that lane remained blocked at the end of the dispense move.

**Common Causes:**

- Broken case (collapsed, torn cardboard)
- Ripped cardboard hanging down
- Crooked or skewed case
- Wrong case dimensions (didn't fault on storing)
- Cases poorly referenced with irregular gaps

**Solution:**

1. Inspect the lane photocell area
  2. Identify what is blocking the photocell
  3. Correct the issue:
    - Remove damaged cardboard or case parts
    - Straighten crooked case
    - Verify dispensed case is properly positioned on sequence conveyor
  4. Check other cases in buffer lane:
    - Ensure proper positioning
    - Verify adequate gaps
    - Confirm ready to dispense next case
  5. Clear photocell completely
  6. Verify on Inventory Graph
  7. Clear alert
  8. Monitor next few dispenses from this lane
- 

## 121 - Photocell above lane never detected the case during a storing move

### **Explanation:**

InnoPick attempted to store a case, but the photocell was never blocked during the move, indicating there was no case present.

### **Most Common Causes:**

- No case present at all
- Inventory count was wrong
- Bad recovery from previous infeed issue
- Case removed and not replaced

### **Less Common Causes:**

- Case snagged or displaced
- Case in wrong position to be picked up by pop-up

- Sequence conveyor offset by one or more positions

**Solution:**

**If case is simply missing:**

1. Determine correct product from alert details
2. Find replacement case
3. Place on sequence conveyor in expected position
4. Verify Inventory Graph matches
5. Clear alert

**If case is displaced:**

1. Locate the case
2. Position correctly for pop-up to engage
3. Retry the store move
4. Clear alert if successful

**Important:** See [Alert Guidelines](#) - this is often a symptom of improper previous recovery.

---

123 / 124 - Photocell above lane never detected the case during a dispense move

**Explanation:**

No case detected during dispense move when one was expected.

**Common Causes:**

- Case is missing (inventory count was wrong)
- Case fell off conveyor
- Case caught on something
- Wrong lane actuated
- Previous storage fault not properly resolved

**Solution:**

1. Check inventory count for this lane
2. Locate the missing case or find replacement
3. Place case in the storage lane at the front
4. Jog lane forward to dispense case onto sequence conveyor
5. Verify case position matches Inventory Graph
6. Update lane inventory count if necessary
7. Clear alert

**Important:** See [Alert Guidelines](#) - often indicates inventory tracking issue.

---

## 130 - Could not Execute a Sequence Conveyor Move. Timed Out Waiting for a previous Dispense Move to Finish

### **Explanation:**

The sequence conveyor was ready to move but had to wait too long for a buffer lane dispense operation to complete, causing a timeout.

### **Common Causes:**

- Lane dispense move taking excessive time
- Mechanical issue slowing lane operation
- Pop-up or clutch not disengaging properly
- Communication delay between systems
- Lane conveyor running slowly

### **Solution:**

1. Check which lane was dispensing when timeout occurred
2. Inspect that lane for:
  - Mechanical binding or slowness
  - Pop-up not retracting fully
  - Clutch not disengaging cleanly

3. Test lane operation manually if needed
  4. Verify lane motor and drive functioning properly
  5. Clear alert
  6. Monitor lane performance
  7. Contact maintenance if issue persists
- 

## 152 - PopUp stayed down during store (300)

### **Explanation:**

Pop-up assembly was detected in incorrect position (down) when it should have been up during a store operation.

### **Common Causes:**

- Pneumatic system issue
- Pop-up mechanism mechanical failure
- Position sensor malfunction
- Obstruction preventing pop-up from rising

### **Solution:**

1. Verify pneumatic components:
  - Air pressure adequate
  - Solenoid valves functioning
  - Air lines not blocked or leaking
2. Check mechanical components:
  - Pop-up shaft moves freely
  - No obstructions
  - Linkage intact
3. Verify position sensors:
  - Clean sensor faces
  - Check sensor alignment

- Verify sensor wiring
4. Remove any obstructing objects
  5. Test pop-up operation manually if safe
  6. Clear alert and test
  7. Contact maintenance if issue persists
- 

### 153 - PopUp stayed down during dispense (301)

**Explanation:**

Pop-up assembly was detected in incorrect position during dispense operation.

**Common Causes & Solution:**

See alert 152 above - same troubleshooting applies.

---

### 154 - PopUp stayed up after dispense (301)

**Explanation:**

Pop-up assembly did not retract after completing dispense operation.

**Common Causes & Solution:**

See alert 152 above - same troubleshooting applies, but focus on retraction mechanism and down position sensor.

---

### 155 - PopUp never went down before store (300)

**Explanation:**

Pop-up assembly failed to fully retract before a store operation was attempted.

**Common Causes & Solution:**

See alert 152 above - same troubleshooting applies, with emphasis on:

- Air cylinder air release adjustment
  - Solenoid valve operation
  - Down position sensor verification
- 

### 156 - PopUp never went down before dispense (301)

**Explanation:**

Pop-up assembly failed to retract before dispense operation.

**Common Causes & Solution:**

See alert 152 above - same troubleshooting applies.

---

### 310 - Clutch Fault: Mechanical engagement was not successfully completed after 15 attempts

**Explanation:**

The clutch assembly failed to engage properly after multiple attempts.

**Common Causes:**

- Mechanical failure in clutch assembly
- Intermittent mechanical issue
- Misalignment
- Worn components
- Obstruction

**Solution:**

1. Visually inspect clutch assembly if accessible
  2. Retry the move in Automatic mode while observing clutch
  3. Look for:
    - Mechanical binding
    - Worn parts
    - Obstructions
    - Misalignment
  4. If engagement seems to work sometimes, note conditions
  5. Document behavior for maintenance
  6. Mechanical intervention may be necessary
  7. Consider deactivating lane if repairs cannot be immediate
  8. Resume production on other lanes
  9. Contact maintenance
- 

## 320 - Clutch Fault: Calculation Error in Mechanical Engagement Position

### **Explanation:**

The system calculated an incorrect position for clutch mechanical engagement, indicating a software or calibration issue.

### **Common Causes:**

- Clutch calibration drift
- Position calculation error in software
- Encoder feedback issue
- Mechanical wear affecting expected positions
- System parameters need updating

### **Solution:**

1. Document current clutch position and behavior
2. Note if this is first occurrence or recurring

3. Check clutch engagement manually if safe
  4. Verify encoder feedback is working
  5. May require clutch recalibration
  6. Contact system expert or maintenance
  7. System parameters may need adjustment
  8. Clear alert after issue resolved
- 

333 - Warning: Actual sensor transition was not seen, but system is assuming clutch was engaged properly

**Explanation:**

Proximity sensor did not detect expected transition, but system continued based on assumptions.

**Common Causes:**

- Sensor misalignment
- Intermittent sensor issue
- Mechanical timing variation
- Sensor malfunction

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

334 - Clutch Fault: Prox sensor off after mechanical transition. Probably clutch mechanism bounce

**Explanation:**

Proximity sensor turned off after mechanical transition, likely due to clutch mechanism bouncing.

**Common Causes:**

- Mechanical bounce in clutch engagement

- Worn clutch components
- Improper clutch timing
- Sensor positioning

**Solution:**

1. Verify clutch assembly condition
  2. Check for worn components
  3. May require mechanical adjustment
  4. Contact maintenance if repeats
  5. Clear alert
- 

### 340 - Clutch Fault: Mechanical disengagement was not successfully completed after 15 attempts

**Explanation:**

Clutch assembly failed to disengage properly after multiple attempts.

**Common Causes & Solution:**

Similar to alert 310, but focus on disengagement mechanism:

1. Inspect clutch release mechanism
  2. Verify return spring operation
  3. Check for binding during retraction
  4. Look for obstructions
  5. Contact maintenance
  6. May need to deactivate lane
- 

### 345 - Warning: Clutch Disengaged the Mechanism but the Prox Sensor is still Off

**Explanation:**

Clutch successfully disengaged mechanically, but the proximity sensor still indicates "off" state when it should be "on".

**Common Causes:**

- Sensor misalignment
- Sensor malfunction
- Wiring issue
- Sensor dirty or obstructed
- Mechanical position slightly off from sensor detection range

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

### 350 - Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move

**Explanation:**

The clutch proximity sensor unexpectedly changed state during a buffer lane operation when it should have remained stable.

**Common Causes:**

- Vibration affecting sensor
- Loose sensor mounting
- Intermittent wiring connection
- Mechanical movement or flex during operation
- Sensor malfunction

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

### 371 - Auxiliary Encoders (at Outfeed & Infeed) are not Aligned to each other on the Buffer Shaft

**Explanation:**

The two auxiliary encoders that monitor the buffer shaft position are showing misalignment, indicating potential mechanical slippage or encoder calibration drift.

**Common Causes:**

- Coupling slip on one or both encoders
- Mechanical slippage in shaft system
- Encoder calibration drift
- Belt or chain stretch
- Mounting looseness

**Solution:**

1. Stop production on affected level
  2. Visually inspect auxiliary encoder couplings
  3. Check for loose mounting
  4. Check shaft for mechanical issues
  5. Contact maintenance - this indicates potential mechanical problem
  6. Clear alert after mechanical verification
- 

## 372 - Auxiliary Encoder C (Infeed side) is not Aligned to Motor Encoder on Buffer Shaft

**Explanation:**

The auxiliary encoder on the infeed side of the buffer shaft is out of alignment with the main motor encoder, indicating slippage.

**Common Causes:**

- Coupling slip on infeed encoder
- Mechanical slippage between motor and encoder
- Encoder mounting looseness

- Belt or coupling issue

**Solution:**

1. Contact maintenance
  2. Inspect infeed-side auxiliary encoder coupling
  3. Check encoder mounting
  4. Verify mechanical connection to shaft
  5. Check for any slip indicators on coupling
  6. Clear alert after inspection and correction
- 

### 373 - Auxiliary Encoder D (Outfeed side) is not Aligned to Motor Encoder on Buffer Shaft

**Explanation:**

The auxiliary encoder on the outfeed side of the buffer shaft is out of alignment with the main motor encoder, indicating slippage.

**Common Causes & Solution:**

Same as alert 372, but for outfeed-side encoder.

---

### 374 - Auxiliary Encoders Offsets have reached the Cumulative threshold

**Explanation:**

The cumulative offset errors between encoders have exceeded acceptable threshold, indicating ongoing slippage or calibration drift.

**Common Causes:**

- Repeated small slippages accumulating over time
- Gradual mechanical wear
- Encoder drift

- Inadequate coupling tension
- System running near limits

**Alert Maintenance Team when this fault occurs**

- It is safe to clear and resume operation
  - If the fault returns soon after, a mechanical inspection of shaft couplings and encoder attachment hardware will be necessary.
- 

## 375 - Auxiliary Encoder C (Infeed side) Coupling Slip was Detected on Buffer Shaft

**Explanation:**

Actual slippage detected in the coupling connecting the infeed-side auxiliary encoder to the buffer shaft.

**Common Causes:**

- Loose coupling
- Worn coupling
- Inadequate coupling clamping force
- Shaft wear
- Excessive torque or shock loads

**Solution:**

1. Contact maintenance
2. Inspect couplings for:
  - Tightness
  - Wear
  - Damage
  - Proper seating
3. Tighten or replace coupling as needed
4. Clear alert after mechanical repair
5. Monitor closely after resuming

---

## 376 - Auxiliary Encoder D (Outfeed side) Coupling Slip was Detected on Buffer Shaft

### **Explanation:**

Actual slippage detected in the coupling connecting the outfeed-side auxiliary encoder to the buffer shaft.

### **Common Causes & Solution:**

Same as alert 375, but for outfeed-side encoder coupling.

---

## 377 - Buffer Motor Gearbox / Shaft Slip Detected. Critical Fault. Inspect Mechanics.

### **Explanation:**

Slippage detected between the buffer motor gearbox and the shaft - a critical mechanical fault indicating major component failure or imminent failure.

### **Common Causes:**

- Gearbox internal failure
- Coupling failure
- Mounting looseness
- Major mechanical breakdown

### **Solution:**

1. **Stop production**
2. **DO NOT attempt to resume** without maintenance inspection
3. Contact maintenance urgently
4. Inspect for:

- Gearbox condition
- Shaft coupling integrity
- Any signs of mechanical failure

5. Clear alert only after repairs completed and verified

---

378 - Buffer Motor Gearbox / Shaft and Coupling Slip Detected. Critical Fault.  
Inspect Mechanics.

**Explanation:**

Multiple slip points detected simultaneously - gearbox, shaft, and coupling - indicating severe mechanical problem or system-wide failure.

**Common Causes:**

- Multiple component failures
- Catastrophic mechanical event
- System overload
- Major mechanical breakdown
- Multiple loose connections

**Solution:**

1. **Stop production**
2. **DO NOT attempt to resume** without maintenance inspection
3. Contact maintenance and supervision urgently
4. Full mechanical inspection required for:
  - Gearbox
  - All couplings
  - Shaft integrity
  - Mounting systems
  - Drive components
5. Clear alert only after complete repairs and testing

---

## 400 & 401 - Communication loopback timeout (Sys\_Flt)

### **Explanation:**

Communication timeout in system loopback check.

### **Common Causes:**

- Network issue
- Communication module fault
- Intermittent connection problem

### **Solution:**

1. If this is the first occurrence, safe to clear and resume
  2. If it happens repeatedly, contact support
  3. Most likely indicates network infrastructure issue
- 

## 402 - Timeout Clutch Fault - Waiting for Clutch to be Disengaged to Resume (Sys\_Flt)

### **Explanation:**

System timed out while waiting for a clutch to fully disengage before resuming operation, indicating clutch is stuck or slow to respond.

### **Common Causes:**

- Clutch mechanically stuck in engaged position
- Pneumatic system slow or weak
- Clutch return mechanism failure

### **Solution:**

1. Check clutch physical position

2. Check for mechanical binding
  3. Contact maintenance if issue persists
  4. Clear alert after clutch verified disengaged
- 

## 512 - Infeed middle photocell was not blocked during an infeed case move

### **Explanation:**

The infeed cycle started and initial photocell was blocked by incoming case, but the second photocell (infeed mid photocell) was never blocked.

### **Common Causes:**

- Case jammed between upstream conveyor and InnoPick infeed
- Case turned 90 degrees
- Case fell or was displaced
- Case too short to reach second photocell

### **Solution:**

1. Locate the case
2. **Important:** System expects a case on infeed belts since cycle started
3. Clear any jamming condition
4. Place the proper expected case correctly on infeed belts
5. Position case carefully to avoid photocell fault during storage
6. Verify case orientation
7. Check case dimensions match expected product
8. Clear alert
9. Resume operation

See [Alert Guidelines](#) for additional infeed troubleshooting steps.

---

## 515 - Unexpected case entering onto the infeed belts

### Explanation:

The first infeed photocell was blocked when InnoPick was not expecting to receive a case.

### Common Causes:

- Back-to-back cases without gap
- Upstream conveyor motor advanced case too far
- Debris blocking photocell
- Extra case sent by upstream system
- Tracking error in upstream system

### Solution:

1. **Key Point:** InnoPick is NOT expecting this case
2. Remove whatever is blocking the photocell:
  - If extra case: Remove from infeed belt
  - If debris: Clear obstruction
  - If case sent too far: Pull back or remove
  - **If removing an extra case:**
    - Adjust count in upstream system
    - Document the discrepancy
    - Verify replenishment quantities
  - Once photocell is clear, operator can resume production
  - **Do not** leave unexpected case on infeed belt
  - Clear alert
  - Coordinate with upstream system operator

See [Alert Guidelines](#) for complete infeed procedures.

---

## 516 - Infeed Case's Length Too Long

### Explanation:

The measured case length was greater than expected length plus tolerance during infeed transition.

### How Measurement Works:

- Case length measured as it transitions from infeed belts to first sequence conveyor position (position 0)
- Compared to configured product length
- Fault if outside tolerance (too long)

### Common Causes:

- Wrong product/case
- Case dimensions incorrectly configured in system
- Turned case (length and width swapped)
- Damaged case extended beyond normal dimensions
- Flaps or damage extending case length

### Solution:

1. Inspect the case that triggered the fault
2. Determine if:
  - Wrong product → Replace with correct case
  - Turned case → Orient properly (length should be longer dimension)
  - Case damaged/extended → Replace with proper case
  - Dimensions wrong in system → Check [Products Page](#)
  - Verify case matches expected product for this replenishment
  - Place correct case on infeed
  - Adjust replenishment quantity if case was rejected
  - Clear alert

See [Alert Guidelines](#).

---

## 517 - Infeed Case's Length Too Short

### **Explanation:**

The measured case length was less than expected length minus tolerance during infeed transition.

### **How Measurement Works:**

- Case length measured as it transitions from infeed belts to first sequence conveyor position (position 0)
- Compared to configured product length
- Fault if outside tolerance (too short)

### **Common Causes:**

- Damaged or collapsed case
- Wrong product/case
- Case dimensions incorrectly configured
- Turned case

### **Solution:**

1. Inspect the case that triggered the fault
2. Determine if:
  - Case is damaged → Replace with proper case
  - Wrong product → Replace with correct case
  - Turned case → Orient properly
  - Verify case matches expected product for this replenishment
  - Check product dimensions in [Products Page](#)
  - Place correct case on infeed
  - Adjust replenishment quantity if case was rejected
  - Clear alert

See [Alert Guidelines](#).

---

## 518 - Infeed Case (on position 0) is too far downstream

### **Explanation:**

The infeed case on position 0 (first sequence conveyor position) is positioned too far downstream from where it should be.

### **Common Causes:**

- Case advanced too far during infeed transition
- Timing error in infeed cycle
- Case slipped or was pushed downstream
- Conveyor speed mismatch
- Case measurement started late

### **Solution:**

1. Visually verify case position on position 0
2. Check if case can be adjusted to correct position
3. If case is only slightly off, may be safe to continue
4. If significantly misplaced:
  - Remove case
  - Replace with properly positioned case
  - See [Case Centering Guide](#)
5. Clear alert
6. Monitor subsequent cases for repeat
7. Contact maintenance if issue recurs

---

## 519 - Infeed Case (on position 0) is too far upstream

See Fault 518 above.

---

## 520 - Infeed Fault: Previous Length Check was not Completed

### **Explanation:**

A new infeed case arrived before the previous case's length measurement was completed, indicating timing or sequencing issue.

### **Common Causes:**

- Cases too close together at infeed
- Previous length check taking too long
- Infeed cycle timing error

### **Solution:**

1. Inspect sensors for proper alignment
2. Clear alert
3. Monitor subsequent infeed moves
4. Contact maintenance if alert re-occurs

---

## 525 - Pop-up shaft reached final upper limit - probably did not engage the pop-up mechanism

### **Explanation:**

The steel finger measuring pop-up height tripped the uppermost limit photocell, which is higher than the pop-up should normally reach.

### **Common Causes:**

- Air cylinder air release adjustment too slow
- Solenoid valve malfunctioning
- Pop-up mechanism not engaging properly

- Mechanical binding or misalignment

**Solution:**

1. Check air cylinder settings:
    - Verify air release adjustment
    - Ensure proper timing
  2. Test solenoid valve operation
  3. Inspect pop-up mechanism:
    - Verify engagement
    - Check for mechanical issues
  4. Adjust timing if needed
  5. Contact maintenance for pneumatic system tuning
  6. Clear alert after resolving
- 

555 - E-Stop condition active - safety is disarmed (door opened or E-Stop pressed)

**Explanation:**

A safety device is not in the proper 'safe' condition. This puts all levels into E-Stop status (Dark Red).

**Common Causes:**

- E-Stop button pressed
- Safety door opened
- Safety interlock open
- Safety device malfunction
- Wiring issue to safety system

**Solution:**

1. **Identify the E-Stop condition:**

- Check all E-Stop buttons → ensure all are pulled out (released)
- Check all safety doors → ensure all are fully closed
- Check safety interlocks → verify all are engaged
- Check guard position switches → ensure guards are in place

## **2. Clear the E-Stop condition:**

- Release/rotate E-Stop buttons to reset
- Close all doors completely
- Verify interlocks are engaged
- Check safety system indicator lights

## **3. After E-Stop is cleared:**

- Press Clear Alerts button
- All levels should return to Manual mode (Yellow)
- Perform visual inspection before resuming
- Put levels back into Automatic mode (Green)

## **4. If E-Stop won't clear:**

- One or more safety devices still not satisfied
- Systematically check each safety device
- Look for damaged switches or wiring
- Contact maintenance if cause not obvious

## **5. Resume operation:**

- Only after all safety conditions are satisfied
- Verify safe to restart
- Monitor closely on restart

**Safety Note:** Never bypass E-Stop or safety interlocks. If safety device seems to be malfunctioning, resolve the issue properly or contact maintenance.

---

## 556 - Open Door Preventing return to RUN Status (Door# 1-4)

### **Explanation:**

A specific safety door (numbered 1-4) is open and preventing the system from returning to RUN status.

**Common Causes:**

- Door not fully closed
- Door latch not engaging
- Door switch malfunction
- Door misaligned
- Safety interlock issue

**Solution:**

1. Locate the problem door
  2. Ensure door is fully closed
  3. Check door latch engages properly
  4. Verify door is properly aligned
  5. Check door safety switch operation
  6. Clear alert after door is secured
  7. If door repeatedly shows open when closed, contact maintenance
- 

## 570 - UPS power supply on battery (power loss)

**Explanation:**

Loss of main power. InnoPick PC is now running on UPS battery backup.

**Common Causes:**

- Building power failure
- Electrical panel issue
- Power supply fault

**Solution:**

1. Investigate cause of power loss

2. Restore main power as soon as possible
  3. UPS battery has limited runtime
  4. If power not restored quickly, perform orderly shutdown
  5. Contact facility maintenance
- 

## 571 - Mico Circuit Tripped (24V Short-Circuit)

### **Explanation:**

The Mico circuit breaker has tripped, indicating a short circuit in the 24V system.

### **Common Causes:**

- Short circuit in 24V wiring
- Damaged cable or connector
- Water or moisture in electrical components
- Failed 24V device
- Pinched or crushed wire

### **Solution:**

1. **DO NOT immediately reset** - find cause first
2. Visually inspect 24V wiring for:
  - Damaged insulation
  - Pinched wires
  - Water or moisture
  - Burn marks
3. Check recently moved or adjusted equipment
4. Disconnect suspect circuits if safe to do so
5. After finding and correcting short:
  - Reset Mico circuit breaker
  - Power up system
  - Clear alert

6. Contact maintenance if cause not obvious

7. **Never bypass** Mico protection

---

## 572 - Air pressure switch off - no air in system

### **Explanation:**

Loss of air pressure in the pneumatic system.

### **Common Causes:**

- E-Stop engaged (normal when doors open)
- Compressor shut down
- Air line leak or blockage
- Pressure switch malfunction

### **Solution:**

1. Check if E-Stop condition is active (doors open, E-Stop pressed)
  2. If E-Stop is cause:
    - Close all security doors
    - Clear E-Stop
    - Power up system
  3. If fault persists after clearing E-Stop:
    - Check compressor status
    - Inspect for air leaks
    - Verify air pressure gauge readings
    - Contact maintenance if issue not obvious
  4. Clear alert once air pressure is restored
- 

## 573 - Input/Output Murr module fault (read variable Murr\_Flt)

**Explanation:**

Loss of communication with Murr Modules (electrical modules mounted on front of InnoPick for every 8-lane section).

**Common Causes:**

- Communication disruption
- Module hardware fault
- Wiring issue
- E-Stop condition

**Solution:**

1. Ensure all security doors are closed
  2. Verify E-Stop is cleared
  3. Power up the system
  4. If fault persists:
    - Perform visual inspection of all Murr Modules for affected level
    - Check BUS Node Modules
    - Compare with modules from other levels
    - Look for any irregularities (LEDs, physical damage)
  5. Contact system expert if issue continues
  6. Document specific module or level affected
- 

## 575 - 12-Volt Power Supply for Auxiliary Encoders Not OK

**Explanation:**

The 12V power supply that powers the auxiliary encoders is not functioning properly or voltage is out of range.

**Common Causes:**

- 12V power supply failure

- Voltage out of specification
- Wiring issue to encoders
- Circuit breaker or fuse blown
- Power supply overload

**Solution:**

1. Check 12V power supply status indicators
  2. Measure voltage if possible (should be ~12VDC)
  3. Check circuit breakers/fuses for 12V circuit
  4. Inspect wiring to auxiliary encoders
  5. Look for damaged cables or connectors
  6. Contact maintenance - encoder operation critical
  7. May need power supply replacement
  8. Clear alert after power restored and verified
- 

## 576 - Warning: UPS Battery Faulted

**Explanation:**

The UPS (Uninterruptible Power Supply) battery has a fault condition, indicating the battery is failing or has failed.

**Common Causes:**

- Battery end of life
- Battery cell failure
- Internal battery fault
- Charging system failure
- Battery disconnected

**Solution:**

1. Check UPS status indicators
2. System can continue running on main power

3. **However:** No backup power protection available
  4. Schedule UPS battery replacement urgently
  5. Avoid intentional power interruptions until battery replaced
  6. Contact maintenance or UPS service provider
  7. Document battery age and fault details
  8. Clear alert after battery replaced and tested
- 

## 577 - Warning: UPS Battery Low

### **Explanation:**

The UPS battery charge level is low, reducing available backup runtime.

### **Common Causes:**

- Recent power outage depleted battery
- Battery aging and reduced capacity
- Charging system not fully recharging battery
- Battery nearing end of life
- UPS continuously on battery power

### **Solution:**

1. Check if main power is currently available
  2. If on main power, battery should recharge automatically
  3. Allow several hours for full recharge
  4. If battery remains low after recharge time:
    - Battery may be failing
    - Charging system may have issue
    - Contact maintenance
  5. Monitor battery status
  6. Consider battery replacement if issue persists
  7. Alert may clear automatically after battery recharges
-

## 660 - Motion Control Following Error - Critical Fault

### **Explanation:**

A motion control following error has occurred, meaning the actual position of a motor-driven axis has deviated too far from the commanded position. This is a critical fault indicating loss of position control.

### **Common Causes:**

- Mechanical binding or obstruction
- Motor or drive fault
- Excessive load on axis
- Broken belt or coupling
- Drive tuning issue
- Encoder failure

### **Solution:**

1. Identify which axis faulted (see specific axis alerts 661-663, 667)
  2. Check for mechanical obstructions or binding
  3. Verify motor and drive operation
  4. Check belts, chains, couplings for breaks
  5. Verify encoder feedback working
  6. Contact maintenance - inspection suggested before restarting
  7. Clear alert
- 

## 661 - Motion Control Following Error - Axis A (Infeed Conveyor) - Critical Fault

### **Explanation:**

Following error on Axis A (Infeed Conveyor), indicating loss of position control on the infeed system.

### **Common Causes & Solution:**

See alert 660 above, with specific focus on:

- Infeed conveyor belt condition
  - Infeed motor and drive
  - Mechanical binding at infeed
  - Encoder on infeed axis
- 

## 662 - Motion Control Following Error - Axis B (Sequence Conveyor) - Critical Fault

### **Explanation:**

Following error on Axis B (Sequence Conveyor), indicating loss of position control on the main sequence conveyor.

### **Common Causes & Solution:**

See alert 660 above, with specific focus on:

- Sequence conveyor chain condition
  - Sequence conveyor motor and drive
  - Mechanical binding on sequence conveyor
  - Encoder on sequence conveyor axis
- 

## 663 - Motion Control Following Error - Axis C (Buffer Conveyor) - Critical Fault

### **Explanation:**

Following error on Axis C (Buffer/Lane Conveyor), indicating loss of position control on the buffer lane system.

### **Common Causes & Solution:**

See alert 660 above, with specific focus on:

- Buffer lane belt/chain condition

- Buffer motor and drive
  - Mechanical binding in buffer lanes
  - Encoder on buffer axis
- 

## 667 - Motion Control Following Error - Axis A B & C (All) - Critical Fault

### **Explanation:**

Following error on all axes simultaneously (Infeed, Sequence, and Buffer conveyors), indicating a system-wide motion control failure.

### **Common Causes:**

- Motion controller fault or crash
- Power loss to all drives
- Emergency stop activation
- Communication loss to all drives
- System-wide mechanical seizure (unlikely)
- Motion controller program error

### **Solution:**

1. Check motion controller status
  2. This is likely a controller or system-level fault, not mechanical
  3. Contact system expert
  4. May require controller restart or program reload
  5. Verify all mechanical systems after restoration
  6. Clear alert only after full system verification
- 

## 671 - Galil Program Thread Command Error

### **Explanation:**

An error occurred in the Galil motion controller program thread, indicating a command execution problem or program fault.

**Common Causes:**

- Invalid command in Galil program
- Program logic error
- Variable out of range
- Array bounds exceeded
- Syntax error in downloaded program

**Solution:**

1. Note any recent program changes
  2. Check Galil controller for error code using:
    - TC1 (Tell Code)
    - MG\_TC (Message Tell Code)
    - MG\_ED0 (The line number of the last error)
    - MG\_ED1 (The thread where the error occurred)
  3. Document the specific error code
  4. Contact system expert or programmer
  5. May require program correction and reload
  6. Verify program version matches expected
  7. Clear alert after program issue corrected
  8. Test operation carefully after program changes
- 

## 701 - Clutch Prox / Sensor Not On Before Starting Routine I/O Check

**Explanation:**

Before starting the routine I/O check, the clutch proximity sensor was not in the expected "on" state, indicating the clutch may not be in the proper starting position.

**Common Causes:**

- Clutch not in home/rest position
- Proximity sensor misaligned
- Sensor malfunction
- Clutch mechanism stuck in wrong position
- Previous move not completed properly

**Solution:**

1. Check physical clutch position
  2. Verify clutch is in expected rest position
  3. Check proximity sensor alignment
  4. Test sensor operation
  5. May need to manually cycle clutch to home position
  6. Verify clutch returns properly after moves
  7. If the issue cannot be resolved quickly, it is possible to [disable the lane](#) and ignore its sensors.  
This allows the fault to be cleared and production to resume.
  8. Clear alert after clutch position verified
  9. Contact maintenance if sensor or clutch issue persists
- 

## 702 - Clutch Prox Not Off after Clutch Activation - Routine I/O Check

**Explanation:**

During routine I/O check, after the clutch was activated (engaged), the proximity sensor did not turn off as expected.

**Common Causes:**

- Sensor stuck or malfunctioning
- Clutch not moving fully when activated
- Mechanical binding preventing full travel
- Sensor misaligned
- Wiring issue

**Solution:**

1. Observe clutch operation during activation
  2. Verify clutch moves fully when activated
  3. Check proximity sensor for proper operation
  4. Verify sensor alignment and mounting
  5. Check wiring and connections
  6. May need sensor adjustment or replacement
  7. Contact maintenance
  8. Clear alert after verification
- 

## 703 - Clutch Prox Not On after clutch De-activation - Routine I/O Check

**Explanation:**

During routine I/O check, after the clutch was deactivated (disengaged), the proximity sensor did not turn on as expected.

**Common Causes:**

- Sensor malfunction
- Clutch not returning to rest position
- Return spring weak or broken
- Mechanical binding
- Sensor misaligned

**Solution:**

1. Observe clutch operation during deactivation
2. Verify clutch returns fully to rest position
3. Check return spring operation
4. Check proximity sensor operation
5. Verify sensor alignment

6. Look for mechanical binding
  7. Contact maintenance
  8. Clear alert after clutch verified returning properly
- 

## 706 - Missing Y Pop-Up Flag (H) after Lift - Routine I/O Check

### **Explanation:**

During routine I/O check, after the pop-up lifted, the "Y" flag sensor (H - intermediate height) did not detect the expected flag position.

### **Common Causes:**

- Pop-up not lifting to expected height
- Sensor malfunction
- Flag missing or damaged
- Sensor misaligned
- Pneumatic pressure insufficient

### **Solution:**

1. Check pop-up operation and lift height
  2. Verify air pressure adequate
  3. Check "H" height sensor alignment
  4. Verify flag is present and undamaged
  5. Test sensor operation
  6. May need pneumatic or sensor adjustment
  7. Contact maintenance
  8. Clear alert after pop-up verified reaching proper height
- 

## 707 - Z Pop-Up Flag (HH) Present after Lift - Routine I/O Check

**Explanation:**

During routine I/O check, after pop-up lift, the "Z" flag sensor (HH - upper limit) is detecting the flag when it shouldn't, indicating pop-up may be going too high.

**Common Causes:**

- Pop-up lifting too high
- Air cylinder over-extending
- Mechanical stop not functioning
- Sensor misaligned or malfunction
- Pneumatic pressure too high

**Solution:**

1. Check pop-up maximum lift height
  2. Verify mechanical stops functioning
  3. Check air pressure - may be too high
  4. Verify "HH" sensor alignment
  5. Check for over-travel conditions
  6. May need pneumatic adjustment
  7. Contact maintenance
  8. Clear alert after pop-up height verified correct
- 

## 708 - Missing W Pop-Up Flag (LL) After Pop-Up De-activation - I/O Check

**Explanation:**

During routine I/O check, after the pop-up was deactivated (lowered), the "W" flag sensor (LL - lower limit) did not detect the flag, indicating pop-up may not be fully retracted.

**Common Causes:**

- Pop-up not fully retracting
- Air cylinder not fully extending downward

- Mechanical obstruction
- Sensor misaligned
- Flag damaged or missing

**Solution:**

1. Verify pop-up fully retracts when deactivated
  2. Check for mechanical obstructions
  3. Verify air exhaust functioning properly
  4. Check "LL" sensor alignment and operation
  5. Verify flag present and undamaged
  6. May need pneumatic or mechanical adjustment
  7. Contact maintenance
  8. Clear alert after pop-up verified retracting fully
- 

## 709 - Buffer Lane Photocell Blocked during Routine I/O Check

**Explanation:**

InnoPick continuously monitors every lane's photocell and clutch proximity sensor inputs. This fault indicates one sensor was in an unexpected state.

**Common Causes:**

*For Photocells:*

- Broken or damaged case
- Case inducted or dispensed too close to photocell
- Random debris (plastic, cardboard)
- Malfunctioning photocell sensor
- Case positioned incorrectly

*For Clutch Proximity Sensors:*

- Proximity sensor in wrong state

- Clutch assembly not in expected position
- Sensor misalignment
- Wiring issue

**Solution:**

1. Note which lane the alert specifies
  2. Inspect that lane:
    - Check photocell area for obstructions
    - Look for broken case parts
    - Verify clutch position (if clutch prox fault)
  3. Clear the obstruction or correct the issue
  4. Verify sensor is now in correct state
  5. Clear alert
  6. Monitor to ensure issue doesn't recur
  7. If the issue cannot be resolved quickly, it is possible to [disable the lane](#) and ignore its sensors. This allows the fault to be cleared and production to resume.
- 

## 715 - Case Detected at Far Back of Buffer Lane

**Explanation:**

A case has been detected at the far back position of a buffer lane, which may indicate the lane is full or a case has moved too far back.

**Common Causes:**

- Lane is full to capacity
- Case pushed too far during storage
- Case slipped backward in lane
- Photocell at back of lane detecting case

**Solution:**

1. Visually inspect the affected lane

2. Remove extra cases or jammed cases
  3. Update lane inventory count if necessary
  4. Clear alert
  5. Monitor lane operation
- 

## 719 - Clutch Prox Sensor Off during Routine I/O Check

### **Explanation:**

During routine I/O monitoring, the clutch proximity sensor is in the "off" state when it should be "on", or vice versa.

### **Common Causes & Solution:**

See alert 709 above - same troubleshooting applies for clutch proximity sensors.

---

## 725 - Photocell above Lane was Blocked on Activated Lane

### **Explanation:**

When a lane was activated (clutch engaged, ready to operate), the photocell above the lane was already blocked, which is unexpected and may indicate a case is in wrong position.

### **Common Causes:**

- Case positioned too far forward in lane
- Case or debris blocking photocell
- Previous dispense did not complete properly
- Case too tall or irregular
- Sensor malfunction

### **Solution:**

1. Inspect the affected lane

2. Check case position in lane
  3. Verify no debris blocking photocell
  4. Ensure cases are properly positioned with gaps
  5. Check if case dimensions match configuration
  6. Adjust case position if needed
  7. Clear photocell obstruction
  8. Clear alert
  9. Monitor subsequent operations on this lane
- 

## 781 - Infeed Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

### **Explanation:**

The infeed motion control thread took too long to complete, exceeding the timeout threshold. This indicates a performance issue or system overload.

### **Common Causes:**

- High system load
- Motion controller processing delay
- Communication delay
- Complex motion calculation taking too long
- Controller near processing capacity

### **Solution:**

1. Document what was happening when timeout occurred
  2. May be transient - safe to clear if isolated occurrence
  3. If repeated, contact system expert
  4. Clear alert
  5. Monitor for recurrence
-

## 782 - Sequence Conveyor Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

### **Explanation:**

The sequence conveyor motion control thread took too long to complete, exceeding the timeout threshold.

### **Common Causes & Solution:**

See alert 781 above - same troubleshooting applies for sequence conveyor axis.

---

## 783 - Buffer Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

### **Explanation:**

The buffer lane motion control thread took too long to complete, exceeding the timeout threshold.

### **Common Causes & Solution:**

See alert 781 above - same troubleshooting applies for buffer/lane axis.

---

## 791 - Motion control (Galil) thread crashed while main thread was waiting - critical fault

### **Explanation:**

The motion controller instruction thread has crashed on the indicated level. This is a serious error that should be adequately documented.

### **Common Causes:**

- Software exception
- Motion controller communication failure

- Serious system fault

**Solution:**

1. Contact System Expert to follow the steps below:

**2. Capture diagnostic data BEFORE clearing the alert:**

- **IMPORTANT:** DMC-4040 motion controllers do NOT store error history
- Once alert is cleared, error information is lost forever
- System expert should capture information from GalilTools immediately:

```
TC1      # Current error code
TB        # Status byte
TS        # System status
SC        # Stop code
MG _SCA   # Axis A stop code
MG _SCB   # Axis B stop code
MG _SCC   # Axis C stop code
```

**3. If Alert 791 has already been cleared:**

- Error information is no longer available
- Document circumstances and await recurrence

**4. After capturing data, clear the alarm**

**5. Perform quick visual check:**

- Compare affected level to Inventory Graph
- Verify case positions match expected

**6. When resuming production:**

- Pay close attention to affected level
- An instruction may have been missed or duplicated
- Sequence conveyor may be offset by 1 position
- May have extra or missing case

**7. Monitor closely** for several cycles

---

900 - Warning: Back Edge of Case was Adjusted after the Case Stored

**Explanation:**

After a case was stored in a lane, the system detected the back edge position was not where expected and made an adjustment to the tracking data.

**Common Causes:**

- Case compressed or shifted during storage
- Case dimensions slightly different than expected
- Normal variation in case positioning
- Gap adjustment in lane

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

## 901 - Warning: Back Edge of Case was Adjusted Outward after Case Dispense (Back Edge not Seen)

**Explanation:**

After dispensing a case, the back edge was not detected where expected, and the system adjusted the position outward (further into the lane).

**Common Causes:**

- Case back edge not clearly detected
- Case compressed in lane
- Photocell timing issue
- Normal variation

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

## 902 - Warning: Back Edge of Case was Adjusted Outward after Case Dispensed

**Explanation:**

After dispensing a case, the system adjusted the tracked position of the back edge outward (further into the lane).

**Common Causes & Solution:**

Similar to alert 901 - normal system adjustment.

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

## 903 - Warning: Back Edge of Case was Adjusted Inward after Case Dispensed

**Explanation:**

After dispensing a case, the system adjusted the tracked position of the back edge inward (closer to lane entrance).

**Common Causes:**

- Case extended further than expected during storage
- Normal variation in case positioning
- Gap adjustment
- System self-correction

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

## 904 - Warning: Pulled Buffer Lane Cases Inward after Case Store (Photocell was Blocked)

**Explanation:**

After storing a case, the photocell was blocked (indicating insufficient gap to next case), so the system adjusted by pulling all cases in the lane inward to create proper spacing.

**Common Causes:**

- Case stored closer to previous case than expected

- Normal gap adjustment
- Case compressed during storage
- System maintaining proper spacing

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

905 - Warning: Pulled Buffer Lane Cases Inward after Case Dispense (Photocell was Blocked)

**Explanation:**

After dispensing a case, the photocell was still blocked, indicating the next case was too close, so the system adjusted by pulling remaining cases in the lane inward.

**Common Causes:**

- Next case too close after dispense
- Normal gap adjustment
- System maintaining proper spacing
- Cases compressed in lane

**Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane**

---

## Outfeed Alerts

Expecting a case but there is no case present at the outfeed

**Explanation:**

InnoPick expects a case on the last sequence conveyor position, but no case is detected there.

**Common Causes:**

- Case never dispensed from lane

- Case fell off somewhere
- Case caught on something earlier in sequence
- Motion controller thread crashed (see alert 791)
- Sequence conveyor offset by one position
- Inventory tracking error

**Solution:**

1. Review Inventory Graph - what case is supposed to be there?
2. Inspect the outfeed area and upstream positions
3. Determine why the case is not present:
  - **Case is one spot back:** May need to jog conveyor forward
  - **Case never dispensed:** Check source lane, may need to manually dispense
  - **Case caught upstream:** Clear obstruction and position case correctly
  - **Case missing entirely:** Find replacement case
4. **Place needed case on last position:**
5.
  - Must be the correct product
  - Must match what Inventory Graph expects
  - Must block the last sensor on InnoPick
6. Verify Inventory Graph matches physical reality
7. Clear alert
8. Resume operation
9. Monitor closely to ensure tracking is correct

**Important:** See [Alert Guidelines](#) - this usually indicates a tracking issue.

---

Not expecting a case but there is a case at the outfeed

**Explanation:**

A case is detected at the last position when InnoPick was not expecting one there.

**Common Causes:**

- Infeed case not stored properly (see infeed alerts)
- Issue not resolved properly after previous fault
- Case advanced without being tracked
- Manual intervention error

**Solution:**

**1. Verify the case truly should not be there:**

- Check Inventory Graph
- Review recent infeed activity
- Check if case belongs to current sequence

**2. Remove the unexpected case:**

- Take case off last position
- Do not leave it in place

**3. Determine where case should go:**

- Look at which lanes are receiving current replenishment
- Compare lane inventory numbers with physical count
- Check [Replenishments Page](#)

**4. Two options for the case:**

- *Option A - Short the lane:*
  - If lane is inaccessible or case not needed urgently
  - Reduce lane count by 1
  - Update in [Lane Inventory Page](#)
- *Option B - Add to back of lane:*
  - If lane is accessible
  - Case may be needed soon
  - Manually place at back of lane
  - Update inventory count if needed

**5. Clear alert**

**6. Resume operation**

**7. Monitor to prevent recurrence**

**Important:** See [Alert Guidelines](#) - usually caused by improper infeed issue resolution.

---

## Related Topics

- [Alert Guidelines](#) - General principles for successful alert resolution
  - [Alerts Page](#) - Viewing and managing alerts
  - [Common Issues](#) - Frequently encountered problems
  - [Case Replacement](#) - Procedures for replacing cases
- 

**Navigation:** [← Alert Guidelines](#) | [Next: Common Issues →](#)

# Deadlocks

[Home](#) > [Troubleshooting](#) > **Deadlocks**

---

## Overview

This section provides guidance for resolving deadlock situations.

A deadlock is when InnoPick is not able to fit all the cases that are due to enter the infeed, and it cannot make room by outputting cases because the next case needed for the sequence is coming after those cases which cannot fit.

In other words, a deadlock is when a level of InnoPick is too full and becomes stuck.

---

## Types of Deadlocks

There are two types of deadlocks:

1. **Preview deadlock (forecast)** - warns the operator that a deadlock will occur unless something is done to avert it.
  2. **Just-in-Time deadlock** - the system is in a deadlock state and cannot proceed without an operator intervention.
- 

## Preview Deadlock

| Level | Message                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Level 5: Caught deadlock 'Level 5: DeadLock detected with given sequence after processing 436 output cases, 451 input cases. The product that is needed to service the next output case '7b546749-be54-4457-9735-ed8ad28454a1  118010574289 145811691' is '49309'. Last input case attempted to process was '7b546749-be54-4457-9735-ed8ad28454a1  118010574289 145811691' with product reference '49309'. If this happens during preview, we still have a chance to avoid the deadlock by manually reassigning this product to a different level than 5. Aborting!': Unable to auto-resolve. This is critical. Halting system. (RunInSimulationMode = True) |

"Level 2. Deadlock detected with given sequence after processing 940 output cases, 850 input cases."

The following is an example alarm message for a preview deadlock:

- *[Level 5] Level 5: Caught deadlock 'Level 5: DeadLock detected with given sequence after processing 111 output cases, 166 input cases. The product that is needed to service the next output case '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' is '47692'. Last input case attempted to process was '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' with product reference '47692'. If this happens during preview, we still have a chance to avoid the deadlock by manually reassigning this product to a different level than 5. Aborting!': Unable to auto-resolve. This is critical. Halting system. (RunInSimulationMode = False)*
- 

#### **Key Points:**

- The deadlock is detected after processing a number of output and input cases
  - This forecasts a future deadlock if nothing changes
  - Manual reassignment of the key product to a different level can prevent it
  - The system automatically enters manual mode to allow operator intervention
- 

#### **Understanding the Causes of Preview Deadlocks**

Preview deadlocks can be generated by one or more factors.

- There are too many products assigned to a single level of InnoPick, and the system has not found any way to avoid a deadlock situation.
- A replenishment has come in out of order (before the next priority replen), meaning it has to be stored completely in order for the priority replen can be brought in.
- The number of enabled buffer lanes was reduced since the initial case sequence calculation.
- A replenishment was shorted, but the shorted cases were needed to complete the case sequence. Now a new replen has been generated but the system has to store the active

replans that are in the pipeline before it.

---

### Resolving a Preview Deadlock

- Note the product # and use the [Products Page](#) to re-assign it to another level.
  - After this, the system will re-calculate the case sequence and may re-generate a preview deadlock fault depending on the context.
  - Continue re-assigning the problematic product (as listed in the preview deadlock alert message) until no new preview deadlock alerts are generated.
- 

## Just-In-Time Deadlocks

A Just-In-Time deadlock occurs when a level of InnoPick is full and cannot make even one move. This can happen if:

- A preview deadlock is ignored / not resolved and the system is allowed to continue operating.
- Some unexpected change has occurred, and InnoPick can no longer continue on that level, such as the disabling of a lane that InnoPick was expecting to be able to use.

Example of a Just-in-Time deadlock alert message:

- *[Level 5] Warning: Just In Time deadlock detected on Level 5. The product that is needed to service the next output case '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' is '47692'. Please fix manually!!!*
- 

### Resolving a Just-In-Time Deadlock

The simplest way to resolve a Just-In-Time deadlock is to:

1. Identify some lanes on that level with low inventory (1-10 cases)

2. Makes sure those cases are not immediately going to be needed (using the [Case Sequence Page](#))
3. Empty those lanes, making them available.

Alternatively, another option is to:

1. Identify which product InnoPick needs in order to resume outputting cases (this is listed in the Alert message).
2. Filter the [Case Sequence Page](#) to see how many cases of this product are needed.
3. Manually load this number of cases of the required product in an empty lane on that level of InnoPick.
4. Edit the inventory for that lane and save the changes, using the [Inventory Page](#).
5. Upon saving the changes to the inventory change, the system should recalculate the case sequence for that level and production should resume.
6. If needed, use the "Recalculate Level" [Recovery Action](#).

**Navigation:** [← Common Issues](#) | [Case Replacement](#) [→](#)

# Case Replacement Procedures

[Home](#) > [Troubleshooting](#) > **Case Replacement**

---

## Overview

This page provides procedures for replacing broken, missing, or incorrect cases during InnoPick operation. Proper case replacement is critical to maintaining accurate inventory tracking and preventing cascading faults.

---

## When Case Replacement is Needed

Common scenarios requiring case replacement:

- **Broken case:** Case damaged, collapsed, or torn
  - **Missing case:** Case fell off, removed, or never present
  - **Wrong product:** Incorrect SKU inducted
  - **Quality issue:** Case rejected for damage or defects
  - **Operator error:** Case removed during troubleshooting
- 

## General Principles

### Before Replacing Any Case

1. **Understand why the case needs replacement**

- What happened to the original case?
- Is this an isolated incident or pattern?
- Will the same issue affect the replacement?

## **2. Determine the correct product**

- Check what product is expected
- Verify SKU from alert message or Inventory Graph
- Confirm with [Replenishments Page](#) or [Case Sequence Page](#)

## **3. Know where to place the replacement**

- Sequence conveyor vs. storage lane
  - Specific position on conveyor
- 

# Scenario 1: Replace Case Currently Being Inducted

## **Situation:**

- Case is currently part of an active replenishment
- Case is already in InnoPick
- Case is broken, too damaged, or wrong product

## **Procedure**

### **1. Obtain replacement case:**

- Take one from upstream conveyor system
- Ensure it's the correct product

### **2. Update upstream system:**

- Delete 1 Case Data on upstream conveyor HMI
- This prevents tracking mismatch

### **3. Short the replenishment in InnoPick Manager:**

- Go to [Replenishments Page](#)
- Find the active replenishment

- Use **Alter Quantity** action (three dots menu)
- Reduce quantity by 1

#### 4. Place replacement case:

- Position on the case exactly where the broken was (or was supposed to be)
- Ensure proper orientation

#### 5. Resume operation:

- Clear any alerts if present
  - Verify the case stores correctly
  - Monitor the transition to the next replenishment
- 

## Scenario 2: Damaged Inbound Case on Sequence Conveyor is Not Needed for Client Pallet

### Situation:

- Case is on the sequence conveyor and is going to be stored
- No replacement cases are readily available
- Case will NOT be needed for upcoming orders
- This can be determined if: There are no additional replenishment scheduled for this product (filter for this product & level in the Replenishments page to determine this).

### If no other replenishment scheduled:

- InnoPick can function without this case
- Case can be shorted rather than replaced
- **Replacement is optional**

## Procedure

### Short the Case

#### 1. Allow the storage move to fault:

- **Alert 121 (photocell never detected case)** will occur

- System expects to store a case but finds none

## 2. Jog lane so it is ready to dispense

- Use [Manual motion control > Lane Operations](#)
- Jog lane forward so existing cases move up
- Front case can now be dispensed

## 3. Short the lane inventory:

- Go to [Lane Inventory Page](#)
- Find the affected lane
- Reduce quantity by 1

## 4. Clear alerts and resume

---

# Scenario 3: Replace Missing Outfeed Case

## Situation:

- Alert: "Expecting a case but there is no case present at the outfeed"
- Case should be at last InnoPick position but is missing
- See [Alert Reference](#) for detailed alert resolution

## Procedure

### 1. Identify the expected case:

- Check [Inventory Graph](#)
- Note product SKU and pallet ID
- Determine from which lane it should have come

### 2. Determine why case is missing:

- Was it never dispensed from lane?
- Did it fall off somewhere?
- Was it incorrectly removed?
- Is sequence conveyor offset by one position?

### **3. Obtain replacement case:**

- Must be exact product that was expected
- Check case label/markings

### **4. Place case at outfeed position:**

- Must block the last sensor on InnoPick
- Position correctly for downstream equipment
- Verify orientation

### **5. Verify Inventory Graph:**

- Should now show filled square at last position
- Matches physical case

### **6. Clear alert and resume**

### **7. Monitor closely:**

- Verify subsequent cases are tracking correctly
  - Watch for signs of offset sequence conveyor
- 

## Special Considerations

### Quality Control

If replacing due to quality issue:

- Note the specific problem
- Report pattern of defects to supervision
- May indicate supplier issue

### Inventory Accuracy

After any replacement:

- Verify all counts are correct
- Check [Lane Inventory](#)

- Confirm [Replenishments Page](#) quantities
  - Ensure [Inventory Graph](#) matches reality
- 

## Preventing Case Replacement Needs

### Best Practices

#### Visual Inspections:

- Check case quality of the cases in InnoPick
- Monitor cases as they induct

#### Proper Handling:

- Ensure upstream handling doesn't damage cases
- Ensure upstream handling properly identifies and rejects damaged or turned cases

#### Maintain Tracking Accuracy:

- Always follow [Alert Guidelines](#)
  - Verify Inventory Graph after every alert
- 

## Related Topics

- [Alert Reference](#) - Specific alerts requiring case replacement
  - [Alert Guidelines](#) - Proper alert recovery procedures
  - [Lane Inventory](#) - Updating lane contents
  - [Replenishments Page](#) - Adjusting quantities
-

# Case Centering

[Home](#) > [Troubleshooting](#) > **Case Centering Adjustments**

## Overview

InnoPick requires precise parameterization in order for cases to be accurately centered all the way from the infeed to the last buffer lane.

This page walks through the proper process for ensuring good case centering.

---

## Step-by-Step Sequence for Proper Case Centering

### 1. Validate Sequence Conveyor Chain Tension

- It's not uncommon to have to adjust the case centering parameters after tensioning sequence conveyor chains.
- Therefore, if it is needed, it's best to do that first.

### 2. Adjust the "Center Placement in Millimeters" Parameter

## Levels Configuration

Level Number: 1

Initialization String : 1:(0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 11 0 0 12 13 14 15 0 0 16 17 18 19 0 0 0 20 21 22 0 0 23 24 25 26 0 0 27 28 29 30 0 0 31 32 33 34 0 0 35 36 37 38 0 0 39 40 41 0 0 42 43 44 45 0)

Motion Controller Address : 10.7.5.141

Zone Number : 1

Lane Number:

00000000000000001100121314150016171819000202122002324252600272829300031323334003536373800039404100424344450

Merge A

Merge B

Lane Pitch In Millimeters

558.87

Center Placement in Millimeters

70

Center offset In Millimeters

-5

Length offset In Millimeters

0

Target offset In Millimeters

0

Send variables to Galil

On the **Administration > Setup InnoPick > Site Levels** page, adjust the "**Center placement in millimeters**" parameter so that cases are properly centered in front of the first buffer lane (the one closest to the infeed).

### Procedure:

1. Stop InnoPick when a case is positioned in front of the first lane (closest to the infeed).
2. Calculate the adjustment required to be centered
  - **Example:** The case is too far in the downstream direction, by 10mm
3. Modify the "**Center placement in millimeters**" by **-10**, and let a new case enter to re-validate
  - **⚠ Note:** The change will only be applied to the next case that is inducted on the infeed section of the upstream conveyor
4. Continue until cases are properly centered in front of the first lane

### 3. Test Alignment Along the Entire Length of InnoPick

Next, place a perfectly centered case in front of the first lane, and manually jog the sequence conveyor until the case is in front of the last lane (nearest the outfeed).

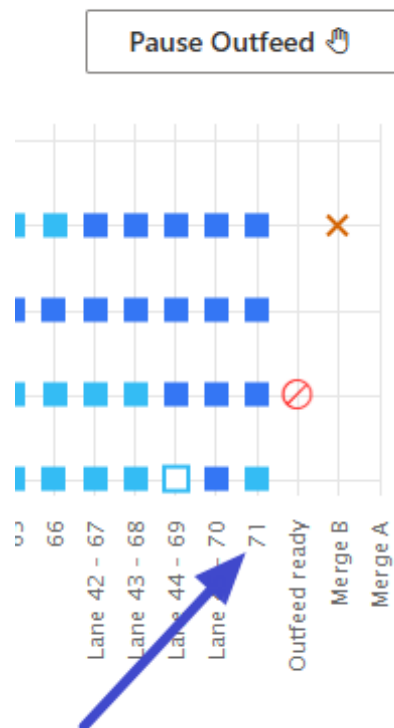
[Click here for information on manually jogging InnoPick](#)

**Important:** When you jog, make sure that the value used for the jogging (on the Manual Motion Controls page) is the same as the "**Lane Pitch in Millimeters**" parameter on the Site Levels page.

- It is usually necessary to input the correct distance from the Lane Pitch parameter for this exercise, in order to recreate the actual movement distances seen in production.

### 4. Measure the Necessary Adjustment

- Go to the case stopped in front of the last buffer lane.
- Measure the adjustment necessary for it be properly centered, and divide the adjustment by the number of positions on the sequence conveyor (visible on the inventory graph).



- In the example above, there are 71 positions.

## 5. Calculate the Lane Pitch Correction

### Example:

- Need to adjust the position by **12mm toward upstream** (case is too far toward downstream)
- The InnoPick has **71 positions**
- Calculation:  **$12\text{mm} \div 71 \text{ positions} = 0.17\text{mm}$**

## 6. Apply the Correction

Adjust the "**Lane Pitch in Millimeters**" parameter by the result of your calculation in Step 5.

### Important Note:

- If the adjustment is towards the upstream, subtract the value from the current Lane Pitch distance. This will reduce the size of the standard move and result in the case moving a smaller distance, ending up more upstream.
- If the adjustment is towards the downstream, add the value to the current Lane Pitch distance. This will increase the size of the standard move and result in the case moving a greater distance, ending up further downstream.

### Example (continued):

- If our calculation resulted in a value of 0.17mm
- And the Lane Pitch parameter is currently at **558.80mm**
- We will subtract the calculated value from the parameter value:
- **$558.80 - 0.17 = 558.63\text{mm}$**

## 7. Final Validation

Validate the new value by repeating the test and ensuring that the case ends up properly centered in front of the last lane (near the outfeed).

---

## Parameter Summary

| Parameter                              | Location                                        | Function                                                                                                                                                                        |
|----------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Center Placement in Millimeters</b> | Administration > Setup InnoPick > Site Levels   | Used to set initial center position, which is validated in front of the <b>first</b> lane                                                                                       |
| <b>Lane Pitch in Millimeters</b>       | Administration > Setup InnoPick > Site Settings | Defines the standard sequence conveyor move distance, used for maintaining centering along the <b>entire length</b> of InnoPick, and validated in front of the <b>last</b> lane |

---

## Calculation Formula

Lane Pitch Adjustment = Total Error (mm) ÷ Number of Positions

### Direction of adjustments:

- Case too far **downstream** → **Negative** (-) adjustment
- Case too far **upstream** → **Positive** (+) adjustment

**Navigation:** ← [Case Replacement](#) | [Disabled Lane](#) →

# Disabled Lane

[Home](#) > [Troubleshooting](#) > **Disabled Lane**

---

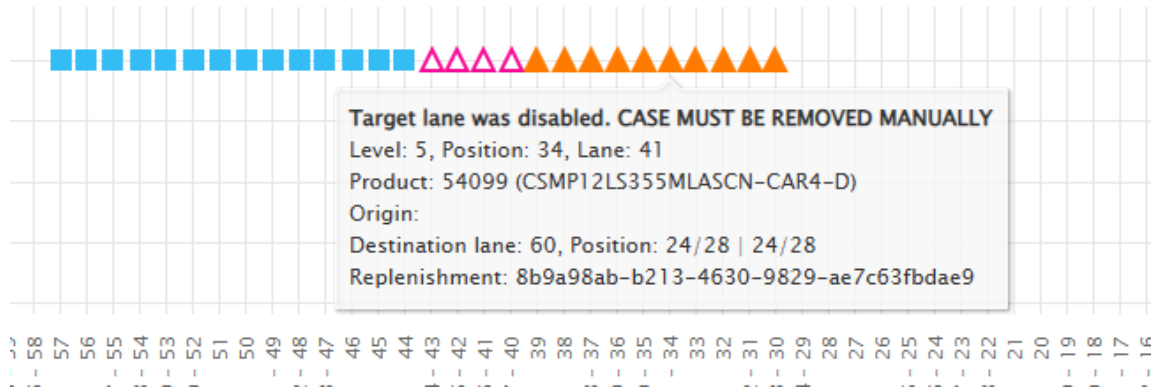
## Overview

If a mechanical issue renders a lane inoperable during production, it may be necessary to disable a lane that InnoPick is expecting to be able to use. This troubleshooting guide covers these scenarios.

---

## Inbound Cases

- Once replenishment cases are on the sequence conveyor, they are assigned to a particular buffer lane, and this destination cannot be changed.
- If that lane suffers a mechanical breakdown and cannot store those cases, then the operator must:
  - Disable the lane in question
  - Manually remove the cases on the sequence conveyor that are destined for that lane.
- Once the lane is disabled, the affected inbound cases will be highlighted and a message will appear when you hover over them:
  - **"Target lane was disabled. CASE MUST BE REMOVED MANUALLY"**
- Failure to remove these cases by hand will certainly cause subsequent issues



## Outbound Cases

- Similarly, once an outbound case has been scheduled on the sequence conveyor (appearing as an empty square), the lane cannot be re-assigned.
- If the source lane suffers a mechanical breakdown, the operator must:
- Disable the lane in question
- Manually place the scheduled outbound cases on the correct sequence conveyor positions
- Failure to place the scheduled outbound cases on the correct sequence conveyor positions will result in faults when those cases are not detected as expected on the last position of InnoPick

**Navigation:** [← Case Centering](#) | [Reference](#) [→](#)

# Reference

[Home](#) > **Reference**

---

## Overview

This page provides quick reference information for InnoPick Manager, including status indicators, alert code summaries, and other frequently-needed reference data.

---

## Reference Pages

### Alert Codes Quick Reference

Condensed summary of all alert codes for quick lookup:

- Alert code number
- Brief description
- Link to detailed resolution in Alert Reference

| Error Code          | Description                                                              |
|---------------------|--------------------------------------------------------------------------|
| <a href="#">98</a>  | ServoDrive Faulted while Moving (Sys_Flt)                                |
| <a href="#">99</a>  | ServoDrive Time-out Fault occurred during Start-up (Sys_Flt)             |
| <a href="#">109</a> | Photocell above lane was still Blocked after the end of the Storing Move |
| <a href="#">113</a> | Buffer lane tried to Dispense a Case but the Case was Too Far back       |

|     |                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------|
| 115 | Buffer lane tried to Dispense a Case but the Case was Too Close to the front                         |
| 117 | Buffer lane tried to Dispense a Case but the Case was Too Far back                                   |
| 119 | Photocell above lane still Blocked after Buffer Lane Dispensed a Case                                |
| 121 | Photocell above lane never Detected the Case during a Storing Move                                   |
| 123 | Photocell above lane never Detected the Case during a Dispense Move                                  |
| 124 | Photocell above lane never Detected the Case during a Dispense Move                                  |
| 130 | Could not Execute a Sequence Conveyor Move. Timed Out Waiting for a previous Dispense Move to Finish |
| 152 | The Pop-Up Mechanism stayed down when trying to Store a Case                                         |
| 153 | The Pop-Up Mechanism stayed down when trying to Dispense a Case                                      |
| 154 | The Pop-Up Mechanism stayed Up after Dispensing a Case                                               |
| 155 | The Pop-Up Mechanism never went Down (from a previous Move) before a Storing Move                    |
| 156 | The Pop-Up Mechanism never went Down (from a previous Move) before a Dispense Move                   |
| 310 | Clutch Fault: Mechanical Engagement was not Successfully Completed after 15 Attempts                 |
| 320 | Clutch Fault: Calculation Error in Mechanical Engagement Position                                    |
| 333 | Warning: Actual Sensor Transition was Not Seen. System is Assuming Clutch was Engaged Properly       |
| 334 | Clutch Fault: Prox Sensor Off after Mechanical Transition. Probably Clutch Mechanism Bounce          |

|     |                                                                                             |
|-----|---------------------------------------------------------------------------------------------|
| 340 | Clutch Fault: Mechanical Disengagement was not Successfully Completed after 15 Attempts     |
| 345 | Warning: Clutch Disengaged the Mechanism but the Prox Sensor is still Off                   |
| 350 | Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move              |
| 371 | Auxiliary Encoders (at Outfeed & Infeed) are not Aligned to each other on the Buffer Shaft  |
| 372 | Auxiliary Encoder C (Infeed side) is not Aligned to Motor Encoder on Buffer Shaft           |
| 373 | Auxiliary Encoder D (Outfeed side) is not Aligned to Motor Encoder on Buffer Shaft          |
| 374 | Auxiliary Encoders Offsets have reached the Cumulative threshold                            |
| 375 | Auxiliary Encoder C (Infeed side) Coupling Slip was Detected on Buffer Shaft                |
| 376 | Auxiliary Encoder D (Outfeed side) Coupling Slip was Detected on Buffer Shaft               |
| 377 | Buffer Motor Gearbox / Shaft Slip Detected. Critical Fault. Inspect Mechanics.              |
| 378 | Buffer Motor Gearbox / Shaft and Coupling Slip Detected. Critical Fault. Inspect Mechanics. |
| 400 | Communication Loopback TimeOut (Sys_Flt)                                                    |
| 401 | Communication Loopback TimeOut (Sys_Flt)                                                    |
| 402 | Timeout Clutch Fault - Waiting for Clutch to be Disengaged to Resume (Sys_Flt)              |
| 512 | Infeed Middle Photocell was not Blocked during an Infeed Case Move                          |
| 515 | Unexpected Case entering onto the Infeed Belts                                              |
| 516 | The Length of the Infeed Case (on position 0) is too long: Expected mm. Measured mm.        |
| 517 | The Length of the Infeed Case (on position 0) is too short: Expected mm. Measured mm        |

|     |                                                                                       |
|-----|---------------------------------------------------------------------------------------|
| 518 | Infeed Case (on position 0) is too far downstream by {4}mm                            |
| 519 | Infeed Case (on position 0) is too far upstream by {4}mm                              |
| 520 | Infeed Fault: Previous Length Check was not Completed                                 |
| 525 | Pop-up Shaft Reached Final Upper Limit - Probably did not Engage the Pop-Up Mechanism |
| 555 | E-Stop Condition Active - Safety is Disarmed (Door Opened or E-Stop Pressed)          |
| 556 | Open Door Preventing return to RUN Status (Door# 1-4)                                 |
| 570 | UPS Power Supply on Battery (Power Loss)                                              |
| 571 | Mico Circuit Tripped (24V Short-Circuit)                                              |
| 572 | Air Pressure Switch Off - No Air in System                                            |
| 573 | Input/Output Murr Module Fault (Read Variable Murr_Flt)                               |
| 575 | 12-Volt Power Supply for Auxiliary Encoders Not OK                                    |
| 576 | Warning: UPS Battery Faulted                                                          |
| 577 | Warning: UPS Battery Low                                                              |
| 660 | Motion Control Following Error - Critical Fault                                       |
| 661 | Motion Control Following Error - Axis A (Infeed Conveyor) - Critical Fault            |
| 662 | Motion Control Following Error - Axis B (Sequence Conveyor) - Critical Fault          |
| 663 | Motion Control Following Error - Axis C (Buffer Conveyor) - Critical Fault            |
| 667 | Motion Control Following Error - Axis A B & C (All) - Critical Fault                  |

|     |                                                                                             |
|-----|---------------------------------------------------------------------------------------------|
| 671 | Galil Program Thread Command Error                                                          |
| 701 | Clutch Prox / Sensor Not On Before Starting Routine I/O Check                               |
| 702 | Clutch Prox Not Off after Clutch Activation - Routine I/O Check                             |
| 703 | Clutch Prox Not On after clutch De-activation - Routine I/O Check                           |
| 706 | Missing Y Pop-Up Flag (H) after Lift - Routine I/O Check                                    |
| 707 | Z Pop-Up Flag (HH) Present after Lift - Routine I/O Check                                   |
| 708 | Missing W Pop-Up Flag (LL) After Pop-Up De-activation - I/O Check                           |
| 709 | Buffer Lane Photocell Blocked during Routine I/O Check                                      |
| 715 | Case Detected at Far Back of Buffer Lane                                                    |
| 719 | Clutch Prox Sensor Off during Routine I/O Check                                             |
| 725 | Photocell above Lane was Blocked on Activated Lane                                          |
| 781 | Infeed Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault            |
| 782 | Sequence Conveyor Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault |
| 783 | Buffer Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault            |
| 791 | Motion Control (Galil) Thread Crashed while Main Thread was Waiting - Critical Fault        |
| 900 | Warning: Back Edge of Case was Adjusted after the Case Stored                               |
| 901 | Warning: Back Edge of Case was Adjusted Outward after Case Dispense (Back Edge not Seen)    |
| 902 | Warning: Back Edge of Case was Adjusted Outward after Case Dispensed                        |

|     |                                                                                      |
|-----|--------------------------------------------------------------------------------------|
| 903 | Warning: Back Edge of Case was Adjusted Inward after Case Dispensed                  |
| 904 | Warning: Pulled Buffer Lane Cases Inward after Case Store (Photocell was Blocked)    |
| 905 | Warning: Pulled Buffer Lane Cases Inward after Case Dispense (Photocell was Blocked) |



## Quick Reference Tables

### Level Status Colors

| Color    | Status    | Meaning                   | Action Available            |
|----------|-----------|---------------------------|-----------------------------|
| Green    | Automatic | System operating normally | Can put in Manual           |
| Yellow   | Manual    | System paused             | Can put in Auto             |
| Red      | Fault     | Active alert              | Must resolve alert          |
| Dark Red | E-Stop    | Safety system engaged     | Clear E-Stop condition      |
| Blue     | Manual    | Communication Loss        | Re-establish communications |



### Alert Severity Levels

| Severity | Description | Typical Response                 |
|----------|-------------|----------------------------------|
| Warning  | Minor issue | Monitor, may not stop production |

|                 |                |                                                                               |
|-----------------|----------------|-------------------------------------------------------------------------------|
| <b>Error</b>    | Standard Alert | Production stopped on that level, buzzer sounds, requires attention.          |
| <b>Critical</b> | Major issue    | Production stopped on all levels, buzzer sounds, requires immediate attention |



### Common Alert Categories

| Category                | Alert Codes                       | Description                  |
|-------------------------|-----------------------------------|------------------------------|
| <b>System</b>           | 99, 400-401, 570, 572-573, 791    | System-level faults          |
| <b>Safety</b>           | 555                               | E-Stop and safety interlocks |
| <b>Storage/Dispense</b> | 109, 113, 115, 117, 119, 121, 123 | Lane operations              |
| <b>Pop-Up</b>           | 152-156, 525                      | Pop-up mechanism             |
| <b>Clutch</b>           | 310, 333-334, 340                 | Clutch assembly              |
| <b>Infeed</b>           | 512, 515, 517                     | Inbound case issues          |
| <b>Photocell/Sensor</b> | 709, 719                          | Sensor monitoring            |
| <b>Outfeed</b>          | (no code)                         | Output positioning           |

See [Alert Reference](#) for complete list with detailed explanations.



### User Roles and Permissions

| Role           | Access Level | Capabilities                            |
|----------------|--------------|-----------------------------------------|
| <b>Admin</b>   | Full         | All features, settings, user management |
| <b>Manager</b> | Operational  | Operations, not site config             |
| <b>Clerk</b>   | Production   | Monitor and operate, limited config     |
| <b>Guest</b>   | View-only    | Browse screens, no changes              |

See [User Manager](#) for details.

---

## Common Tasks Quick Reference

| Task                       | Where to Go                                           |
|----------------------------|-------------------------------------------------------|
| Check system status        | <a href="#">Home Page</a>                             |
| View current alerts        | <a href="#">Alerts Page</a>                           |
| Check replenishment status | <a href="#">Inventory &gt; Replenishments</a>         |
| View lane contents         | <a href="#">Inventory &gt; Lane Inventory</a>         |
| Track specific case        | <a href="#">Case Sequence Page</a>                    |
| Change system settings     | <a href="#">Administration</a>                        |
| Resolve an alert           | <a href="#">Troubleshooting &gt; Alert Reference</a>  |
| Replace a case             | <a href="#">Troubleshooting &gt; Case Replacement</a> |

---

## InnoPick Terminology

| Term                     | Definition                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------|
| <b>Accumulation</b>      | Buffer space between InnoPick and merge or palletizer                                      |
| <b>Case Sequence</b>     | Order in which cases are scheduled to exit InnoPick                                        |
| <b>Clutch</b>            | Mechanical assembly that engages lane with pop-up                                          |
| <b>Dispense</b>          | Moving case from storage lane to sequence conveyor                                         |
| <b>Infeed</b>            | Where cases enter InnoPick from upstream conveyor system                                   |
| <b>Inventory Graph</b>   | Visual display of case positions on sequence conveyors                                     |
| <b>Lane</b>              | Storage conveyor that holds buffer inventory                                               |
| <b>Level</b>             | Complete operational unit with sequence conveyor and storage lanes                         |
| <b>Outfeed</b>           | Where cases exit InnoPick to downstream equipment                                          |
| <b>Pop-Up</b>            | Assembly that lifts cases from the sequence conveyor, used for both storing and dispensing |
| <b>Replenishment</b>     | Quantity of product sent to replenish InnoPick                                             |
| <b>Sequence Conveyor</b> | Main conveyor that indexes cases position by position                                      |
| <b>SKU</b>               | Stock Keeping Unit (product code)                                                          |

---

# System Architecture Summary

## Case Flow

1. Cases arrive at **Infeed** from depalletizer
  2. Cases advance on **Sequence Conveyor**
  3. Cases **Store** into **Storage Lanes** via pop-up and clutch
  4. Cases remain in lanes until needed
  5. Cases **Dispense** back to **Sequence Conveyor**
  6. Cases advance to **Outfeed**
  7. Cases exit to downstream palletizer
- 

## Related Topics

- [Getting Started](#) - For new users
  - [Daily Operations](#) - Routine procedures
  - [Main Screens](#) - Detailed feature reference
  - [Troubleshooting](#) - Problem resolution
- 

**Navigation:** [← Disabled Lane](#) | [Home](#) [→](#)